B787 IOE Workbook

Answered, in flashcard form

Source: B787 OE Workbook, Version 3, June 2026.

Workbook questions	**342**
Answered from the manuals	**330**
Not answered in our manuals	**12**
Critical Observable Items	**120**
Drill cards / walkthrough items	**323 / 19**
Manuals used	**FOM Rev 125.1 (8/12/26), FCOM R10, QRH R7**
Reference	**ICAO NAT Doc 007 V.2026-1, where our manuals are silent**
Compiled	**August 2026**

How to read this document

Each question carries the answer in green, and nothing else. That green line is what you should be able to say cold. Key numbers are set larger so they land at a glance. Everything under the answer is the backup: the full answer, the manual section, a verbatim quote where one exists, and a note.

Questions are grouped by topic in roughly the order you meet them on a trip, preflight to shutdown, with the long-haul topics in the middle. The tag above each question carries the grade sheet item it came from.

An answer in amber is not published anywhere in the FOM, FCOM or QRH. It names the manual, card or system that holds it, or says plainly that no requirement exists. Nothing was guessed.

Two questions ask for the flight deck door entry code and the crew rest door lock combination. Those are deliberately not recorded. They are security sensitive and do not belong in a study document.

Numbers to Know

The facts on these cards worth knowing cold before the first leg.

How many may use the OFCR?	**2 occupants**
FD Pro updates completed?	**Yes, synchronized within 24 hours of starting a trip**
Are there high mins Captain restrictions?	**Yes, relieved by Exemption 5549**
B043 vs B044 flight plan?	**B043 international fuel reserve, B044 planned redispatch**
Domestic dispatch fuel requirement?	**Destination, alternate, plus 45 minutes**
When do you declare MAYDAY FUEL?	**When you will land with less than 30 minutes**
Max crosswind for a HUD takeoff?	**20 knots crosswind**
Severe turbulence penetration speed?	**290 KIAS below FL250, 310 or .84 above**
Minimum altitude for autopilot engagement?	**200 ft AGL after takeoff**
How far do you stand off severe cells?	**20 nm laterally; clear tops by 5000 ft**
Lowest visibility for CAT I?	**1800 RVR**
Lowest visibility for CAT II?	**1000 RVR**
Lowest visibility for CAT II SA?	**RVR 1200 (350 m), DH not below 100 ft**
Max breakaway thrust N1?	**40% N1 maximum for breakaway**
Pushing back late. When is a PA required?	**At scheduled departure time, then every 15 minutes**
Max taxi speed?	**30 kts straight and smooth, 10 kts congested**
Normal 180 degree turning radius?	**144 feet wing tip radius; 154 feet pavement needed**
Pivot 180 degree turning radius?	**150 foot runway with light differential braking and thrust**
Engine cool requirement before shutdown?	**At least 3 minutes at taxi thrust**
Which radio for ATC, which for ramp control?	**ATC on #1/L VHF, ramp control on #2/R**

What is a Runway Condition Code?	**The TALPA number, 0 through 6**
When do you make cold weather corrections on approach?	**Airport temperature 0 degrees C or colder, when charted**
What are icing conditions?	**OAT or TAT 10C or below with visible moisture/contamination**
When must engine anti-ice be on?	**Whenever icing conditions exist or are anticipated, above -40C**
Max altitude with flaps extended?	**20,000 ft**
Normal rotation rate?	**2 to 2.5 degrees per second**
Normal takeoff rotation target pitch?	**15 degrees pitch attitude**
Rotation target pitch after an engine failure at V1?	**12 to 13 degrees, 2 to 3 below normal**
What are the PM callouts during an RTO?	**SPEEDBRAKES UP, REVERSERS NORMAL, 60 KNOTS, plus omitted items**
How do you find max rate of climb speed?	**Approximately VREF 30 plus 140 knots until 0.84 Mach**
Transition altitude and transition level?	**18,000 ft MSL in the US, charted elsewhere**
Any Dispatch call required after takeoff, before the EEP?	**Yes. 30 to 60 minutes after takeoff**
ETOPS diversion speed?	**.84M/290 KIAS, the one-engine-inoperative cruise speed**
Is your ETOPS distance 120 or 180?	**180 minutes MDT; as dispatched on the Release**
What altitudes are RVSM airspace?	**FL290 through FL410**
Diverting without a clearance. What altitude do you plan?	**Below FL290**
How early do you log on to CPDLC leaving the contiguous 48?	**About 30 minutes prior to departure**
How far can you offset with SLOP?	**Up to 2 nm right of centerline. Never left**
What is a TA?	**Traffic advisory: traffic predicted to conflict in 20-48 seconds**
What is an RA?	**Resolution advisory: conflict predicted in about 15-35 seconds**

What is MedLink?	**Contract ER physicians giving 24-hour inflight medical advice**
How do you make a MedLink call from the aircraft?	**SATCOM to (602) 282-4910, or ARINC phone patch**
What altimeter setting defines a level?	**Standard, 29.92 in Hg or 1013.25 hectopascals**
When do you stow the thrust reversers?	**Start at 60 knots, reverse idle before taxi speed**
Factored vs unfactored landing data?	**Factored adds margin. The 787 adds 15 percent**

Contents

Emergency Equipment (2)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

What flight deck emergency equipment must you check?

Fire extinguisher, crash axe, overhead door, plus smoke and escape gear

Nine items, all on the Preliminary Preflight. Then emergency descent devices, protective breathing equipment, gloves, AVSAX, life vest and flashlights, each stowed.

> **FCOM NP.21.1 Preliminary Preflight Procedure - Captain or First Officer**
>
> *"Fire extinguisher ....Checked and stowed Crash axe ....Stowed Flight deck overhead door .... Closed and latched Emergency Descent Devices ....Stowed PBE ....Stowed Gloves ....Stowed November 1, 2024 787 FCOM - HAWAIIAN AIRLINES NP."*
>
> The overhead door is checked closed and latched, not just present. It is your escape route out the top of the flight deck.
>
> *Workbook wording: What flight deck emergency items are required to be checked?*

Cabin Emergency Equipment Locations (Megaphone, ELT, etc.)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where are the passenger cabin emergency equipment locations?

FCOM Emergency Equipment section, passenger cabin diagram

FCOM Chapter 1 Section 45 covers emergency equipment, and the passenger cabin location diagram sits at FCOM 1.45.8, immediately after the flight deck diagram at 1.45.7 and the symbols page at 1.45.6. The ELT text at FCOM 1.45.1 explicitly refers you to that diagram for ELT locations.

> **FCOM 1.45.8 Emergency Equipment Locations - Passenger Cabin**
>
> *"ELTs are installed in the passenger cabin, as shown in the Emergency Equipment Locations – Passenger Cabin diagram."*
>
> It is a graphic, so text search will not find individual items. The flight deck companion diagram is FCOM 1.45.7 and the symbol legend is FCOM 1.45.6. Crew rest equipment is separate: OFCR at FCOM 1.46.17, OFAR at FCOM 1.47.13.
>
> *Workbook wording: Where can the "Emergency Equipment Locations – Passenger Cabin" be found?*

Security and Doors (4)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where is the Jumpseat and Rest Facility Briefing Card?

Carried on the electronic flight bag; not required for flight

The FOM lists the Jumpseat Card on the electronic flight bag and marks it not required for dispatch. The Captain still owes the occupant a briefing: seat and harness, communication gear, exits, emergency equipment, sterile flight deck, and flight deck door security. The card only helps you cover it.

> **FOM 5.1.22.1.2.3 Flight Deck Jumpseat - Occupant Briefing**
>
> *"The Jumpseat Briefing Card is not required for flight, but it may be used to assist this briefing:"*
>
> On the 787, if two Overhead Flight Attendant Rest bunks are used as alternate pilot bunks, a separate laminated briefing card is supplied to the Cabin Crew.
>
> *Workbook wording: Where is the Cockpit Jumpseat and/or Rest Facility Briefing Card located?*

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

How do you escape through the 787 overhead door?

Remove the cover, rotate the lock handle, then use descent devices

The door is hinged to fall inward and hang from the ceiling, so stay clear as it drops. It only opens on the ground with the airplane depressurized. Slide the descent device compartment latch open, pull a device out, and hold its handle through every step until you are on the ground.

> **FCOM 1.40.19 Flight Deck Overhead Door**
>
> *"The flight deck overhead door is used for emergency egress using the installed descent devices."*
>
> Climb out using the first observer seat, the observer console, the fold-down step in the bulkhead, and the hinged area of the door.
>
> *Workbook wording: How would you operate/escape through the 787 Flight Deck Overhead Door?*

Flight Deck Door/Overhead Door Operation (787)

1.2.1 PERFORM PREFLIGHT PREPARATION

Where is the 787 overhead door described?

FCOM Doors systems description, Flight Deck Overhead Door

FCOM 1.40.19 gives the description and operation. The door is for emergency egress using the installed descent devices and can only be opened on the ground with the airplane depressurized. Remove the protective cover and rotate the door lock handle; it swings down inside the flight deck on hinges.

> **FCOM 1.40.19 Flight Deck Overhead Door (controls at FCOM 1.30.30)**
>
> *"Flight Deck Overhead Door The flight deck overhead door is used for emergency egress using the installed descent devices. The door can only be opened on the ground with the airplane depressurized."*
>
> Two places: FCOM 1.30.30 Flight Deck Overhead Door and Vent for the cover, handle and components, and FCOM 1.40.19-1.40.21 for description, operation and the egress route. FCOM NP.21.1 requires it closed and latched at preflight.
>
> *Workbook wording: Where can you find the 787 forward overhead door description, operation, components and egress procedures?*

Flight Deck Entry Procedures/Security

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Flight deck door procedure with no flight attendants?

Yes. Captain covers cabin duties; door closed, critical phases

FOM 16.1.11 states that when Flight Attendants are not on board a Company-operated flight, the Captain must ensure the cabin duties normally done by F/As are accomplished, and refers to fleet-specific manuals for procedures. The WARNING requires galley carts to be properly secured and the Flight Deck Door closed during critical phases of flight.

> **FOM 16.1.11 Pilot Duties With No Flight Attendants**
>
> *"properly secured and the Flight Deck Door is closed during critical phases of flight."*
>
> FOM 15.2.6.2 sends you to Pilot Duties With No Flight Attendants at 16.1.11, which then refers to the fleet manuals for the actual procedure.
>
> *Workbook wording: Are there special Flight Deck Door Procedure on Flights with No Flight Attendants?*

Crew Rest (5)

Crew Rest Procedures/Door Operation

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where are the crew rest requirements?

Flight Operations Manual chapter 5, Crew Rest Facility

Use is limited to Flight Crew on the Hawaiian seniority list, one occupant per bunk, no food or drink except bottled water. It cannot be occupied from pushback to 10,000 ft above field elevation, or from 45 minutes before landing to block in.

FOM 5.1.23 Crew Rest Facility

> *"The purpose of the crew rest facility is to provide a quiet place for the crew to rest. It is not a lounge, a meal area, or a place to stow personal carry-on items."*

The 787 primary facility is the Overhead Flight Crew Rest, maximum two occupants. The Overhead Flight Attendant Rest is the backup, limited to two labeled alternate pilot bunks for one segment.

Workbook wording: Where can the Crew Rest procedures/requirements be found?

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

What is the OFCR/OFAR door lock combination?

Not recorded. Security sensitive

FCOM 1.46 (OFCR) and FCOM 1.47 (OFAR) describe the entrance enclosure keyless lock: a door handle, a button keypad and a reset button. To unlock you enter the airline-programmed code and rotate the handle either direction; releasing the handle re-latches the bolt so the door can be closed without retracting it.

FCOM 1.46 Overhead Flight Crew Rest - Door Lock Operation (also FCOM 1.47 OFAR)

> *"To unlock the door, enter the airline-programmed code and rotate the door handle in either direction to retract the door bolt. When the door handle is released, the bolt automatically returns to the latched position allowing the door to be closed without retracting the bolt."*

The FCOM only describes the keyless lock as taking an 'airline-programmed code' and never prints the digits. Actual combinations are Sensitive Security Information under FOM 15.1 Security Policy and are not in the FCOM, FOM or QRH.

Workbook wording: 787 - What is the OFCR/OFAR Door lock combination?

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

How many may use the OFCR?

2 occupants

The primary 787 crew rest facility is the Overhead Flight Crew Rest (OFCR); the backup is the Overhead Flight Attendant Rest (OFAR). If flight time or FDP requires it the Company may provide up to two bunks in the OFAR, labeled alternate pilot bunks, with a laminated briefing card supplied to the Cabin Crew.

FOM 5.1.23.7 787 Crew Rest Facility

> *"No more than 2 occupants are authorized to use the OFCR at any time."*

The OFCR seat, bunks and common area may not be occupied during taxi, takeoff or landing. 787s departing a domicile or outstation with an Augmented, Double Augmented or Double Crew must carry at least 4 prepackaged bedding packets (sheet, blanket, pillow and case).

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

What is the flight deck door entry code?

Not recorded. Security sensitive

FOM 15.2.6.3 states the keypad control device shall not be used for normal entry and that entry attempts using the keypad shall be denied by the Flight Crew. FOM 15.2.6.3 also warns that an unexpected keypad activation signal must be treated as a Level 4 hijack or attempted hijack in progress until the Flight Attendants say otherwise.

FOM 15.2.6.3 Keypad Control

> *"The keypad control device shall not be used for normal entry. Entry attempts using the keypad shall be denied by the Flight Crew. For A330F exception, see 15.2.6.10 – Flight Deck Access without Functioning Service Interphone."*

The code is not published in FOM, FCOM or QRH; it is SSI (FOM 15.1.1.1 lists security access codes as SSI). Note the policy point: the keypad is not for normal entry and keypad attempts must be denied.

Workbook wording: What is the Flight Deck Door entry code?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Which PA reaches the cabin but not crew rest?

Individual passenger address areas on the cabin interphone directory page

The passenger address mic switch on the audio control panel keys every area at once, crew rest included. To leave the bunks out, go through the cabin interphone directory or the flight deck handset and pick only the cabin areas you want.

> **FCOM 5.30.2 Passenger Address System**
>
> *"Pushing a PA mic switch on an audio control panel provides direct access to all PA areas."*
>
> The directory carries CREW REST as its own entry, separate from the passenger address areas. That is the one you avoid selecting when the off duty pilot is asleep.
>
> *Workbook wording: What selection should you use to make a PA throughout the cabin, but NOT to the crew rest facilities?*

EFB and FD Pro (22)

EFB Review

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

EFB IOS current and updated?

Yes, iOS at or above Manual Currency minimum version

The EFB operating system and every required app must be at or above the minimum version listed on the Manual Currency page. The CrossCheck app shows the current versions. Updates are not required or encouraged while on a trip.

> **FOM 2.5.1.2 EFB, App, and Document Currency**
>
> *"The iOS and all required apps shall be at or above the minimum version number listed on Manual"*
>
> Check CrossCheck before you leave home, not at the gate.
>
> *Workbook wording: EFB IOS - Current/updated.*

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro updates completed?

Yes, synchronized within 24 hours of starting a trip

Within 24 hours of starting a trip or training event, all flight-related apps must be synchronized so charts and databases are current, and all apps must meet minimum currency. Required memos, revisions, and safety flashes must also be read and acknowledged.

> **FOM 2.5.1.4 EFB - Prior to Flight**
>
> *"all flight-related apps must be synchronized to ensure all charts and databases are current"*
>
> Never run app or iOS updates over Flight Deck internet.
>
> *Workbook wording: FD Pro: updates completed.*

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: open Weather and the Weather Layers pane

Weather Company layers drawn on the enroute map

Per Jeppesen's official FliteDeck Pro X material, not the company manuals. The Weather button opens the Weather Layers pane, and the layers are Weather Company data overlaid directly on the enroute map. The greyed info buttons give each layer's legend and valid time on the device.

> **Jeppesen FliteDeck Pro X Fact Sheet**
>
> *"Weather Layers, provided by The Weather Company, contribute to improved flight safety and operational efficiency by enabling flight deck crew to make better decisions based on better weather information depicted on their enroute map."*
>
> Company manuals do not cover app operation. Confirm pane layout on your own EFB during OE.
>
> *Workbook wording: FD Pro: "Weather" (bottom/left of screen), "Weather Layers" pane/"I" (greyed-out "info" button/s, RT...*

CRITICAL OBSERVABLE ITEM
1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Select the layers you want on the Weather Layers pane

Tap each weather type to display it

From Jeppesen's official FliteDeck Pro iOS user guide, not the company manuals. With weather on, you tap the weather types you want and they render over the enroute map. Layer names and grouping in the pane vary with app version and airline weather subscription.

> **Jeppesen FliteDeck Pro User Guide**
>
> *“Tap the weather type that you want to view.”*
>
> Wording is from the Jeppesen iOS guide for an earlier app version. Verify the Pro X pane on the device.
>
> *Workbook wording: Select desired layer/s on “Weather Layers” pane...*

CRITICAL OBSERVABLE ITEM
1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

REFRESH on the Weather Layers pane

Forces an update; weather auto checks every six minutes

Per Jeppesen's official FliteDeck Pro iOS user guide, not the company manuals. Displayed weather is checked against the Jeppesen weather server automatically about every six minutes, and the Refresh control forces an immediate manual update of the selected weather.

> **Jeppesen FliteDeck Pro User Guide**
>
> *“Weather information for the selected weather type is automatically checked every six minutes. You can also manually update a selected weather type by tapping the Refresh button at the bottom of the Weather popover.”*
>
> Interval is from the Jeppesen iOS guide for an earlier app version. Confirm current behavior on the device.
>
> *Workbook wording: “REFRESH” (bottom/right) on “Weather Layers” pane...*

CRITICAL OBSERVABLE ITEM
1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: verify the correct aircraft

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

> **FD Pro Pubs Quick Reference (Comply365)**
>
> *Workbook wording: FD Pro: Verify correct “aircraft” on top/center of screen...*

CRITICAL OBSERVABLE ITEM
1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: flight plan upload demo

Loads origin, destination, alternates, and full route

Per Jeppesen's official FliteDeck Pro user guide, not the company manuals. An uploaded flight plan from the company planning system brings in origin, destination, alternates, and the full route string with any user waypoints, selected from the flight plan list in the app.

> **Jeppesen FliteDeck Pro User Guide**
>
> *“A flight plan route includes origin, destination, any alternate airports, and the route description, including any user waypoints.”*
>
> Demonstrate the actual upload on the device during OE.
>
> *Workbook wording: FD Pro Flight plan upload demo...*

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

FD Pro: transfer a flight plan to another EFB

Both EFBs need Allow Flight Sharing on

Per Jeppesen's official FliteDeck Pro user guide, not the company manuals. You share the active flight from the flight info area, pick the other pilot's device from the nearby list, and both EFBs must be running the app with Allow Flight Sharing enabled.

> **Jeppesen FliteDeck Pro User Guide**
>
> *“To share flight information between two iPads, switch Allow Flight Sharing to ON for both iPads.”*
>
> Guide text is for an earlier iOS app version. Confirm the Pro X share flow on the device.
>
> *Workbook wording: FD Pro Flight plan transfer to another EFB demo...*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

How do you know the nav database is current?

Check the ACTIVE date range on the FMS IDENT page

The identification page shows the effective date range of the active navigation database, with the inactive cycle on the line below. Preflight has you cycle the active database to erase old pilot entries, then confirm the ACTIVE range covers today.

> **FCOM 11.40.4 Identification Page**
>
> *“Verify that the navigation database ACTIVE date range is current.”*
>
> If a database expires in flight, keep using it. The date can only be changed on the ground, after landing.
>
> *Workbook wording: How can you tell if the Nav database is current?*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Can you change the nav database?

Yes. Swap in the inactive cycle on the IDENT page

Select the inactive date range to copy it into the scratchpad, then select the ACTIVE line to move it up. The old active date drops to the inactive line. Ground only, and it erases all previously entered route data, so reload the route afterward.

> **FCOM 11.40.4 Identification Page**
>
> *“If the active navigation database is out of date, it can be changed to the inactive navigation database.”*
>
> The REF NAV DATA page only displays waypoint, navaid, airport and runway data. It does not change the database.
>
> *Workbook wording: Can you change the Nav database? How?*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

How do you know the AMM database is current?

CrossCheck app lists required versions; sync all apps before the trip

Our manuals give no moving map specific expiry display. Within 24 hours of starting a trip every app on the electronic flight bag must meet minimum currency and be synchronized so all charts and databases are current. CrossCheck and the Manual Currency page show which versions are required.

> **FOM 2.5.1.4 EFB - Prior to Flight**
>
> *“all apps on the EFB must meet minimum currency requirements, all flight-related apps must be synchronized to ensure all charts and databases are current, and all required operational memos, manual revisions/bulletins, and safety flashes must be read and acknowledged or signed as applicable.”*
>
> The airport moving map rides inside Jeppesen FD Pro. If that app will not run, the Jeppesen Trip Kit is the approved backup.
>
> *Workbook wording: 787 – How can you tell the AMM database is current?*

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Must both pilots have the FD Pro taxi chart out?

Yes, both pilots reference airport diagrams during taxi

Both pilots are required to reference the appropriate airport diagram during taxi and follow the aircraft's progress on it against the ATC clearance. The taxiing pilot should stay primarily heads-up while doing so. If a mount is broken the EFB may still be used to keep the diagram in view.

> **FOM 5.3.4.3 During Taxi**
>
> *"Both pilots shall reference the appropriate airport diagrams during taxi."*
>
> If any doubt about the clearance, stop the aircraft and ask.
>
> *Workbook wording: Do both crewmembers must have the FD Pro TAXI chart out and available during surface movements?*

Cold Weather Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

SureWx with active precipitation or a contaminated surface?

In the SureWx guide and the FOM

SureWx holdover-time workflow is vendor material. The governing policy is the pretakeoff contamination check requirement. Verify the app steps in the SureWx guide and the de-icing policy in FOM Chapter 9 and the fleet cold weather supplement.

> **SureWx guide / FOM Ch 9**
>
> *Workbook wording: While using SureWx app, if/when there is active precipitation or during contaminated surface operations,...*

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

SureWx holdover time expired. What check is required?

Pretakeoff contamination check within five minutes of takeoff

This is FAA regulation, not company text. Once any holdover time is exceeded, including a SureWx LWE time, takeoff is permitted only after a pretakeoff contamination check confirming critical surfaces are free of frost, ice, and snow. The check must be completed within five minutes of beginning takeoff.

> **14 CFR 121.629**
>
> *"check to make sure the wings, control surfaces, and other critical surfaces, as defined in the certificate holder's program, are free of frost, ice, and snow. It must be conducted within five minutes prior to beginning take off."*
>
> FAA winter guidance adds that in the demanding cases the check is visual and tactile on the critical surfaces. Company deicing detail lives in the FODM and the SureWx guide.
>
> *Workbook wording: While using SureWx app and the holdover time has expired, what check is required prior to takeoff?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Where is the AIREP form in Comply365?

In Comply365

Document location inside Comply365 changes with releases. Verify in the live app rather than trusting a written path.

> **Comply365**
>
> *Workbook wording: Where in Comply365 can the AIREP form be found?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Do you complete an AIREP every flight?

Not stated in our manuals. Comply365 form

AIREP completion and retention, CPDLC request types, and plot saving or submission are company workflow and aircraft menu content. FOM 5.7.3.15 requires the entire cleared route be loaded for ORCN segments but does not mandate saving plots after every flight. Verify in Comply365 and on the aircraft.

> **Comply365 / FCOM Ch 11 / FOM 5.7.3.15**
>
> *Workbook wording: Do you need to complete the AIREP form for each flight?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Do you save the AIREP after each flight?

Not stated in our manuals. Comply365 form

AIREP completion and retention, CPDLC request types, and plot saving or submission are company workflow and aircraft menu content. FOM 5.7.3.15 requires the entire cleared route be loaded for ORCN segments but does not mandate saving plots after every flight. Verify in Comply365 and on the aircraft.

> **Comply365 / FCOM Ch 11 / FOM 5.7.3.15**
>
> *Workbook wording: Do you need to save the AIREP form after each flight?*

CPDLC Menus

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What can you request over CPDLC?

Altitude, route, speed, and voice contact requests

The ATC menu downlink pages let you build level, route, speed, clearance, when can we expect, and voice contact requests, plus free text. Position reports also go over the link. ATC replies by uplink and the crew accepts, rejects, or sends standby.

> **FCOM 5.40 ATC Datalink**
>
> *“route changes, speed and vertical clearances, and voice contact requests can be sent or received as appropriate”*
>
> Level and In Trail Procedure requests are built automatically when created from the ITP display.
>
> *Workbook wording: What types of requests can be made through CPDLC?*

FD Pro ICAO Procedures

1.1.1.16.4 KNOW INTERNATIONAL PROCEDURES (AS APPLICABLE) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Where in FD Pro are the ICAO country procedures?

Pubs, then the region's Airway Manual, State Rules and Procedures

Open Pubs in FD Pro, pick the region, open that region's Jeppesen Airway Manual, then State Rules and Procedures for the country. The FOM sends you there for country specific holding, and its regional chapters use the same path for Canada, Pacific, and Latin America material.

> **FOM 5.5.15.4 International Holding Procedures**
>
> *“For country-specific holding procedures and airspeeds, see that country’s State Rules and Procedures under the applicable region’s Jeppesen Airway Manual in Jeppesen FD Pro on the EFB.”*
>
> FOM chapter 20 lists suggested FD Pro paths for the Pacific region item by item.
>
> *Workbook wording: Where in FD Pro can you find the ICAO (country) procedures for destinations that we serve?*

1.1.1.16.4 KNOW INTERNATIONAL PROCEDURES (AS APPLICABLE) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Where in FD Pro are the ICAO emergency procedures?

Region specific Emergency section of the Jeppesen Airway Manual

Under Pubs in FD Pro, open the region's Jeppesen Airway Manual and go to the Emergency section. The FOM points there for ICAO communication failure procedures, which differ by country. Review it for each international destination before you need it in the air.

FOM 11.1.4.2 Loss of Communication in International Airspace

> *"found in the Jeppesen Airway Manual, region-specific Emergency section."*

Workbook wording: Where in FD Pro can you find the ICAO (country) emergency procedures for destinations that we serve?

1.1.1.16.4 KNOW INTERNATIONAL PROCEDURES (AS APPLICABLE) WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the FD Pro Pubs Quick Reference guide

In the FD Pro guide in Comply365

This is a Jeppesen FD Pro X or EFB workflow step, not FOM/FCOM content. Verify against the FD Pro Pubs Quick Reference guide in Comply365 and demonstrate on the actual device during OE. FOM 5.7.3.15 does require the ENTIRE cleared route be loaded into the approved charting app for ORCN segments.

FD Pro Pubs Quick Reference (Comply365)

Workbook wording: Review FD Pro Pubs Quick Reference guide in Comply365...

De/Anti-Ice Procedures

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Discuss the SureWx app

Gives liquid water equivalent based holdover times, FAA authorized

The app reports holdover times built from liquid water equivalent, a direct measure of the water content in frozen precipitation. Those times are more precise and often longer than METAR based tables, and they shall be used unless clear visual evidence shows the data is wrong. This includes heavy snow and freezing rain.

FOM 9.3.1.1 Liquid Water Equivalent (LWE)

> *"If your HOT is derived from LWE, it is FAA-authorized and shall be used unless there is clear, visual evidence that the data is inaccurate"*

The FOM calls it the HOTs app. The flight log deicing block says use the HOTs app; the listed backup is printed holdover times.

Workbook wording: Discuss SureWx app...

Dispatch and Release (13)

Dispatch Release: Review all sections

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

What is in the dispatch release package?

Dispatch release, flight plan, weather package, and NOTAMs

Seven items: the dispatch release including the minimum equipment list, the flight plan, the aerodrome forecast, the routine weather report, special weather and significant meteorological reports with turbulence, Class D notices, and flight data center notices. The load closeout is also required before departure.

FOM 8.5.1 Flight Information Packet/Dispatch Release Package

> *"Dispatch produces the Flight Information Packet/Dispatch Release Package containing required and supplemental data for flight release."*

Your signature on the dispatch release says you reviewed all of it. Before pushback also confirm tail number, equipage and maintenance status match the release.

Workbook wording: What documents are included in the Dispatch/Release package? ("Lucky 7"; in order or printing)

CRITICAL OBSERVABLE ITEM

1.1.2.4 KNOW OPERATIONAL CONTROL (DISPATCH) REQUIREMENTS

Who has operational control of a flight?

The PIC and the Dispatcher, jointly

The PIC and the Aircraft Dispatcher are jointly responsible for preflight planning, delay, and the Dispatch Release, and jointly determine the suitability of weather, field and traffic conditions, airway facilities, and the filed route. Neither can dispatch alone; both must agree.

FOM 2.1.1 Joint Responsibility

> *"jointly responsible for the preflight planning, delay, and Dispatch Release of a flight"*

Regulatory basis 14 CFR 121.533/.535/.537/.551/.557/.627 and Ops Spec A008. Captain-side point: joint responsibility is not shared authority in flight. Once airborne the PIC retains final authority under 2.1.2.

CRITICAL OBSERVABLE ITEM

1.1.2.4 KNOW OPERATIONAL CONTROL (DISPATCH) REQUIREMENTS

What requires Dispatch notification in flight?

Any abnormality or irregularity

Flights experiencing abnormalities or irregularities, including but not limited to those listed in 7.2.3 Pilot Reports Quick Reference, shall contact Dispatch and report the details. Notification is mandatory; the timing flexes to safety of flight. If time does not permit contact in flight, contact shall be made after gate arrival.

FOM 11.1.2 Abnormal Operations Dispatch Notification

> *"Contact Dispatch" implies that while notification is mandatory, such communications should take place as safety of flight considerations permit."*

Rev 125.1 revised 7.2.3 to state the table is not all-inclusive and that the Company may request safety reports for items not listed when additional information is needed. Separately 11.1.3 covers Cabin notification, and 11.2.11 gives the NTSB briefing format for the Flight Attendants.

Workbook wording: What items require Dispatch notification during a flight?

Load Closeout/TPR Update and review procedure

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Who reads the load closeout data?

The First Officer reads the count and jumpseat numbers

Passenger count verification is part of the load closeout procedure. The Captain checks them against the Flight Deck Report. Any discrepancy goes to Load Planning before you go.

FOM 5.2.17.1 Passenger Count Verification

> *"The First Officer reads the passenger count and jumpseat numbers from the Load Closeout."*

The closeout passenger count drives performance. The Flight Deck Report count is what the agents and flight attendants work from, so the two must agree.

Workbook wording: Who reads the closeout data?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Who verifies the closeout, and when?

The Captain, before pushback

The Captain verifies the numbers match the Flight Deck Report, and as Pilot in Command must receive and review the closeout prior to pushback. If a delay or a return to the gate changes passengers or loading, get a revised closeout before takeoff.

FOM 5.2.19 Load Closeout/SABLE Weight and Balance Loadsheet

> *"The Flight Crew will normally receive the Load Closeout via ACARS and the Pilot-In-Command (PIC) must receive and review it prior to pushback."*

On the 787 the closeout normally arrives by ACARS, and the final takeoff performance request uplinks zero fuel weight and takeoff data with it.

Workbook wording: Who verifies it? When?

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

What closeout items must be verified?

Takeoff weight, weight and balance, passenger count, dangerous goods notice

Compare the planned takeoff weight on the dispatch release against the actual takeoff weight on the closeout. Crosscheck the weight and balance and performance numbers, including the flight management system entries, for reasonableness. Verify the passenger and jumpseat count, and confirm the dangerous goods entry is accurate.

> **FOM 5.2.19 Load Closeout/SABLE Weight and Balance Loadsheet**
>
> *"The crosscheck includes comparing the planned Takeoff Weight on the Dispatch Release against the Actual Takeoff Weight indicated on the Load Closeout or SABLE Weight and Balance Loadsheet."*
>
> If the release and closeout takeoff weights differ, apply the dispatch release weight tolerance rules before accepting the numbers.
>
> *Workbook wording: List several items that are required to be verified.*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

No TPR. How do you get takeoff performance?

Use the release Takeoff and Landing Report, or call Dispatch

The Takeoff and Landing Report comes on the dispatch release and flight plan. It is the third choice, behind the final and then the preliminary takeoff performance request. Dispatch can also compute an updated report and send it to the printer, by email, or read it to you.

> **FCOM PI.18.1 Takeoff**
>
> *"Dispatcher: An updated TLR can be calculated and sent to the aircraft printer, email, or relayed verbally to the flight crew."*
>
> Do not build takeoff data with the onboard performance tool. The FCOM prohibits it. The FOM allows new data by ACARS or from Dispatch.
>
> *Workbook wording: How do you obtain takeoff performance if a TPR is unavailable?*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Can you use the TLR for takeoff data?

Yes. It is the takeoff data on the Dispatch Release

FOM 5.4.5 requires takeoff data be available and checked for each takeoff, and names where it comes from: the Flight Information Packet carries the TPR, the Dispatch Release carries the TLR. So the TLR is a legitimate source of takeoff data, not a workaround. Validation criteria live in the fleet manuals, and invalid data is replaced by ACARS or Dispatch.

> **FOM 5.4.5 Takeoff Data**
>
> *"Flight Information Packet (TPR)/Dispatch Release (TLR)"*
>
> Takeoff data must be available and checked for every takeoff. If the data is invalid against the fleet validation criteria, get new data by ACARS or from Dispatch, who has the online performance tool.
>
> *Workbook wording: Can we use the TLR for takeoff information? When would we use it?*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

TLR concerns with temp and altimeter?

Dispatch's assumed temperature and altimeter, not actual conditions

The Takeoff and Landing Report is built on the conditions the dispatcher expected at dispatch and the weight the dispatcher planned. If the actual temperature or altimeter setting has moved, those numbers no longer match the day. Send a fresh takeoff performance request or get updated data from Dispatch.

> **FCOM PI.18.2 Hierarchy Of Takeoff Performance**
>
> *"The TLR is considered the tertiary data source for takeoff performance. The TLR includes the environmental conditions the dispatcher considered at dispatch, and the takeoff weight planned by the dispatcher."*
>
> Numeric validation tolerances are not in the FOM or FCOM. They sit in the AeroData Performance Handbook on the electronic flight bag. Get the exact temperature and altimeter limits from the check airman.
>
> *Workbook wording: Any concerns when using TLR for takeoff information? Temp and altimeter.*

Cruise Altitude Considerations

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Is there a tropopause chart in the release package?

No chart. TROP is a Navigation Log column per waypoint

The Navigation Log columns on the flight plan include TROP, the tropopause level at each waypoint, alongside MORA, TTK, NM, IAS, Mach, FL and TEMP (FOM 8.5.7). That per-waypoint field, not a separate chart, is the released tropopause data. FOM 8.5.1 lists the required contents of the Flight Information Packet and does not include a tropopause chart.

> **FOM 8.5.7 Dispatch Release Sample (HA) - Navigation Log Legend**
>
> *"MORA Minimum Off Route Altitude TROP Tropopause TTK True Track to WPT NM Nautical Miles to WPT IAS Indicated Airspeed to WPT ."*
>
> FOM 8.5.1 lists everything in the Flight Information Packet (release, flight plan, TAF, METAR, SIGMETs and turbulence products, NOTAMs); no tropopause chart is among them. Graphical hazard products from the weather vendor are covered in FOM 9.8.1.
>
> *Workbook wording: Do we have a chart in our dispatch release package that shows the level(s) of the tropopause with...*

SATCOM / HF Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What prompts a SATCOM call to Dispatch?

Pilot discretion; any required report to Dispatch

FOM 7.1.7 makes satellite voice to Dispatch or MedLink purely the pilot's call, so the circumstance is judgment driven: use it when the message is too complex, too urgent, or too consequential for ACARS text. What is not discretionary is the contact itself.

> **FOM 7.1.7 SATCOM (if installed)**
>
> *"Satellite voice communication with Dispatch or MedLink is at the pilot's discretion."*
>
> The mandatory triggers are in FOM 7.1.12 Reports to Dispatch: 3000 lbs fuel deviation (787), 100 nm lateral, 4000 ft altitude, 15 minute arrival delay, weather, moderate or greater turbulence, safety of flight, abnormalities, diversions, mechanical irregularities, ETOPS reroutes.
>
> *Workbook wording: What circumstance precipitates a SATCOM call to Dispatch?*

Analysis of Airport Conditions

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Are there high mins Captain restrictions?

Yes, relieved by Exemption 5549

FOM 5.6.17 cross-references 3.16 Exemption 5549 for high minimums Captain limitations. A newly qualified Captain operates to increased minimums until the experience requirement is met, and Exemption 5549 is the FAA relief that modifies how those high-minimums restrictions apply.

> **FOM 5.6.17 / FOM 3.16 Exemption 5549**
>
> *"See 3.16 – Exemption 5549 for high minimums Captain limitations."*
>
> Read FOM 3.16 for the exact hour threshold and the precise relief Exemption 5549 grants. This is directly relevant to you as a new Captain and is a near-certain oral question, so get the numbers from the PDF rather than from memory.
>
> *Workbook wording: Are there restrictions for High Mins Captains?*

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is FAA Exemption 5549?

Relief allowing High Minimums Captains to fly published minima

Exemption 5549 lets a High Minimums Captain fly to published CAT I minima instead of the increased high-minimums values, provided the approach is flown by the Captain with an autopilot coupled to DH or missed approach (HA wording, FOM 3.16.1).

> **FOM 3.16 Exemption 5549**
>
> *"High Minimums Captains may be dispatched using Exemption 5549 and must comply with the restrictions listed below."*
>
> A High Minimums Captain has under 100 hours PIC in type; without the exemption they add 100 ft and 1/2 sm to published minimums, but no less than 300-1 (FOM 3.17). OE flight hours do not count toward the 100 hours.

Fuel Planning (17)

Dispatch Release: Review all sections

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Where is preflight planning and the international release?

FOM Dispatch chapter, Flight Information and Redispatch section

FOM Chapter 8 (Dispatch) holds the planning material: 8.1 authorized operations and airports, 8.2 alternates, 8.3 navigation, 8.4 weather, and 8.5 flight information documents and redispatch. FOM 8.5.1 defines the Flight Information Packet/Dispatch Release Package and makes the PIC responsible for having it prior to departure from an originating station.

> **FOM 8.5 Flight Information - Documents and Redispatch (8.5.1 Flight Information Packet/Dispatch Release Package)**
>
> *"Dispatch produces the Flight Information Packet/Dispatch Release Package containing required and supplemental data for flight release. The PIC shall review and ensure the following information (as applicable) is in their possession prior to departure from an originating station:"*
>
> The HA international release example and its legend are FOM 8.5.7 and 8.5.7.1. Preflight duties themselves are FOM 5.2; theater-specific international items are FOM chapters 19-24.
>
> *Workbook wording: FOM Preflight Planning and International Dispatch Release...*

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

B043 vs B044 flight plan?

B043 international fuel reserve, B044 planned redispatch

B043 is the Ops Spec authorizing special fuel reserves in international operations, where the 10 percent applies only to planned oceanic/remote continental time and the final reserve is 45 minutes at normal cruise consumption.

> **FOM 8.3.2.3 and FOM 8.5.9 Redispatch Planned**
>
> *"Ops Spec B043"*
>
> Both reduce carried fuel but by different mechanisms: B043 narrows the window the 10 percent applies to; redispatch (B044) shortens the leg the 10 percent is computed against. 8.5.9 lists Ops Specs A002, A004, A008, B044 and B050.
>
> *Workbook wording: What is the difference between a BO44 vs B043 flight plan?*

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Standard release vs redispatch?

Redispatch runs two plans, so you carry less fuel

A standard release dispatches you to the destination on one plan. Planned redispatch is an FAA-authorized flag procedure that minimizes the fuel needed to meet the 10 percent requirement. You operate with two plans: the original dispatch to the intermediate destination, and a second plan from a planned redispatch waypoint to the intended destination.

FOM 8.5.9 Redispatch Planned

> *"The terms of original Dispatch Release must be adhered to until the Captain and the Dispatcher concur on the terms of the redispatch."*

Terminology to have cold: Intended Destination is the ultimate destination from the redispatch waypoint; Intermediate Destination is the one on the original release; the Redispatch Point is the common point on both flight plans where the continue decision is made.

Workbook wording: Discuss the differences between a standard dispatch release vs a Re-Dispatch/Re-Release...

Cruise Altitude Considerations

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Why does the tropopause level matter for cruise?

It sets air temperature; descending below it warms fuel

FOM 9.3.2.3 is the only place the manuals tie tropopause height to cruise altitude selection, and it does so through fuel temperature. Fuel drifts toward TAT at roughly 3 degrees C per hour, up to 12 in extreme conditions. On long cold-soak sectors the tropopause level tells you where warmer air is.

FOM 9.3.2.3 Fuel Freeze Considerations

> *"For instance, a 4000 ft descent below the tropopause can result in approximately a 7°C increase in TAT."*

Tropopause height per waypoint (TROP) is printed on the Navigation Log. FOM 8.3.3 also notes ozone forms primarily above the tropopause, which is why ozone converters are fitted.

Workbook wording: Why is it important to consider the level of the tropopause regarding your cruise altitude?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What do you expect crossing the tropopause?

Temperature stops falling; an inversion may warm you climbing

Near and through the tropopause the normal lapse rate breaks down. Temperature stops decreasing with altitude and can invert, so a climb may actually warm TAT and help a cold fuel problem. A 4000 ft descent below the tropopause is worth about 7 degrees C of TAT (FOM 9.3.2.3).

FOM 9.3.2.3 Fuel Freeze Considerations

> *"Occasionally, a temperature inversion at the Tropopause may occur, and, in this case, climbing will result in warmer temperatures and be beneficial for fuel warming."*

The FOM only addresses the tropopause in temperature/fuel-freeze terms (9.3.2.3) and ozone terms (8.3.3). Turbulence and wind shear expectations near the tropopause are not stated in the FOM, FCOM or QRH.

Workbook wording: What can expect when cruising near, or crossing through, the tropopause?

Analyze Fuel Factors for Diversion

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

What drives normal fuel planning?

Weather, traffic delays, MEL penalties, turbulence and cruise altitude

Six factors. MEL and configuration deviation list limits and penalties. Turbulence and cruise altitude. One instrument approach with a possible missed approach at destination. Anything else that could delay landing.

FOM 8.3.1.1 Fuel Planning Factors

> *"Wind, severe weather, and other forecast weather conditions, and • Anticipated traffic delays, and • MEL/CDL limitations and penalties, and • Turbulence/cruise altitude, and • One instrument approach and possible missed approach at destination, and • Any other conditions that may delay landing."*

If you and the Dispatcher disagree on extra fuel, you have the final call. Agree before fueling starts, and get changes to Dispatch at least 30 minutes before pushback.

Workbook wording: What items must you keep in mind when doing fuel planning (normal flight)?

CRITICAL OBSERVABLE ITEM
5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Domestic dispatch fuel requirement?

Destination, alternate, plus 45 minutes

Flights must be dispatched with enough fuel to fly to the airport to which dispatched. Add fuel to fly to and land at the most distant alternate, plus 45 minutes at last cruise altitude consumption rate. With no alternate required, the reserve is 55 minutes at that rate instead.

> **FOM 8.3.2.1 Domestic Operations**
>
> *"With no alternate, fly for 55 minutes at last cruise altitude consumption rate."*
>
> 14 CFR 121.639. Both reserves are at LAST CRUISE ALTITUDE consumption rate, not holding. Applicable areas at 8.1.1.1.
>
> *Workbook wording: What are the dispatch fuel requirements for domestic operations?*

CRITICAL OBSERVABLE ITEM
5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Flag dispatch fuel requirement?

Destination, 10 percent of trip time, alternate, 30 minutes holding

Flag dispatch fuel is enough to reach the destination, then 10 percent of the total time from departure to landing at destination. Add fuel to the most distant alternate if one is required. Finish with 30 minutes at holding speed 1500 ft above the destination or that alternate.

> **FOM 8.3.2.2 Flag Operations**
>
> *"Fly to and land at the most distant alternate airport, if one is required, then • Fly for 30 minutes at holding speed at 1500 ft above the destination airport or alternate if one is required."*
>
> The 10 percent here is on total flight time. That is what separates it from the B043 special reserve, where the 10 percent counts only oceanic and remote continental time.
>
> *Workbook wording: What are the dispatch fuel requirements for Flag operations?*

CRITICAL OBSERVABLE ITEM
5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Supplemental dispatch fuel requirement?

Domestic or Flag rules, whichever applies

Supplemental (non-scheduled) operations are authorized within the areas approved for enroute operations and are conducted under Domestic or Flag rules per Ops Spec A030 or Exemption 11047 (FOM 8.1.1.3). The Dispatcher must notate on the release whether the flight is being operated as Part 121 Domestic or Flag (FOM 8.5.2). Fuel then comes straight from FOM 8.3.2.1 or 8.3.2.2.

> **FOM 8.1.1.3 Supplemental Operations / FOM 8.5.2 Dispatch Release**
>
> *"Supplemental (non-scheduled) flights shall be dispatched using either Domestic or Flag rules as applicable."*
>
> No separate supplemental fuel rule.
>
> *Workbook wording: What are the dispatch fuel requirements for Supplemental operations?*

CRITICAL OBSERVABLE ITEM
5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

B043 special fuel requirement?

Destination, 10 percent of oceanic time, alternate, 45 minutes cruise

Fuel to the destination, then 10 percent of the planned time in oceanic and remote continental airspace. Add fuel to the most distant alternate if one is required. Finish with 45 minutes at normal cruise consumption, figured at the weight and altitude you would have at top of descent.

> **FOM 8.3.2.3 Special Fuel Reserves in International Operations**
>
> *"10% of the planned time in oceanic/remote continental airspace, then • Fly to and land at the most distant alternate airport, if one is required, then • Fly for 45 minutes at normal cruising consumption."*
>
> On the 787 release these show in the fuel summary as 10% RSV and 45@CRZ. A standard flag release shows 30@1500 instead.
>
> *Workbook wording: What are the special fuel requirements for a B043 flight plan?*

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

What is final reserve fuel?

30 minutes holding at 1500 ft, or 45 minutes under B043

Final reserve is the fuel that must remain untouched at the end of the planned profile. For Flag operations it is 30 minutes at holding speed at 1500 ft above the destination or alternate. Under the B043 special international reserve it is 45 minutes at normal cruising consumption computed at the flight's weight and altitude at top of descent.

> **FOM 8.3.2.2 / 8.3.2.3**
>
> *"Fly for 45 minutes at normal cruising consumption."*
>
> Tie this to 11.2.6: if it becomes apparent you will land with less than 30 minutes of fuel, declare MAYDAY MAYDAY MAYDAY FUEL. Final reserve is a planning number; the 30-minute emergency-fuel trigger is an in-flight number.
>
> *Workbook wording: What is Final Reserve Fuel?*

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

When do you ask ATC about the delay?

When delay plus divert could land you under 30 minutes

FOM 11.2.5 reinforces the same posture: crews should be proactive in assessing estimated arrival fuel and, if delays are anticipated that could lead to minimum fuel, should evaluate and inquire well in advance.

> **FOM 11.2.6 Emergency Fuel**
>
> *"The PIC should request ATC delay information if it is anticipated that an airborne delay followed by a diversion could result in landing with less than a 30-minute fuel supply."*
>
> Inquire early. By the time you can declare minimum fuel you have already committed to an airport and given up the divert option.
>
> *Workbook wording: When should you plan to "inquire" from ATC the current delay situation when unanticipated...*

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

When do you declare minimum fuel?

Once committed to one airport and able to accept no delay

Minimum fuel means the fuel supply has reached a state where the flight can accept little or no delay. Declare it only after committing to land at a specific airport. Do not declare it at the destination while a divert to the alternate is still available. It is an advisory, not an emergency, and grants no priority.

> **FOM 11.2.5 Minimum Fuel**
>
> *"should only be made when the flight has committed to land at a specific airport"*
>
> The FOM also directs crews to be proactive and inquire well in advance if delays could lead to minimum fuel.
>
> *Workbook wording: When should you declare Minimum Fuel?*

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

When do you declare MAYDAY FUEL?

When you will land with less than 30 minutes

The PIC shall declare MAYDAY MAYDAY MAYDAY FUEL when it becomes apparent that landing will occur with less than a 30-minute supply of fuel, to ensure priority handling. The 30-minute supply is based on holding fuel flow or higher, as required by configuration, at 1500 ft above field elevation.

FOM 11.2.6 Emergency Fuel

> *"the PIC shall declare, "MAYDAY, MAYDAY, MAYDAY, FUEL" to ensure priority handling."*

Note shall, not should. Thirty minutes is a floor, not a target. Contrast with 11.2.5 minimum fuel, which is advisory and grants no priority.

Workbook wording: When should you declare "MAYDAY, MAYDAY, MAYDAY, FUEL"?

Assess Diversion and Alternate Airport Weather Conditions and Minima

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

How do you get updated enroute and descent winds?

Push WIND DATA REQUEST on the route data page

Get to the route data page from the legs page, then select WIND DATA REQUEST. It sends a datalink request and loads winds and the descent forecast into the flight computer. You can enter up to four altitudes on any wind page first to shape what comes back.

FCOM 11.42.42 Route Data Page

> *"transmits a datalink request for wind and descent forecast data. Flight crew may enter up to four altitudes on any wind page to qualify the request."*

Enter winds nearest the airplane first. The first entry is applied to every waypoint, so entering downtrack winds first corrupts the winds ahead of you.

Workbook wording: How (aircraft specific) can you get updated enroute/descent winds?

Fuel Management

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Where is fuel planning in the FOM?

Dispatch chapter, under Planning Considerations and Requirements

Fuel planning sits in the Dispatch chapter of the Flight Operations Manual, under Planning Considerations and Requirements. Fuel Planning covers the factors you must consider, departure fuel coordination with Dispatch, and tanker fuel. The next block, Fuel Requirements, carries the domestic, flag and international reserve numbers.

FOM 8.3.1 Fuel Planning

> *"When computing required fuel for a flight, the following factors must be considered:"*

Same chapter also holds departure fuel coordination. If you and Dispatch cannot agree on extra fuel, the Captain has the final say on fuel on board.

Workbook wording: Where in the FOM is the fuel planning section?

Fuel Conservation

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are fuel conservation considerations?

Green Policy; green-symbol fuel efficiency items appear throughout

FOM 1.6.3 Green Policy establishes that green-symbol guidance supports Alaska Air Group's fuel efficiency commitment, with the explicit safety is always first and fuel saving measures must never compromise safety.

FOM 1.6.3 Green Policy

> *"operational decisions that align with Alaska Air Group's commitment to fuel efficiency."*

No section titled Fuel Conservation exists in the FOM. The guidance is marked by a green symbol wherever it applies: single engine taxi in the taxi guidance, cruise and optimum altitude, and route deviation and optimization.

Workbook wording: Where can you find the Fuel Conservation considerations in the FOM?

Weather and Minimums (19)

Dispatch Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Destination alternate weather required to depart?

At or above alternate minimums for the ETA

A flight may not be dispatched unless the weather at the destination alternate is forecast at or above the alternate minimums for the time the aircraft would arrive there. Those minimums come from the Derived Alternate Airport IFR Weather Minimums table in 8.2.2.1 under Ops Spec C055.

FOM 8.2.2.1 Destination Alternate Airport Weather Requirements

> *"A flight may not be dispatched to the destination airport unless the weather conditions at a destination alternate airport are forecast to be at or above the alternate minimums"*

Second alternate rules live in 8.2.2.2 and Exemption 20108 in 8.2.2.3.

Workbook wording: Weather required to be at our destination/destination alternate (Flag) to depart?

Enroute Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Destination alternate weather in flight?

Crew operating minima

Dispatch planning uses the derived alternate minimums table at 8.2.2.1. In flight the standard shifts to the crew's operating minima, meaning what the operating crew needs to conduct the approach given their minima and any applicable MEL. FOM 6.3.5 states plainly that the in-flight weather requirements are less restrictive than the criteria used by Dispatch in planning.

FOM 6.3.5 / 8.2.2.1

> *"Note that these weather requirements are less restrictive than the criteria used by Dispatch in planning."*

The dispatch-versus-inflight distinction is the whole question. Do not apply the 400-and-1 derived alternate minimums to an in-flight divert decision.

Workbook wording: Weather required at destination alternate airport in flight.

Takeoff Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Takeoff weather minimums at the departure airport?

1 sm or RVR 5000; lower than standard per table

Standard takeoff minimums are 1 sm visibility or RVR 5000 for aircraft of two engines or less (FOM 5.4.1.1).

FOM 5.4.1.1 Standard Takeoff Minimums

> *"Defined as 1 sm visibility or RVR 5000 for aircraft having 2 engines or less."*

FOM 5.4.4 authorizes lower than standard down to 500 RVR with the required lighting, and below 500 RVR (787) via the HGS/HUD Takeoff Briefing Card in the PRC. Charted SID/obstacle minimums may be more restrictive than the 10-9A.

Workbook wording: Weather required at departure airport for takeoff.

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Who can fly a low visibility takeoff?

Below 500 RVR, work off the PRC low visibility card

Ops Spec C078 authorizes lower than standard takeoff minimums wherever a procedure allows standard minimums. For the 787, takeoffs below 500 RVR (150 m) require the Takeoff Low Visibility Card in the PRC. At foreign airports published 10-9A minimums may not conform to Ops Specs, and crews are restricted to the higher of the 10-9A or Ops Specs.

FOM 5.4.4 IFR Lower than Standard Takeoff Minimum Requirements

> *"For takeoffs less than 500 RVR (150 m), see Takeoff Low Visibility Card in PRC."*

NEW in Rev 125: 5.4.4 added 787 bannering and the HGS/HUD Takeoff Briefing Card, which is why this now applies to your fleet.

Workbook wording: Any limitations on who can perform a low visibility takeoff?

Approach/Landing Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Visibility drops on approach. Can you continue?

Yes, once the final approach segment has begun

Do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums. However, if a weather report is obtained AFTER the final approach segment has been initiated and it indicates below minimum conditions, the pilot may continue the approach to DA/DH or MDA.

FOM 5.6.11 Visibility Requirements Approach

> *"the pilot may continue the approach to DA/DH or MDA."*

The trigger is the start of the FINAL APPROACH SEGMENT, not the FAF crossing or the clearance. For non-precision and CAT I: TDZ RVR is controlling, Mid and Rollout are advisory, and Mid may substitute if TDZ is unavailable. RVR above 6000 is advisory only.

Workbook wording: If the visibility drops while on the approach, can we continue the approach?

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

Different visibility rules at international airports?

Yes. Some ICAO states allow Converted Meteorological Visibility

Some ICAO countries allow use of Converted Meteorological Visibility (CMV) when RVR is not reported, for approaches that are not low-visibility. The FOM directs you to the Aerodrome Operating Minimums and Converted Meteorological Visibility references in the Jeppesen Publications for further guidance.

FOM 5.6.11 Visibility Requirements Approach

> *"Some ICAO countries allow for use of Converted Meteorological Visibility (CMV) when RVR is not reported for non low-vis approaches."*

Separately, at foreign airports the published 10-9A takeoff minimums may not conform to Ops Specs, and crews are restricted to the HIGHER of the 10-9A or Ops Specs (FOM 5.4.4). Theatre chapters 19 through 24 carry the country-specific items.

Workbook wording: Are there different requirements at some international airports?

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

When do SMGCS procedures apply?

At airports with a published low visibility taxi plan

Airports may develop a low visibility taxi plan such as SMGCS and may publish low visibility taxi procedures. Canadian airports with low visibility taxi plans may produce SMGCS charts in Jeppesen format.

FOM 5.3.11 Taxi Operations Low Visibility

> *"If an airport is equipped with Surface Movement Guidance and Control System (SMGCS), see the AMM Low Vis feature in Jeppesen FD Pro or Jeppesen Low Visibility Taxi Route charts"*

The specific RVR trigger value at which SMGCS becomes active is airport-published on the 10-9 series, not a single FOM number. Canada-specific low visibility taxi plans are at FOM 19.3.1.5.

Workbook wording: When do SMGCS procedures go into effect?

Crosswind Control

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

Max crosswind for a HUD takeoff?

20 knots crosswind

FCOM L.10.9 gives HUD takeoff wind component limits of 25 knots headwind, 15 knots tailwind, and 20 knots crosswind, and restricts low weather minima takeoff to ILS guidance on U.S. Type II or Type III ILS facilities. Compare with the autoland limits on the same page: 25 headwind, 15 tailwind, 25 crosswind but only 15 knots for CAT II/III.

> **FCOM L.10.9 Autoflight - AFM Limitations - Low Visibility (HUD) Takeoff**
>
> *"on HUD takeoff operations: # Headwind 25 knots # Tailwind 15 knots # Crosswind 20 knots"*
>
> This limit applies only when takeoff weather minima are predicated on HUD takeoff operations. It is not the general takeoff crosswind limit, which comes from the TALPA and non-TALPA tables at FCOM L.10.4 through L.10.7.
>
> *Workbook wording: (787) Max Crosswind for a HUD takeoff?*

Climb Airspeed / Mach Control

4.1.1 PERFORM DEPARTURE AND CLIMB

Severe turbulence penetration speed?

290 KIAS below FL250, 310 or .84 above

Severe turbulent air penetration speed in severe turbulence is 290 KIAS below 25,000 feet, and 310 KIAS or.84 Mach, whichever is lower, at and above 25,000 feet.

> **FCOM L.10 Severe Turbulent Air Penetration Speed**
>
> *"290 KIAS below 25,000 feet • 310 KIAS/."*
>
> Memory-marked limitation. Note the crossover is 25,000 ft, not FL250 by pressure altitude convention, and the upper figure is whichever is LOWER of 310 KIAS or.84 Mach. Additional guidance at FCOM SP Chapter 16 Severe Turbulence.
>
> *Workbook wording: What is severe turbulence penetration speed?*

Manual Aircraft Control

4.1.1 PERFORM DEPARTURE AND CLIMB

Minimum altitude for autopilot engagement?

200 ft AGL after takeoff

The autopilot must not be engaged below 200 ft AGL after takeoff. That is a certified limitation, not a technique. Watch it on a low acceleration height departure where the urge is to get the autopilot in early.

> **FCOM L.10.8 Autoflight**
>
> *"The autopilot must not be engaged below a minimum engage altitude of 200 feet AGL after takeoff."*
>
> Coming down the numbers differ. Without LAND 2 or LAND 3 the autopilot must be off below 135 ft AGL. With LAND 2 or LAND 3 and a glideslope steeper than 3.25 degrees, off below 100 ft AGL.
>
> *Workbook wording: Minimum Altitude for autopilot engagement?*

FOM and FCTM guidance on Weather avoidance

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

How far do you stand off severe cells?

20 nm laterally; clear tops by 5000 ft

FOM 9.5.5 sets 20 nm minimum lateral avoidance for any thunderstorm identified as severe or giving an intense radar echo. Overflying in the clear requires 5000 ft of clearance above the tops. With less than that, revert to the 20 nm lateral standard.

FOM 9.5.5 Thunderstorm Avoidance

> *"Avoid by at least 20 miles any thunderstorm identified as severe or giving an intense radar echo. When "in the clear" on top, cloud tops should be cleared by 5000 ft."*

If you cannot clear the tops by 5000 ft, go back to 20 nm lateral avoidance. The FOM does not specify an upwind or downwind side for the deviation, so that preference is not manual sourced.

Workbook wording: How far (in nm's) should you avoid Severe CB's if possible? Which direction? Vertically?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

No deviation clearance. What do you do?

Emergency authority, PAN-PAN three times, offset 300 ft beyond 5 nm

If you cannot continue per clearance due to dangerous weather and no prior clearance can be obtained, the Captain should exercise emergency authority. Communicate the urgency signal PAN-PAN spoken three times, or a CPDLC urgency downlink, as early as possible.

FOM 5.7.4.6 Revised ATC Clearance Cannot be Obtained

> *"If possible, deviate away from an organized track or route system."*

Weather Deviation Table 5.7.4.6(1). Flying EAST (000 to 179 magnetic): deviating LEFT descend 300 ft, RIGHT climb 300 ft. Flying WEST (180 to 359 magnetic): LEFT climb 300 ft, RIGHT descend 300 ft.

Workbook wording: If a weather deviation clearance cannot be attained, what is the procedure to deviate around weather?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

What do you report to Dispatch when deviating?

100 nm lateral, 4000 ft altitude, 15 minute delay, weather

Dispatch wants a call for any significant deviation from the flight plan. That means 100 nm or more off the planned route, 4000 ft or more off planned cruise altitude, anything likely to delay arrival more than 15 minutes such as holding, and enroute changes made for weather.

FOM 7.1.12 Reports to Dispatch

> *"Lateral deviation: 100 nm or more from the planned route. – Altitude deviation: deviation of 4000 ft or more from the planned cruise altitude."*

The same section adds two you will use often. Call when actual fuel differs from planned by 3000 lb on the 787, and any time you are in moderate or greater turbulence.

Workbook wording: What are the four items that are required to be reported to dispatch when deviating?

Weather Considerations

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What weather do you need to begin an approach?

Reported RVR or visibility at or above minimums

The reported weather at the destination must be at or above the visibility minimums prescribed for the approach. Do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums.

FOM 5.6.11 Visibility Requirements Approach

> *"TDZ RVR reports are controlling."*

Visibility is the controlling parameter; ceiling applies only where the procedure requires it. RVR values above 6000 are advisory only. Visibility below 1/2 sm is not authorized and shall not be used for approach minimums.

Workbook wording: What must the Weather be in order to begin an approach? Ceiling and visibility?

Analysis of Airport Conditions

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Lowest visibility for CAT I?

1800 RVR

Standard CAT I is 1800 RVR. It is authorized to approved runways even without touchdown zone or centerline lights, or with those lights inoperative, provided the flight director or autopilot is used in approach mode down to decision altitude.

> **FOM 5.6.16.1 Reduced Landing Minima - 1800 RVR and 200 ft DH**
>
> *“Authorized precision CAT I landing minima as low as 1800 RVR to approved runways without TDZ”*
>
> Reduced minima below standard CAT I are not available to you. Touchdown zone RVR is controlling, and the flight director or autopilot must be used in approach mode to decision altitude.
>
> *Workbook wording: What is the lowest visibility for a CAT I approach?*

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Lowest visibility for CAT II?

1000 RVR

For CAT II the final approach segment may not begin unless the latest controlling RVR reports are at or above authorized minimums. Required airborne and ground equipment must be operational: LOC and GS, an outer marker or DME defining the FAF, HIRL, and an approach light system (FOM 5.6.17.2).

> **FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations**
>
> *“Landings and subsequent ground operations for CAT II operations to 1000 RVR must be conducted in accordance with the airport’s low visibility operations plan with reference to the Low Visibility Taxi Charts or the Jeppesen Low Visibility AMM.”*
>
> The controlling RVR table lives on the approach briefing cards in the PRC, not in the FOM - FOM 5.6.11 says so explicitly and FOM 5.6.14 notes CAT II/III minima are NOT tailored on the Jeppesen charts.
>
> *Workbook wording: What is the lowest visibility for a CAT II approach? (C060)*

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Lowest visibility for CAT II SA?

RVR 1200 (350 m), DH not below 100 ft

The MEL defines CAT II SA lowest minima as RVR 1200, or 350 meters, with a DH no lower than 100 ft, and autoland is required. Lowest authorized touchdown, mid, and rollout RVR are 1200, 600, and 300. Charted minima for the runway may be more restrictive.

> **MEL 05-10-01B Lower Landing Minimums Autoland Status**
>
> *“A precision approach to no less than RVR 1200 (350m) and a DH of no less than 100 ft. Autoland”*
>
> Confirm against the Jeppesen chart and the approach briefing card for the specific runway.
>
> *Workbook wording: What is the lowest visibility for a CAT II SA approach? (C060)*

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Lowest visibility for CAT III?

RVR 150 feet for CAT IIIb; CAT IIIc has no limit

This comes from the FAA AIM ILS minimums, not the company manuals. CAT IIIa is RVR not less than 700 feet, IIIb runs down to 150 feet, and IIIc has no RVR limitation. Our authorized CAT III minimum is on the approach briefing card in the Pilot Reference Cards.

> **FAA AIM 1-1-9**
>
> *“Category IIIb. No DH or DH below 50 feet and RVR less than 700 feet but not less than 150 feet”*
>
> The FOM lists CAT III operating conditions but publishes no RVR figure. Brief the number off the approach briefing card, not from memory of the AIM values.
>
> *Workbook wording: What is the lowest visibility for a CAT III approach? (C060)*

8.2.1 PERFORM CAT 1 ILS APPROACH

Inside the FAF. Minimum visibility?

Once the segment has begun, continue to DA or MDA

You may not begin the final approach segment unless the controlling RVR is at or above the authorized minimums and both the airborne and ground equipment are operational. Ground equipment means localizer and glide slope plus the required lighting. Once the segment has begun, you may continue to decision altitude.

> **FOM 5.6.17.2 CAT II/SA CAT II Operating Limitations**
>
> *"The Final Approach Segment (FAS) of an IAP may not begin unless the latest controlling RVR reports for the landing runway are at or above the minimums authorized"*
>
> Note visibility below 1/2 sm is not authorized and shall not be used for approach minimums (5.6.11).
>
> *Workbook wording: What if you are inside the FAF? Minimum visibility?*

Communications and PA (11)

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Where are the FMS/ACARS uplink and setup procedures?

FCOM Communications chapter, Supplementary Procedures, and Performance Inflight AeroData

FCOM SP.5 gives the verification procedures for messages uplinked in COMPANY format. Pre-departure and oceanic clearances are manually compared against the filed flight plan, with voice contact if anything disagrees. D-ATIS altimeter numeric and alpha values must match. Weight and balance and takeoff data are verified per the FOM.

> **FCOM SP.5 Communications - Flight Deck Communications System (Datalink)**
>
> *"The following procedures are one means which may be used to verify Pre-Departure Clearance, Digital-Automatic Terminal Information Service (D-ATIS), Oceanic Clearances, Weight and Balance and Takeoff Data messages transmitted via the COMPANY format."*
>
> Three places to know: FCOM SP.5 for uplink verification policy, FCOM 5.40 MFD Communications Functions for the datalink/ACARS pages and managers, and FCOM PI.19 AeroData for the ACARS takeoff and landing performance requests. FMS setup itself is FCOM NP.21.2 CDU Preflight Procedure.
>
> *Workbook wording: Where are the FMS/ACARS/Uplink and Setup Procedures located?*

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

How do you verify the correct ECL is loaded?

Database revision must match the QRH front page

During the preflight flow you push the CHECKLIST display switch, select RESETS, then Verify/Set AIRLINE DATABASE. Select the AIRLINE DATABASE that has a REVISION matching the ECL Revision listed on the front page of the current approved QRH, then select RESET ALL.

> **FCOM NP.21.18 Preflight Procedure**
>
> *"Select the AIRLINE DATABASE that has a REVISION that matches the "ECL Revision" listed on the front page of the current approved QRH."*
>
> The QRH is the authority here, not the airplane. If the airplane database revision does not match the QRH front page, the ECL is not the approved checklist. RESET ALL after setting it clears any prior crew state.
>
> *Workbook wording: 787 – How do you verify the correct ECL is loaded?*

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Where are the standard customer service PAs?

Flight Operations Manual chapter 7, Standard Customer Service Announcements

Identify yourself as the Captain or First Officer. Keep it plain: no aviation abbreviations, no jargon, no knots, broken clouds, chop or abort. Limit announcements while passengers are asleep. Drop the choice of airlines line on Southeast Alaska flights.

FOM 7.1.21 Standard Customer Service Announcements

> *"Pilots can use the PA system as a customer service tool to enhance safety and leave a positive impression with our customers."*

Workbook wording: What section of the FOM are the Standard Customer Service Announcements (PA's) located?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Which PAs must the flight crew make?

The required PAs, starting with the seat belt sign

Flight Crews are required to make the PAs in 7.1.19 during each flight with passengers or persons on board, acknowledging that flight conditions and priorities may occasionally preclude timely execution in the interest of safety.

FOM 7.1.19 Required Announcements

> *"Flight Crew may use judgment to vary the wording of these announcements except where the announcement is noted as verbatim."*

Workload permitting, pilots announce any change in Seat Belt Sign condition, and the F/As must also announce it regardless of whether the pilots do. The FOM encourages Aloha from the Flight Deck and Mahalo on Hawaii flights.

Workbook wording: Which announcements are required to be made by the Flight Crew?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

How do you make a PA on the headset?

Push and hold the ACP PA mic switch

With a headset or boom mic, the PA is keyed from the audio control panel. Push and hold the PA mic switch. That extinguishes the MIC light for any other transmitter and keys the announcement from that station to all PA areas (FCOM 5.10.2).

FCOM 5.10.2 Audio Control Panel (ACP) - PA Mic Switch

> *"6 PA Mic Switch Push and hold – • the MIC light for any other transmitter extinguishes • keys passenger address announcement from this station to all PA areas 7 Receiver Lights Illuminated (green) – indicates the respective receiver volume control is manually selected on."*

The ACP PA mic switch is direct access to ALL PA areas and takes top cabin PA priority, above priority all-area announcements and above handset announcements (FCOM 5.30.2). Monitor the PA by pushing the PA receiver volume control on the ACP.

Workbook wording: How do you make a PA using a headset (which button/transmit switches)?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

How do you make a PA on the flight deck handset?

Select the PA dial code, push handset PA push-to-talk

Take the handset off the cradle. Select the PA area with the numeric keys, the step up/step down switches, or the dial code select switch. Then push and hold the handset PA push-to-talk switch, which connects the handset microphone to the selected PA area and works only in PA mode (FCOM 5.10.19).

FCOM 5.10.19 Handset (aft aisle stand panel)

> *"2 Handset PA Push-To-Talk Switch Push – • connects the handset microphone to the selected PA area • only used in the PA mode January 31, 2023 787 FCOM - HAWAIIAN AIRLINES 5."*

Select PA CALL from the TCP speed dial list or enter/step to the PA area dial code on the handset, then key the handset PA push-to-talk.

Workbook wording: How do you make a PA using the flight deck handset?

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Where is the list of cabin communication codes?

Printed on the flight deck handset, the two digit station codes

The two digit station codes are printed right on the flight deck handset. You can also step through them on the cabin interphone directory and subdirectory pages, or type a code into the scratchpad and select MAKE CALL. An invalid code returns INVALID ENTRY.

> **FCOM 5.30.4 Cabin Interphone Main Menu**
>
> *"A list of the two digit station codes is located on the handset."*
>
> Hawaiian builds the directory, so it is not reproduced in the manual. Speed dial covers the common ones: passenger address, all call, purser, forward galley, ground crew.
>
> *Workbook wording: (787) Where can you find a list of Cabin Communication Codes?*

16.1.2.1.3 KNOW ACARS PROCEDURES

Review the ACARS menus and functions

Six top level menus reached by the COMM display switch.

Push COMM and the main menu gives six picks: ATC, Flight Information, Company, Review, Manager, and New Messages. Flight Information holds clearance, weather and taxi requests. Company holds the airline messages. Review and New Messages gray out when there is nothing to show.

> **FCOM 5.40.3 MFD Communications Functions, Communications Menus**
>
> *"Selecting ATC, FLIGHT INFORMATION, COMPANY, REVIEW, MANAGER, or NEW MESSAGES selection:"*
>
> White text on gray means selectable, cyan on black means inhibited, and three dots after an item mean another menu sits below it.
>
> *Workbook wording: Review ACARS menus and quiz them on where to find specific functions to demonstrate proficiency.*

16.1.2.1.3 KNOW ACARS PROCEDURES

Where is the directory of ACARS functions?

The FCOM Communications chapter, MFD Communications Functions

The datalink and ACARS function set is documented in the FCOM communications chapter, under MFD Communications Functions. Its table of contents lists each function by page: communications menus, command key functions, message list, air traffic control datalink and the downlink pages.

> **FCOM 5.40 Communications - MFD Communications Functions (see 5.TOC table of contents)**
>
> *"The MFD communications functions are used to control datalink features. Datalink messages not processed by the FMC are received, accepted, rejected, reviewed, composed, sent, and printed using communications functions on the MFD."*
>
> The Communications chapter table of contents (5.TOC.2 through 5.TOC.4) is the closest thing to a directory: it indexes every menu and page from Communications Menus through Manager Functions and New Messages.
>
> *Workbook wording: Where can you find a list/directory of ACARS FUNCTIONS?*

16.1.2.1.3 KNOW ACARS PROCEDURES

Where is the spare ACARS paper?

Not published in our manuals. Ask the check airman.

Neither the FOM nor the FCOM gives a stowage location for spare printer paper. The FOM does make it a flight deck consumable you may replace yourself, with no tools. After you change a roll, make an Info to Maintenance entry so the spare gets restocked.

> **FOM 12.2.6 Consumables Replacement, Flight Deck**
>
> *"Consumable items such as light bulbs, ACARS printer paper, tissue boxes, and trash bags are provided to the Flight Crew and may be replaced by the Flight Crew when they are expended during normal operations."*
>
> *Workbook wording: Where is the spare ACARS paper located on the flight deck?*

16.1.2.1.3 KNOW ACARS PROCEDURES

Where are the instructions to change ACARS paper?

Not published in our manuals. Ask the check airman.

No paper loading steps appear in the FOM or the FCOM. The FCOM gives you the hardware only: a cover release lever that opens the printer cover, a paper door lever, and a slew switch that advances paper. The PAPER light means out of paper or the loading door open.

FOM 12.2.6 Consumables Replacement, Flight Deck

> *"(HA) If ACARS printer paper is replaced, the Flight Crew should enter an "Info to Maintenance" logbook entry to ensure spare printer paper is restocked."*

Workbook wording: Where are the instructions on how to change the ACARS paper?

Pushback and Start (17)

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Demonstrate a late pushback, go-around and engine failure PA

Use FOM sample PAs: delay, go-around, emergency equipment

Late pushback: at departure time either push or talk, give the reason and an approximate length, then update every 15 minutes (FOM 7.1.21.1). Go-around: FOM 7.1.21 says a short PA should be made when workload and ATC coordination permit, and below 10,000 ft, on an extended downwind or in holding, you may state the reason and your intentions.

FOM 7.1.21 Standard Customer Service Announcements and FOM 7.1.22 Emergency Vehicles Requested Sample PA

> *"• Missed Approach/Go-Around – Should a go-around be necessary, a short PA announcement should be made when workload and ATC coordination permits."*

No verbatim engine failure PA in the FOM. The closest scripted text is FOM 7.1.22 (emergency equipment standing by) and the NTSB brief examples at FOM 11.2.11, one of which is an engine shutdown and return.

Workbook wording: Demonstrate a late pushback PA, a go around PA, and engine failure PA...

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START

What check is called to the flight deck before start?

Walkaround Check complete, verbally from the Ground Crew

FOM 5.2.20 requires, before movement, that the Flight Crew check all door indications to confirm closed and receive a verbal confirmation from the Ground Crew that the Walkaround Check is complete.

FOM 5.2.20 Cabin Safety Check Prior to Taxi or Pushback

> *"to confirm they are closed and receive a verbal confirmation from the Ground Crew that the Walkaround Check is complete. If installed, the Installed Physical Secondary Barrier (IPSB) must be verified stowed and latched."*

Two separate confirmations happen here: the A Flight Attendant/FFA says 'Cabin Secure' and closes the flight deck door, and the Ground Crew verbally confirms the Walkaround Check. The crew must also check all door indications and verify the IPSB stowed and latched if installed.

Workbook wording: What "check" must be communicated to the Flt Deck prior to engine start?

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Review the engine failure procedure

Rotate at VR to 12 to 13 degrees, fly V2 to V2+15

FCTM technique. With an engine failure at or after V1, rotate smoothly at VR to a 12 to 13 degree target, about half a degree per second slower than normal, then fly V2 to V2+15. Memory items wait until gear up, flight path controlled, and at least 400 feet above ground level.

FCTM 3.34 Takeoff - Engine Failure

> *"the target pitch attitude is approximately 2° to 3° below the normal all engine pitch attitude resulting in a 12° to 13° target pitch"*

Flight controls add most of the rudder for you; you still add a small amount.

Workbook wording: Review Engine failure procedure...

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START

Where do you find the EFP for a runway?

Takeoff performance report item 28; also AeroData special engine out procedures

Item 28 of the takeoff performance report gives the engine failure procedure for that runway, and item 27 gives the engine out flap retraction altitude. The printed AeroData special engine out procedures for departure and destination are your backup copy. Brief both against the runway and intersection you are actually using.

> **FCOM PI.19.17 Takeoff Performance Report**
>
> *"27. Engine out flap retraction altitude 28. Engine failure procedure November 1."*
>
> The engine out procedure is runway specific and can change when the intersection or thrust mode changes. Re-read item 28 on every new performance report, not just the first one.
>
> *Workbook wording: Where can you find the EFP for a specific runway? Anywhere else?*

1.3.2 PERFORM FCOM PROCEDURES PRIOR TO ENGINE START

Where is the departure briefing guidance?

Yes. FOM Threat-Based Departure Briefing/TPC

FOM 5.2.7 says the PF initiates the departure briefing prior to pushback, interactive, concise and big picture. The PM identifies relevant threats first, using the Crew Briefing Reference Card threat list and the 10-7 airport threats as needed. The PF then adds any others.

> **FOM 5.2.7 Threat-Based Departure Briefing/TPC**
>
> *"Prior to pushback, the PF initiates the departure briefing. The departure briefing should be interactive, concise, and focused on the big picture. Pilots jointly identify any relevant threats, followed by the PF presenting the departure plan and any other considerations."*
>
> TPC = Threats (PM and PF), Plan (PF), Considerations (PF). The threat list is on the Crew Briefing Reference Card plus airport threats in the 10-7. The arrival mirror image is FOM 5.5.16 Threat-Based Approach Briefing/TPC.
>
> *Workbook wording: Do we have any guidance regarding the Departure briefing? Where can it be found?*

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE

How do you talk to the ground crew before pushback?

Flight interphone headset, nose gear jack, standard phraseology

Normal means is the flight interphone. The ground crew plugs into the flight interphone jack on the nose landing gear (FCOM 5.30.1) and the flight deck keys INT on the ACP or control wheel. FOM 5.2.24 scripts the exact phraseology for brake release, cleared to start, and cleared to disconnect.

> **FOM 5.2.23 Pushback / 5.2.24 Pushback Interphone Communication**
>
> *"Interphone communication between Flight Deck and pushback crew is normally required. Standard phraseology shall be used."*
>
> FCOM NP.21.30 Pushback or Towing Procedure makes it a Captain item: establish communications with ground handling personnel. The jack itself is described in FCOM 5.30.1.
>
> *Workbook wording: How do you communicate with your ground crew prior to pushback? (Cockpit/Ground Communications)*

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE

How do you alert the ground crew?

Flight interphone; CONNECT INTERPHONE hand signal if unavailable

Ground-to-flight-deck alerting is well documented: a ground crew call gives an aural chime, the GROUND CALL EICAS communications message, and a CALL light on the ACP transmitter select switch (FCOM 5.30.1).

> **FOM 5.3.12 Hand Signals**
>
> *"CONNECT INTERPHONE; indicates the need for direct contact between the Flight Crew and ground personnel."*
>
> The 787 FCOM shows no flight deck switch dedicated to calling the ground crew. The FLIGHT DECK CALL SW on the nose wheel well panel (FCOM 8.10.5) runs ground-to-flight-deck only.
>
> *Workbook wording: How do you alert/call the ground crew?*

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE

No interphone for pushback. What are the hand signals?

Set brakes, release brakes, emergency stop, all clear

No interphone means you brief it before anyone moves. Cover the pushback plan and the signals you will use, and be specific about how the ground crew tells you to stop. The signal set itself is the standard marshalling one: set brakes, release brakes, emergency stop, all clear.

FOM 5.2.23.1 Pushback - No Interphone

> *"In the event that interphone communication is not available, the Captain will brief the Ground Crew on expected pushback procedures and review hand signals."*

Start engines doubles as the walkaround complete signal when headset communication is not available.

Workbook wording: If pushing back without an interphone, what are the hand signals?

1.3.3 PERFORM AIRCRAFT PUSHBACK PROCEDURE

Can you be pulled forward with two engines running?

No. Prohibited with both engines running

FOM 5.2.23 states the ground crew policy currently prohibits forward tow operations with both engines running. Shut one down first, or use a single engine taxi procedure if the tow is really a repositioning.

FOM 5.2.23 Pushback

> *"The Ground Crew policy currently prohibits forward tow operations with both engines running."*

This is a ground crew policy limit, not an aircraft limitation. Expect the crew chief to refuse the pull rather than the airplane to stop you.

Workbook wording: Can you have the aircraft pulled forward with two engines running?

Non-Normal Start Operations

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

Which quick action items apply during engine start?

Aborted Engine Start, ENG AUTOSTART, Fire Eng on Ground/Tailpipe

Quick Action Index items that can bite during engine start: Aborted Engine Start L, R (7.1), ENG AUTOSTART L, R (7.3), and ENG LIMIT EXCEED L, R (7.4). For fire: Fire Eng Tailpipe L, R (8.5), FIRE ENG L, R (8.2), and Fire Eng on Ground L, R (Back Cover.2).

QRH Quick Action Index QA.Index.1

> *"*Aborted Engine Start L, R .... 7.1 *AIRSPEED UNRELIABLE."*

No separate 787 hot start or hung start checklist. Autostart manages start faults, and ENG AUTOSTART L, R (QRH 7.3) is the annunciated failure when autostart cannot start the engine.

Workbook wording: Which Quick Action Items (787) may be required during ENG START, e.g. Aborted Engine Start, Hot Start, Hung Start?

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

Is there a QRH procedure for a tailpipe fire?

Yes. Fire Eng Tailpipe L, R, ground only, no fire warning

It is a ground checklist for a tailpipe fire with no engine fire warning. Fuel control switch cutoff, verify the engine fire switch is in, advise the cabin, then run the start selector to START to motor the engine. Advise air traffic control, and return the selector to NORM once the fire is out.

QRH 8.5 Fire Eng Tailpipe L, R

> *"Condition: An engine tailpipe fire occurs on the ground with no engine fire warning."*

Motoring only works with the auxiliary power unit running or two external power sources on. Losing that power source is what turns this into an evacuation decision.

Workbook wording: Is there a QRH procedure for a TAILPIPE FIRE?

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

What is the hand signal for a tailpipe fire?

Horizontal figure eight; other hand points to fire

FOM 5.3.12 defines FIRE as a horizontal figure 8 made by hand or wand, with the other hand or wand pointing at the location of the fire.

FOM 5.3.12 Hand Signals

> *“FIRE; signifies a fire and it’s location to the Flight Crew by indicating a horizontal figure 8, by hand or wand.”*

Same FIRE signal is used for any fire location, not just a tailpipe fire. Pair it with QRH 8.5 Fire Eng Tailpipe L, R.

Workbook wording: What is the hand signal for Tailpipe Fire?

1.3.4.1 COMPLETE NORMAL ENGINE START FCOM PROCEDURE

Is there a QRH procedure for a ground engine fire?

Yes. Fire Eng on Ground L, R

Fire Eng on Ground L, R applies when fire is observed or detected in the affected engine on the ground.

QRH Back Cover.2 Fire Eng on Ground L, R

> *“Condition: One or more of these occur on the ground:”*

It lives on the back cover of the QRH, not in the Fire chapter, and it is cross-referenced from QRH 8.4 and from the Quick Action Index.

Workbook wording: Is there a QRH procedure for an ENG FIRE on the ground?

1.3.5 PERFORM FCOM PROCEDURES FOLLOWING ENGINE START/BEFORE TAXI

What must you consider before pushback and taxi?

NOTAMs, ATIS, closures or construction, and the taxi route

Two things before you move. Review NOTAMs and current ATIS for taxiway or runway closures, construction, or any other airport risk that touches your taxi route. Then confirm both pilots understand the expected route. If either of you is unsure, stop and ask the controller.

FOM 5.3.4.2 Prior to Taxi

> *“• Review NOTAMs and current ATIS for any taxiway or runway closures, construction activity, or other airport risks that could affect the taxi route.”*

Complex taxi instructions have to be written down. Dual acknowledgment is required for runway crossing, hold short, line up and wait, cleared for takeoff and cleared to land.

Workbook wording: What items must be considered prior to pushback/taxi?

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Max breakaway thrust N1?

40% N1 maximum for breakaway

FCOM NP.21.33 lists three engine warm up requirements before takeoff: run the engines at least 3 minutes, use no more than 40% N1 for breakaway, and engine oil temperature above the bottom of the temperature scale. FOM 5.3.2 Taxi Thrust only warns to stay aware of equipment, structures, and aircraft behind you when engines are above idle.

FCOM NP.21.33 Before Takeoff Procedure - Engine warm up requirements

> *“Use no more than 40% N1 for breakaway • Engine oil temperature must be above the bottom of the temperature scale Pilot Flying Pilot Monitoring Set the TAXI light switch ON and the RUNWAY TURNOFF light switches as needed.”*

Same 40% N1 number appears in FCOM SP sand and dust guidance, paired with a 10 knot ground speed cap. FOM 5.3.2 Taxi Thrust carries no number, so the pointer to FOM is wrong.

Workbook wording: Thrust Use – Max breakaway thrust, N1?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

How do you call the cabin on the headset or handset?

Select CAB, dial the station; handset dials its own code

On the headset, select the CAB transmitter switch on the audio control panel, pick the station on the cabin interphone page of the tuning and control panel, then push the mic switch to talk. On the handset, enter the two digit code and push the dial code select switch.

> **FCOM 5.30.2 Cabin Interphone System**
>
> *"Calls can also be answered or placed using the flight deck handset. The desired call location is entered using the dial code step switches or numeric keys. The call is then made by pushing the dial code select switch."*
>
> Pushing the CAB switch twice within one second places a priority call and cuts in on a station already talking. The handset push to talk is only needed for a PA announcement.
>
> *Workbook wording: How do you make an interphone call to the cabin using the headset? Using the flight deck handset?*

10.1.1.1 COMPLETE PARKING FCOM PROCEDURE

No marshaller. Where do you stop?

Stop; marshalling signals must be visible throughout

FOM 5.6.33 lists one marshaller/VDGS operator and two wing walkers as required personnel for the 787, and requires marshalling signals be clearly visible to the Captain at all times under manual procedures.

> **FOM 5.6.33 Parking Requirements (manual or VDGS)**
>
> *"Manual – for standard manual procedures, the marshalling signals shall be clearly visible to the Captain at all times (see 5.3.12 – Hand Signals)."*
>
> The FOM does not name a specific stop point for a missing marshaller.
>
> *Workbook wording: If there is no marshaller or if they are not marshalling you in, where should you stop the aircraft?*

Taxi and Ground (31)

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

Pushing back late. When is a PA required?

At scheduled departure time, then every 15 minutes

FOM 7.1.21.1 states that at the departure time the Captain will either authorize pushback/engine start or notify Operations and the passengers of the delay, giving the reason if known.

> **FOM 7.1.21.1 General - Departure Time Delay Announcement ('push or talk')**
>
> *"• Departure Time Delay Announcement ("push or talk") – At the departure time, the Captain will either authorize pushback/starting engines procedures or notify Operations and the passengers of the delay."*
>
> The 'push or talk' rule is the trigger: at departure time you either authorize pushback or you tell Operations and the passengers.
>
> *Workbook wording: When should you need to make a PA if you are pushing off the gate late? Minutes?*

Taxiing

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Where is taxi guidance in the FOM?

Flight Operations chapter, Taxi section, right after pushback and engine start

Taxi sits in the Flight Operations chapter between pushback and takeoff. It covers when taxi is prohibited, taxi thrust, taxi guidance and speeds, runway incursion prevention, low visibility taxi, hand signals and tarmac delays. It then points you to the fleet manuals for airplane specific technique.

> **FOM 5.3 Taxi**
>
> *"While taxiing in the ramp/gate area, any congested area, where taxi instructions are complex, or during periods of reduced visibility, the Flight Crew shall devote their full attention to taxiing the aircraft."*
>
> Taxi is prohibited when braking action is reported NIL by the airport or another aircraft. That is a company rule, not a technique.
>
> *Workbook wording: Where in the FOM can you find information on taxiing?*

CRITICAL OBSERVABLE ITEM
2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Can the ND airport map be used for navigation?

No, positional awareness aid only, never primary navigation

An FCOM caution prohibits using the airport map as a primary navigation reference; it exists only to aid positional awareness. The FCTM adds a technique warning against fixation on the display during taxi. GPS position must be available for the ND airport map mode.

> **FCOM NP.21 Preflight Procedure**
>
> *"Do not use the Airport Map application as a primary navigation reference. The Airport Map application is designed to aid flight crew positional awareness only."*
>
> Taxi off the Jeppesen airport diagram and the view outside, not the ND.
>
> *Workbook wording: (787) Can the Airport map displayed on the ND be used for navigation?*

CRITICAL OBSERVABLE ITEM
2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Max taxi speed?

30 kts straight and smooth, 10 kts congested

While taxiing in the ramp or gate area, any congested area, where taxi instructions are complex, or during reduced visibility, the Flight Crew shall devote full attention to taxiing the aircraft.

> **FOM 5.3.3 Taxi Guidance**
>
> *"Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and approximately 10 kts in congested areas."*
>
> Note shall, and 30 kts is conditional on straight and smooth, not a blanket allowance. When operationally feasible, single-engine taxi is the standard procedure for fuel efficiency.
>
> *Workbook wording: Max Taxi Speed/s, and Braking technique?*

CRITICAL OBSERVABLE ITEM
2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Why keep groundspeed in your scan?

To hold 30 kts max, about 10 kts congested

FOM 5.3.3 caps taxi speed at 30 kts, only on straight smooth taxiways, and approximately 10 kts in congested areas, and requires full crew attention while taxiing in the ramp, congested areas, complex taxi instructions, or reduced visibility.

> **FOM 5.3.3 Taxi Guidance**
>
> *"Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and approximately 10 kts in congested areas."*
>
> There is no taxi speed readout on the PFD. Ground speed on the ND is the reference, so keep it in the scan. Turn speed matters more than straight line speed for tire scrub and passenger comfort.
>
> *Workbook wording: Emphasize importance of keeping groundspeed in scan...*

CRITICAL OBSERVABLE ITEM
2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Recommended speed before a 90 degree turn?

No turn number published; about 10 knots in congested areas

Our manuals do not give a speed tied to a 90 degree turn. What they do give is a taxi limit: no more than 30 knots, and only on straight smooth taxiways, with roughly 10 knots in congested areas. Turns happen in the congested areas, so 10 knots is the working number.

FOM 5.3.3 Taxi Guidance

> *"Speed shall be limited to not more than 30 kts, achieved only on straight, smooth taxiways, and approximately 10 kts in congested areas."*

The turn specific technique lives in the FCTM ground operations chapter, which is not one of the three manuals above. Confirm the number there before you quote it.

Workbook wording: Review recommended speed prior to entering a 90n turn...

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

When do you use differential braking for hot brakes?

Not published in our manuals. Ask the check airman.

No differential braking technique for uneven brake temperatures appears in the FOM, FCOM or QRH. What is published is the hot brake response: minimize taxiing, leave the parking brake off unless you need it, and expect wheel fuse plugs can melt. Cooling time comes from the brake cooling schedule.

QRH 14.2 BRAKE TEMP

> *"2 Minimize taxiing. 3 Unless needed, do not set the parking brake. 4 Wheel fuse plugs can melt."*

Brake temperatures per wheel are on the gear synoptic. BRAKE TEMP shows at 5.0 or higher and clears below 3.0. The braking technique itself is FCTM ground operations material.

Workbook wording: Demonstrate use of differential braking if brake temperature on one side is significantly hotter than the...

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Gear location and turning radius?

Wing tip swings the widest arc, 144 feet radius

The wing tip sets your obstruction clearance because it swings the largest arc and everything else stays inside it. Wing tip radius is 144 feet, nose radius 110 feet, tail radius 126 feet, at the maximum steering angle of 70 degrees. Gear positions are on the principal dimensions page just ahead of it.

FCOM 1.10.2 Turning Radius

> *"The wing tip swings the largest arc while turning and determines the minimum obstruction clearance path. All other portions of the airplane structure remain within this arc."*

Never turn away from an obstacle within 15 feet of the wing tip or 55 feet of the nose. Turning away is what swings the tail and the far wing into it.

Workbook wording: Location of gear/turning Radius...

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Where is the guidance on oversteering in turns?

FCTM Ground Operations chapter, taxi turning technique

The FCTM covers this as technique. In a turn the main gear tracks inside the nose gear, so the tighter the turn, the farther you steer the nose gear outside your taxi path. The next section gives visual cues for judging turn initiation from the seat.

FCTM 2.8 Turning Radius and Gear Tracking

> *"steer the nose gear outside of the taxi path (oversteer)"*

See also FCTM 2.9 Visual Cues and Techniques for Turning while Taxiing for seat position cues and turn initiation points.

Workbook wording: Where can you find taxi guidance/technique regarding oversteering in turns?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Normal 180 degree turning radius?

144 feet wing tip radius; 154 feet pavement needed

A normal 180 degree turn is a slow continuous turn with minimum thrust on all engines and no differential braking. That gives a 144 foot wing tip radius and needs 154 feet of pavement width. Anything narrower is no longer the normal turn.

FCOM 1.10.2 Turning Radius

> *"Note: Minimum pavement width for 180o turn: 154 Feet 0 Inches (47.0 Meters)."*

Most runways are 150 feet wide, so the normal no brakes turn does not quite fit. Know that before you accept a back taxi and turn.

Workbook wording: What is the normal 180 degree turning radius?

CRITICAL OBSERVABLE ITEM

2.1.1 PERFORM NORMAL TAXI-OUT (TAXI-IN)

Pivot 180 degree turning radius?

150 foot runway with light differential braking and thrust

On a 150 foot wide runway the airplane will not make the 180 with a clean continuous turn. It takes light differential braking and differential thrust to tighten it. Keep it light and keep it slow, and watch the wing tip, not the nose.

FCOM 1.10.2 Turning Radius

> *"Caution: Requires light differential braking/thrust for 180o turns on a 150 foot (45 meter) wide runway."*

The FCOM publishes no separate pivot turn radius, only this caution. Same 15 foot wing tip and 55 foot nose obstacle rule applies. If the shoulder is soft, ask for a tug.

Workbook wording: What is the non-normal (pivot) 180 degree turning radius?

Airport Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Runway change during taxi. What do you do?

New takeoff data, reload FMC departure, re-brief

FOM 5.4.5 requires runway analysis data to be available and checked for each takeoff; data comes from the Flight Information Packet (TPR) or Dispatch Release (TLR), and if invalid, new takeoff data can be obtained via ACARS or from Dispatch.

FOM 5.4.5 Runway Analysis

> *"new takeoff data can be obtained via ACARS or from Dispatch (who has access to the online performance tool)."*

On the 787, if the HUD TAKEOFF runway changes, FCOM NP.21.3 requires both flight director switches OFF, then back ON after the new runway is executed on the CDU.

Workbook wording: If given a change of departure runway during the taxi, how is this accomplished prior to takeoff and what...

Passenger Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Must you stop taxiing if a guest stands up?

No

Depending on the circumstances, it may be safer to continue to taxi.

FOM 5.3.9 Passengers Standing While Taxiing

> *"There is no regulatory requirement that the aircraft stop if a passenger gets out of their seat."*

Captain judgment call. Stopping on an active taxiway can create a bigger hazard than the standing passenger. Coordinate with the Cabin Crew rather than reflexively braking.

Workbook wording: Are we required to stop the aircraft during taxi if a guest stands up?

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Engine failure over the western states. What do you do?

Driftdown on the ETCP route, land nearest suitable airport

FOM 11.2.7 requires landing at the nearest suitable airport based on time considerations, and the Captain must notify ATC and Dispatch as soon as possible and file a safety report. Suitability factors are listed in FOM 11.2.7.1; fuel sufficiency, passenger accommodation, and maintenance availability are explicitly NOT justification to fly past the nearest suitable airport with one engine out.

> **FOM 11.2.7 Engine Shutdown in Flight / FOM 8.3.7 Enroute Terrain Clearance Program (ETCP)**
>
> *"If an engine fails or is shut down in flight, land the aircraft at the nearest suitable airport, based on time considerations, where a safe landing can be made."*
>
> Over the western US the driver is terrain. The Flight Plan carries the ETCP/driftdown solution with decision points and any Driftdown Alternates (FOM 8.3.7); weather for a named Driftdown Alternate must be in the Flight Information Packet.
>
> *Workbook wording: What are the Engine Failure and diversion procedures while flying over the western part of the 48...*

STAR/Airport/Arrival Review

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What do you brief off the STAR, approach and taxi charts?

Route and constraints, minimums, missed approach, landing runway and taxi

The briefing runs threats, plan, then considerations. Off the charts you cover the arrival, approach and approach mode, altitude and speed constraints, minimums, fuel and route to the alternate, the missed approach, the landing runway and its assessment, the exit and taxi, autobrakes, flaps and approach speed.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *"Prior to beginning the descent, the PF initiates the approach briefing. The approach briefing should be interactive, concise, and focused on the big picture."*
>
> The pilot monitoring calls threats first, off the Crew Briefing Reference Card and the 10-7 page. Keep it interactive and big picture, not a recital.
>
> *Workbook wording: What items should be briefed on the STAR/Approach/Taxi charts?*

Diversion Considerations

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What drives a diversion?

Anything degrading a safe landing at destination

If an engine fails or is shut down in flight, land at the nearest suitable airport based on time considerations where a safe landing can be made. Suitability factors include aircraft configuration, weight, systems status and fuel remaining, and wind and weather conditions enroute at the diversion altitude.

> **FOM 11.2.7 / 11.2.7.1 Engine Shutdown in Flight, Nearest Suitable Airport**
>
> *"land the aircraft at the nearest suitable airport, based on time considerations, where a safe landing can be made."*
>
> Nearest Suitable Airport is FAA-approval-required text, so it is regulatory, not company preference. Nothing justifies flying past the nearest suitable airport after an engine shutdown.
>
> *Workbook wording: What factors may precipitate a diversion?*

Stabilized Approach Criteria

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

What is required at each stabilized approach gate?

Not listed in these manuals. The 787 fleet manual holds the gates

The Flight Operations Manual sets the policy and sends you to the fleet manual for the actual gate numbers. What it does state: an approach is stabilized only if every criterion is met, you must normally be aligned with the runway by 500 ft above field elevation, and you do not land from an unstable approach.

> **FOM 5.6.5 Stabilized Approach**
>
> *"An approach is stabilized only if all criteria are met. Normally, the aircraft must be aligned with the runway by 500 ft AFE."*
>
> Either pilot may call the go-around under the no fault policy. Some runways carry a 300 ft alignment exception, shown on the airport 10-7 page. Get the gate numbers from the 787 flight crew training manual or your check airman.
>
> *Workbook wording: What are the Stabilized Approach requirements at each "gate"?*

2.1.1.10 KNOW SINGLE ENGINE TAXI PROCEDURES

Where are the single engine taxi procedures?

Fleet-specific manuals; FOM states the policy only

FOM 5.3.3 makes single-engine taxi the standard procedure when operationally feasible and points to fleet-specific manuals for the procedure. The identical green-policy sentence is repeated at FOM 5.6.22.1 for the taxi-in case.

> **FOM 5.3.3 Taxi Guidance**
>
> *“When operationally feasible, a single-engine taxi is the standard procedure to optimize fuel efficiency. See fleet-specific manuals for further guidance.”*
>
> No single-engine taxi procedure anywhere in the FOM, FCOM or QRH. FOM 5.3.3 and FOM 5.6.22.1 both state the green policy and then defer to the fleet manual (787 FCTM/AOM). Pull the actual procedure from there before OE.
>
> *Workbook wording: Where can you find the single engine taxi procedures? Review any differences in SOP.*

2.1.1.10 KNOW SINGLE ENGINE TAXI PROCEDURES

Engine cool requirement before shutdown?

At least 3 minutes at taxi thrust

FCOM NP.21.53 sets the cooldown at a minimum of 3 minutes at no more than normal all-engine taxi thrust, and this is the clock that governs a single-engine taxi-in shutdown as well as arrival at a short-taxi gate.

> **FCOM NP.21.53 After Landing Procedure**
>
> *“Engine cooldown recommendations: • Run the engines for at least 3 minutes • Use a thrust setting no higher than that normally used for all engine taxi operations Pilot Flying Pilot Monitoring The captain positions or verifies that the SPEEDBRAKE lever is DOWN.”*
>
> It is worded as a recommendation, not a limitation, but it drives the taxi-in plan.
>
> *Workbook wording: What is the engine cool requirements before shutting down an engine?*

2.1.1.10 KNOW SINGLE ENGINE TAXI PROCEDURES

What do you weigh before a single engine taxi?

Not published in our manuals. Ask the check airman.

The FOM makes single engine taxi standard when it is operationally feasible, then sends you to fleet manuals for the rest. Neither the FCOM nor the QRH carries a single engine taxi procedure or a list of factors. Get the items your check airman expects before you brief it.

> **FOM 5.3.3 Taxi Guidance**
>
> *“When operationally feasible, a single-engine taxi is the standard procedure to optimize fuel efficiency.”*
>
> FOM taxi guidance still applies: no more than 30 knots on straight smooth taxiways and about 10 knots in congested areas.
>
> *Workbook wording: What factors should you consider before conducting a single engine taxi?*

10.1.1.1 COMPLETE PARKING FCOM PROCEDURE

Gate arrival procedures

Marshaller or VDGS, two wing walkers, parking brake set

FOM 5.6.33 requires two wing walkers and one marshaller/VDGS operator for the 787. With manual marshalling the signals must be clearly visible to the Captain at all times (hand signals are in FOM 5.3.12).

> **FOM 5.6.33 Parking Requirements (manual or VDGS)**
>
> *“• Two Wing Walkers and One Marshaller/VDGS Operator”*
>
> The 787 requires more people than the narrowbodies: two wing walkers plus one marshaller/VDGS operator. Station exceptions to the two wing walker requirement are published in the airport 10-7.
>
> *Workbook wording: Gate Arrival Procedures...*

10.1.1.1 COMPLETE PARKING FCOM PROCEDURE

What do you check before taxiing into the gate area?

Correct aircraft type displayed on the VDGS

FOM 5.6.33 gives the VDGS sequence: correct aircraft type displayed before entering the gate area, guidance screen with centerline indicator displayed before the nose passes the jet bridge, and stop on WAIT or STOP. If lead-in line visibility is poor at night or in weather, the crew may request manual marshalling from Station Ops.

> **FOM 5.6.33 Parking Requirements (manual or VDGS)**
>
> *"Ensure the correct aircraft type is displayed prior to entering the gate area."*
>
> Also confirm required personnel are in place (two wing walkers plus marshaller/VDGS operator for the 787), that the Circle of Safety is clear, and that the engine cool-down will be satisfied.
>
> *Workbook wording: What should be checked prior to taxiing into the gate area?*

10.1.2 PERFORM POSTFLIGHT

Perform the securing procedure

Run it whenever the flight deck is unattended

FCOM NP.21.56 flow: IRS selectors OFF, BATTERY switch off, FD POWER switch OFF, EMERGENCY LIGHTS switch OFF, PACK switches OFF, EFB PWR switch off (both). Stow the HUD combiner, ensuring it is fully raised and securely latched. The Captain then calls for the SECURE checklist and the FO reads it.

> **FCOM NP.21.56 Secure Procedure**
>
> *"Run the Secure Procedure any time the flight deck will be left unattended."*
>
> Trigger is an unattended flight deck, not end of day. Watch the PECS caution: up to 22 minutes of cool-down may be required before removing power (FCOM SP.6 Electrical Power Down).
>
> *Workbook wording: Perform Securing Procedure (Optional/If not performed, discuss)*

10.1.2 PERFORM POSTFLIGHT

Who starts the post-flight debrief?

PF initiates it and asks PM for feedback

FOM 5.6.39 says the postflight debriefing should be interactive, concise and focused on relevant items, to learn from experiences and facilitate mentoring. The PF initiates it by asking the PM for positive and/or constructive feedback, referencing the Crew Briefing Reference Card.

> **FOM 5.6.39 Debrief**
>
> *"After the completion of each flight, as workload allows, the PF should initiate the postflight debriefing."*
>
> The PF initiates regardless of seat, so as Captain you may be debriefed by your FO after their leg. Use the Crew Briefing Reference Card. Separate from FOM 11.1.12 Postflight Debriefing/Flight Irregularities, which is the irregularity reporting path.
>
> *Workbook wording: Debrief...*

Holding (Procedures, speeds)

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are the US holding speeds?

200 knots below 6000 ft, 230 to 14,000, 265 above

At or below 6000 ft hold at 200 knots or less. Above 6000 ft through 14,000 ft the limit is 230 knots. Above 14,000 ft it is 265 knots. Inbound leg is one minute at or below 14,000 ft and one and a half minutes above.

> **FOM 5.5.15 Holding**
>
> *"Altitude Maximum Airspeed Time (Inbound Leg) 6000 ft or below 200 KIAS Above 6000 ft 230 KIAS 1 minute Through 14,000 ft (210 KIAS)* Above 14,000 ft 265 KIAS 1 1/2 minutes"*
>
> Charts marked 210K in Alaska or New York hold you to 210 knots above 6000 through 14,000 ft, except the Anchorage and Fairbanks approach control areas. Pattern protection assumes 280 knots or Mach 0.8, whichever is lower.
>
> *Workbook wording: Where can you find the holding speeds (FAR) in USA?*

Freighter Operations

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Review QRH procedures without moving switches

Open the electronic checklist, then select CHKL OVRD

Opening a non-normal checklist to read it makes that checklist active. When you finish reading, press CHKL OVRD at the bottom of the page before you close the electronic checklist or move to another one. That forces it back to inactive. No switch has to move to do this.

> **QRH CI.2.2 Checklist Instructions, Using CHKL OVRD**
>
> *"If the flight crew decides not to do the checklist, select the CHKL OVRD key on the bottom of the checklist page before closing ECL and before selecting another checklist."*
>
> Do not use CHKL OVRD if you only want to pause and finish later. Close the page or go to another checklist and it stays active in the queue. Override also wipes collected notes, checklist inhibits and deferred items.
>
> *Workbook wording: Review QRH procedures without moving any switch positions (when time permits). This reviews the...*

Airspace ownership vs control

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

How and when do you request an RCL?

ACARS data link; Gander 60-90 min, Shanwick 30-90 min prior OEP

FOM 22.6.1 is the reason the RCL still exists: except for Shanwick, oceanic clearances are no longer issued in North Atlantic OCAs, but the RCL message is still required for every other OCA except New York East.

> **FOM 22.6.4 Timing for RCL Message**
>
> *"• Gander: 60-90 minutes. Flights departing from airports less than 45 minutes flying time from the OEP should send the RCL 10 minutes prior to start-up. • Shanwick: 30-90 minutes."*
>
> Santa Maria is at least 40 minutes, Bodo at least 20 (FOM 22.6.4). Go to voice only if ACARS is inop, you get RCL REJECTED, or nothing comes back within 15 minutes (FOM 22.6.3). Shanwick telephone fallback is 01294-655100.
>
> *Workbook wording: How do you request an RCL and when?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Transition level by ATC. When do you set QNH?

Descending through the level the controller or ATIS assigns

When the chart says the transition level is by air traffic control, the number comes off the ATIS or from the approach controller. Get it early and put it in the descent brief. Set QNH descending through that level. Standard stays set above it, QNH below the transition altitude.

> **FOM 23.10 Altimetry**
>
> *"Transition altitudes are normally charted, while transition levels normally are on either the ATIS or provided by the approach controller."*
>
> European transition altitudes and levels are lower than in the United States and vary by country. Settings are normally hectopascals. If a controller gives you QFE, ask for QNH and set it in the normal manner.
>
> *Workbook wording: When do you set QNH when transition level states "by ATC"?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

FCO and LHR departures. Who do you call?

Clearance delivery; they give your start-up and takeoff slot times

Target Off-Block Time is when the airline says you will be ready to push. Target Start-Up Approval Time comes back from air traffic control and is when to expect pushback and start clearance. Calculated Takeoff Time is your slot, a window from 5 minutes early to 15 minutes late.

FOM 18.1.25 Airport Collaborative Decision Making (A-CDM)

> *"TOBTs and TSATs are communicated through various methods, including docking guidance systems, clearance delivery, data link, or Operations."*

Be ready at the off-block time. At Rome you report ready no later than 5 minutes past it or you have to request a new slot. If you cannot make the takeoff time, tell air traffic control as early as you can.

Workbook wording: Departure considerations – FCO/LHR. Who are you going to call and what are they giving you?

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Which radio for ATC, which for ramp control?

ATC on #1/L VHF, ramp control on #2/R

FOM 7.1.5 assigns #1/L VHF to primary communications including ATC, Tower, Ground, CTAF and Ramp Control; #2/R to monitoring 121.5 or Company in flight and Company on the ground; and #3/C to ACARS. The solves the ground shuffle reads that #2/R VHF may be used for ramp control when transitioning to and from the movement area.

FOM 7.1.5 VHF Radio Priority

> *"#2/R VHF may be used for ramp control when transitioning to/from the movement area."*

The base rule puts ATC/Tower/Ground/CTAF/Ramp Control on #1/L, 121.5 or Company on #2/R, and ACARS on #3/C. The note lets you park ramp on #2/R so you can hold clearance or ground on #1/L through the handoffs instead of retuning one radio repeatedly.

Workbook wording: Frequency dance on the ground = Clearance-ramp-clearance-ground...

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Diversion airport considerations. Benefits and pitfalls?

Planned ETOPS alternates unless a better one exists

FOM 6.4.1 says the ETOPS alternates picked during flight planning are in some cases the optimum diversion airports, but the Flight Crew in concert with Dispatch may select another if it is more suitable for prevailing operational conditions.

FOM 6.4.1 Diversions

> *"The Flight Crew, in concert with Dispatch, may select another diversion airport if they determine it is more suitable based the prevailing operational conditions."*

The pitfall is in FOM 6.2.1's caution: an adequate airport may not be appropriate for an ETOPS diversion because it may lack rescue and firefighting support or be in bad weather.

Workbook wording: Diversion airport considerations - Benefits/pitfalls of which ones are used.

Cold Weather and Runway Condition (15)

Cold Weather Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Which manual holds the cold weather and de-icing material?

The FODM, plus fleet-specific manuals

FOM 9.3.1 sends you to the fleet-specific manuals and the FODM for cold weather procedures, and FOM 9.3.1.1 sends you to the FODM again for Liquid Water Equivalent holdover times. FOM 16.2.9 names the ASA Deicing Procedures Manual alongside the FODM.

FOM 9.3.1 Cold Weather Preflight Requirements

> *"Comply with the cold weather procedures in the fleet-specific manuals and FODM."*

FOM does not spell out the FODM acronym in text. Aircraft-side cold weather procedures are in FCOM SP.16 Cold Weather Operations; FOM 9.3 is the policy layer only.

Workbook wording: In which manual is most of the cold weather/de-icing information found?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

What are your cold weather ops resources?

Deicing manual (FODM), holdover times app, and fleet-specific manuals

Three sources. The deicing manual, FODM, carries the deice and anti-ice procedures and the printed holdover tables. The holdover times app on the electronic flight bag gives the live numbers. The FCOM adverse weather supplementary procedures carry the airplane specific cold weather steps.

> **FOM 9.3.1 Cold Weather Preflight Requirements**
>
> *"Comply with the cold weather procedures in the fleet-specific manuals and FODM."*
>
> Holdover times built on liquid water equivalent data are FAA authorized and shall be used over METAR or ATIS derived times unless you can see the data is wrong.
>
> *Workbook wording: What are your cold weather ops resources? (Manuals, cards etc.)*

Contaminated Runway Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Wet vs contaminated runway?

Wet is a water film. Contaminated is beyond that

The FOM carries the operational consequence rather than a bare definition: for a Landing Distance Available of less than 7000 ft, do not use WET data. The wet versus contaminated distinction drives which performance table you are allowed to use, which is why the LDA threshold matters more on the line than the textbook definition.

> **FOM 9.1.12.3 Wet vs. Contaminated Data**
>
> *"For LDA less than 7000 ft, do not use WET data."*
>
> The full wet/contaminated definitions and the RCAM/Runway Condition Code tables sit in FCTM and the TALPA material, which are not in our manuals. Verify the definition wording there. Related: FOM 9.1.13 Degraded Braking uses the Can U StoP mnemonic.
>
> *Workbook wording: What is the difference between a wet vs contaminated runway?*

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Braking action report vs RCAM?

PIREP is pilot-reported deceleration; RCAM is an assessment tool

A braking action report is a pilot qualitative PIREP (GOOD, GOOD to MEDIUM, MEDIUM, MEDIUM to POOR, POOR, NIL) and airport personnel cannot issue one. The RCAM is a published FAA or ICAO matrix used primarily by airports to assess conditions and generate a Runway Condition Code, 0 (NIL) through 6 (dry), and correlate it with those same descriptive terms.

> **FOM 9.1.2 Braking Action Report / FOM 9.1.6 Runway Condition Assessment Matrix (RCAM)**
>
> *"A braking action report (PIREP) is a qualitative description of deceleration by pilots such as GOOD, GOOD to MEDIUM, MEDIUM, MEDIUM to POOR, POOR, and NIL."*
>
> Key operational rule in FOM 9.1.2: a PIREP may downgrade expected braking action from the RCAM or FICON, but must never be used to upgrade it.
>
> *Workbook wording: What is the difference between a braking action report vs RCAM?*

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

What is a FICON NOTAM?

NOTAM giving runway condition codes by runway third

A FICON reports condition for touchdown, midpoint, and rollout in the direction of takeoff and landing, listed as runway thirds such as 3/4/2, with an expected braking action expressed as a Runway Condition Code. Where a single report gives multiple conditions along the runway, assume the most restrictive for the whole runway.

FOM 9.1.3 Field Condition (FICON) NOTAM

"A FICON is a specifically formatted NOTAM reporting runway condition for each third of the runway, and giving an expected braking action as a Runway Condition Code (RWYCC)."

Absence of a FICON does not assure the runway is dry; airports are encouraged but not required to issue one for a wet runway.

Workbook wording: What is a FICON (Field Condition) NOTAM?

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

What is a Runway Condition Code?

The TALPA number, 0 through 6

The Runway Condition Code is the TALPA output used to assess a contaminated runway. On the 787 it drives the landing crosswind limitation directly: RwyCC 6 gives 40 kts, 5 (Good) 40, 4 (Good to Medium) 35, 3 (Medium) 25, 2 (Medium to Poor) 17, 1 (Poor) 15, and 0 (Nil) is no operation.

FCOM L.10 Landing Crosswind Limitations TALPA

"The crosswind limitation is determined by entering the table below with Runway Condition Code or Control/Braking Action."

A FICON NOTAM is how the runway condition code reaches you. Reduce the crosswind limit by 5 knots on wet or contaminated runways whenever asymmetric reverse thrust is used.

Workbook wording: What is a runway CC (Runway Condition Code)?

Crosswind Control

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

What is TALPA?

Takeoff and Landing Performance Assessment

TALPA is the framework that translates reported runway surface condition into a Runway Condition Code and a braking action, which then drive performance and limits.

FCOM L.10 Landing Crosswind Limitations TALPA

"The crosswind limitation is determined by entering the table below with Runway Condition Code or Control/Braking Action."

Reduce the crosswind limitation by 5 knots on wet or contaminated runways whenever asymmetric reverse thrust is used. Winds are measured at 33 ft (10 m) tower height and apply to runways 148 ft (45 m) or greater in width.

Workbook wording: (787) TALPA; What is TALPA?

Cold Temperature Altimeter Correction/VNAV Temperature Limitation

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

When do you make cold weather corrections on approach?

Airport temperature 0 degrees C or colder, when charted

Colder than ISA means true altitude is lower than indicated, and the error grows with height above the altimeter reference source and as surface temperature approaches -30 C or colder (FCOM SP.16.10, FOM 9.3.3.1). Within the US and its territories the Jeppesen briefing strip carries a note when cold temperature corrections must be applied.

FCOM SP.16.10 Cold Temperature Altitude Corrections / FOM 9.3.3.1

"Extremely low temperatures create significant altimeter errors and greater potential for reduced terrain clearance."

No correction is needed under ATC radar vectors, when maintaining an assigned flight level, or when reported airport temperature is above 0 C or at/above the procedure's minimum published temperature. Advise ATC of the corrections, and never correct the altimeter barometric reference setting.

Workbook wording: When/why do you make cold weather corrections while on approach?

Centerline Drift / Crosswind Control

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Where is crosswind landing guidance?

FCOM Limitations crosswind tables; FOM Wind Limits

Crosswind limits come off the FCOM table by runway condition code: 40 kt at code 6 and 5, 35 kt at 4, 25 kt at 3, 17 kt at 2, 15 kt at 1, no landing at 0. Subtract 5 kt on a wet or contaminated runway when using asymmetric reverse.

> **FCOM L.10.7 Landing Crosswind Limitations - TALPA**
>
> *“*** Sideslip only (zero crab) landings are not recommended with crosswind components in excess of 25 knots.”*
>
> FCOM L.10.7 carries both the TALPA table (keyed to Runway Condition Code / braking action) and the Non-TALPA table. Landing technique itself (crab, decrab, sideslip) is in the 787 FCTM Landing chapter, which is not one of these three manuals.
>
> *Workbook wording: Where can you find guidance on landing the aircraft in a crosswind?*

De/Anti-Ice Procedures

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What are icing conditions?

OAT or TAT 10C or below with visible moisture/contamination

FCOM SP.16.1 defines icing conditions as OAT on the ground or TAT in flight of 10C or below, plus one more element. That element is visible moisture (clouds, fog with visibility of one statute mile / 1600 m or less, rain, snow, sleet, ice crystals), or ice, snow, slush, or standing water on the surfaces.

> **FCOM SP.16.1 Cold Weather Operations**
>
> *“Icing conditions exist when OAT (on the ground) or TAT (in-flight) is 10°C”*
>
> Two-part test: temperature AND either visible moisture (including fog with visibility 1 SM / 1600 m or less) or ice, snow, slush, or standing water on ramps, taxiways, or runways. Note the ground/air split: OAT on the ground, TAT in flight.
>
> *Workbook wording: What is definition of icing conditions?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are the aircraft de-ice and anti-ice procedures?

FCOM Supplementary Procedures, Adverse Weather; company deicing in FODM

FCOM SP.16 Cold Weather Operations carries the airplane procedures. It runs Preliminary Preflight, Exterior Inspection, Engine Start, Engine Anti-ice On the Ground (SP.16.4), De-icing / Anti-icing (SP.16.6), Fan Ice Removal (SP.16.8), Wing Anti-ice In-flight (SP.16.8), Cold Temperature Altitude Corrections (SP.16.10), and the cold weather Secure Procedure.

> **FCOM SP.3.1 Anti-Ice Operation, referring to SP.16 Cold Weather Operations**
>
> *“Note: (Refer to Cold Weather Operations in Chapter SP, Section 16, for additional information.”*
>
> Split the question: airplane anti-ice system operation is FCOM SP.16 (SP.3 covers only windshield washer/wiper). Ground deicing/anti-icing fluid, holdover times, and the contamination check are FODM and the ASA Deicing Procedures Manual, per FOM 9.3.1 and 16.2.9.
>
> *Workbook wording: Where do you find the aircraft de/anti-Ice procedures?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are airport specific deice procedures?

FODM and the ASA Deicing Procedures Manual

FOM 16.2.9 states plainly that all deice procedures are found in the ASA Deicing Procedures Manual and the FODM, with the Deicing Procedures Manual available through Maintenance Control.

FOM 16.2.9 Deice - Charters

> *"All deice procedures are found in the ASA Deicing Procedures Manual and the FODM ."*

Verified only to this level: the FOM contains no airport-specific deice pages, and states all deice procedures live in the FODM (on the EFB) and the ASA Deicing Procedures Manual, which is obtained or reviewed through Maintenance Control.

Workbook wording: Where do you find specific airport deice procedures?

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

When must engine anti-ice be on?

Whenever icing conditions exist or are anticipated, above -40C

FCOM SP.16.7 requires engine anti-ice AUTO or ON during all flight operations when icing conditions exist or are anticipated, except below -40C SAT. SP.16.4 requires it selected ON and left on for all ground operations in the same conditions, except below -40C OAT, and confirms EAI shows on the primary engine display for both engines.

FCOM SP.16.7 Engine Anti-ice Operation - In-flight (and SP.16.4, On the Ground)

> *"Engine anti-ice must be AUTO or ON during all flight operations when icing conditions exist or are anticipated, except when the temperature is below -40°C SAT."*

Ground and air differ: on the ground it must be selected ON (manual) and remain on, in flight AUTO or ON satisfies the requirement. Cutoff is -40C OAT on the ground, -40C SAT in flight; do not use above 10C OAT/TAT.

Workbook wording: When are you required to turn the engine anti-ice on?

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

When must wing anti-ice be on?

AUTO normally; select ON manually when wing icing possible

FCOM SP.16.8 says the wing anti-ice system is mainly used to prevent icing in flight. The primary method is the automatic ice detection system controlling it throughout the flight envelope. That gives the cleanest airfoil, the least runback ice, and the least thrust and fuel penalty.

FCOM SP.16.8 Wing Anti-ice Operation - In-flight

> *"The secondary method is to select the WING ANTI-ICE selector ON when wing icing is possible and use for anti-icing."*

Wing anti-ice is primarily an in-flight system and is inhibited on the ground below 75 knots (FCOM 30.20). Do not use it above 10C displayed TAT. Ice on window frames, windshield center post, wiper arm, or side windows is the cue for structural icing.

Workbook wording: When are you required to turn on the wing de/anti-ice...

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are cold weather procedures?

Weather chapter, the fleet manuals, and the FODM deicing manual

The FOM does not list cold weather steps. It sends you to the fleet books and to the FODM, which carries the deicing procedures, holdover times and the post-deicing check. For the 787 the procedure itself is in FCOM Supplementary Procedures, Cold Weather Operations.

FOM 9.3 Cold Weather Operations

> *"Comply with the cold weather procedures in the fleet-specific manuals and FODM."*

The Weather chapter still owns the policy side: icing intensity definitions, adverse weather restrictions and liquid water equivalent holdover times. Cold weather approach restrictions and altitude corrections point back to the fleet manuals.

Workbook wording: Where can you find the cold weather procedures in the FOM? Any other manual?

Takeoff and Departure (24)

Threat Based Briefing/TPC

1.3.1 PERFORM FOM TAKEOFF AND DEPARTURE BRIEFING INCLUDING IDENTIFICATION OF OPERATIONAL THREATS

What is the main concept of the TPC briefing?

Threats jointly, then the PF plan, then considerations

TPC replaces a read-the-plate monologue with a threat-led, interactive exchange. The PM opens by naming relevant threats, the PF adds any others, then the PF briefs the plan and the considerations. It should be interactive, concise and focused on the big picture, with management strategies discussed as threats surface.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *"The approach briefing should be interactive, concise, and focused on the big picture."*
>
> Captain-side: an interactive briefing is where you set flight-deck tone. If the FO never speaks during your brief, the T was skipped and you have lost the main safety value of the format.
>
> *Workbook wording: Discuss the main concept of the TPC briefing...*

1.3.1 PERFORM FOM TAKEOFF AND DEPARTURE BRIEFING INCLUDING IDENTIFICATION OF OPERATIONAL THREATS

Who starts the briefing?

The PF, but the PM speaks first on threats

Within the format the PM identifies relevant threats first, then the PF briefs any additional threats followed by the plan and considerations.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *"Prior to beginning the descent, the PF initiates the approach briefing."*

1.3.1 PERFORM FOM TAKEOFF AND DEPARTURE BRIEFING INCLUDING IDENTIFICATION OF OPERATIONAL THREATS

Where is the briefing guidance?

Yes. Threats, Plan, Considerations departure briefing before pushback

The pilot flying starts it before pushback, and it runs Threats, Plan, Considerations. The pilot monitoring names the relevant threats first, using the Crew Briefing Reference Card and the airport 10-7 page. Then the pilot flying briefs taxi and departure runway, performance and configuration, route, and the return plan.

> **FOM 5.2.7 Threat-Based Departure Briefing**
>
> *"Prior to pushback, the PF initiates the departure briefing. The departure briefing should be interactive, concise, and focused on the big picture."*
>
> Considerations is where you assign specific pilot monitoring duties tied to the threats you just named. Recap the critical threats at the end.
>
> *Workbook wording: Do we have any guidance regarding this briefing? Where can it be found?*

Airport Ops

2.1.3 PERFORM FCOM PRE-TAKEOFF PROCEDURES

Can you depart at night with runway edge lights out?

Only at 1/4 sm or RVR 1600, with FODO permission

If ALL runway lights are inoperative, takeoff is not authorized during darkness unless both conditions are met: reported visibility greater than or equal to 1/4 sm or RVR 1600, and permission of the Flight Operations Duty Officer.

> **FOM 5.4.9 Runway Lights Inoperative**
>
> *"Permission of the Flight Operations Duty Officer (FODO)"*
>
> Two different tests depending on all versus partial. The FODO call is only required for ALL lights inoperative. This is a Captain decision item that people commonly get wrong by assuming any light outage grounds the departure.
>
> *Workbook wording: Can we depart at night from a runway where the edge lights are inoperative?*

Crosswind Control

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

Max takeoff crosswind?

Off the takeoff table, less 5 kts above 40 percent N1

Takeoff crosswind limits come from the FCOM L.10 takeoff table, entered using the columns labeled 25 percent MAC and 30 percent MAC, which is why the takeoff table is CG and weight sensitive while the landing TALPA table is not.

> **FCOM L.10 Takeoff Crosswind Guidelines**
>
> *"Reduce all takeoff crosswind guidelines by 5 knots if thrust is above 40% N1 (GE engines) at brake release for takeoff roll."*
>
> Demonstrated crosswinds are for certification only and do not represent operational limitations; the tables are the limits. Do not interchange the takeoff and landing tables, they serve different functions. read the exact takeoff table values off the FCOM L.10 PDF rather than quoting from memory.
>
> *Workbook wording: Max takeoff crosswind limitations?*

Gear/Flap Management

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

Max altitude with flaps extended?

20,000 ft

The maximum altitude with flaps extended is 20,000 feet.

> **FCOM L.10 Limitations, Non-AFM Operational Information**
>
> *"The maximum altitude with flaps extended is 20,000 feet."*
>
> Adjacent limits worth carrying from the same page: prior to takeoff the maximum allowable difference between the Captain's or First Officer's altitude display and field elevation is 75 feet, and takeoff is permitted only in the normal flight control mode.
>
> *Workbook wording: Max altitude with flaps extended.*

Rotation Rate

CRITICAL OBSERVABLE ITEM

3.1.1 PERFORM TAKEOFF

Normal rotation rate?

2 to 2.5 degrees per second

Boeing technique, not a limit. A smooth continuous rotation initiated at VR toward 15 degrees of pitch produces about 2 to 2.5 degrees per second, reaching liftoff attitude in roughly 3 to 4 seconds depending on weight and thrust.

> **FCTM 3.6 Rotation and Liftoff - All Engines**
>
> *"resultant rotation rates vary from 2° to 2.5° per second"*
>
> With an engine inoperative the rate is about a half degree per second slower, 1.5 to 2.5 degrees per second.
>
> *Workbook wording: Normal rotation rate? (degrees/sec)*

Pitch Attitude

CRITICAL OBSERVABLE ITEM
3.1.1 PERFORM TAKEOFF

Normal takeoff rotation target pitch?

15 degrees pitch attitude

Pilot monitoring calls ROTATE at VR. Pilot flying rotates smoothly toward 15 degrees of pitch, then follows the flight director after liftoff and establishes a positive rate of climb. Fifteen degrees is also the standard pitch Boeing uses for a go around.

> **FCOM NP.21.35 Takeoff Procedure**
>
> *"At VR, rotate toward 15° pitch attitude. At VR call "ROTATE."*
>
> Rotate toward 15 degrees, do not snap to it. Rate matters as much as the target, and an over rotation risks a tail strike at heavy weights.
>
> *Workbook wording: What is the normal takeoff rotation target pitch?*

CRITICAL OBSERVABLE ITEM
3.1.1 PERFORM TAKEOFF

Rotation target pitch after an engine failure at V1?

12 to 13 degrees, 2 to 3 below normal

Boeing technique. Still rotate smoothly at VR, but target about 2 to 3 degrees below the normal all engine attitude, so 12 to 13 degrees. Rotation rate is about a half degree per second slower than normal. After liftoff adjust pitch to hold the desired speed.

> **FCTM 3.34 Rotation and Liftoff - One Engine Inoperative**
>
> *"below the normal all engine pitch attitude resulting in a 12° to 13° target pitch"*
>
> The FCTM also states the flight director pitch command is not used for rotation itself.
>
> *Workbook wording: What is the takeoff rotation target pitch attitude after an engine failure at/after V1 and prior to liftoff?*

Callouts

CRITICAL OBSERVABLE ITEM
3.1.1 PERFORM TAKEOFF

What are the PM callouts during an RTO?

SPEEDBRAKES UP, REVERSERS NORMAL, 60 KNOTS, plus omitted items

The first officer verifies each captain action out loud: thrust levers closed, autothrottles disengaged, maximum brakes applied, reverse thrust applied. Then SPEEDBRAKES UP, or SPEEDBRAKES NOT UP if the lever did not come up. Then REVERSERS NORMAL, or the specific no reverser call. Then 60 knots.

> **QRH MAN.1.2 Rejected Takeoff**
>
> *"Verify speedbrake lever UP and call wheel brakes if the AUTOBRAKE "SPEEDBRAKES UP." If speedbrake message is displayed or deceleration is lever not UP call "SPEEDBRAKES not adequate. NOT UP."*
>
> The first officer also calls out any omitted action item, and communicates the reject to the tower and cabin once stopping is assured. The captain owns the stop decision and the announcement afterward.

Noise Abatement Departure Procedures (NADP)

4.1.1 PERFORM DEPARTURE AND CLIMB

What is NADP1?

The close-in noise abatement profile

NADP procedures are governed by Ops Spec C068. Specific requirements, including gradients and altitudes for configuration changes, may be defined in aircraft-specific manuals, the 10-7, or the Jeppesen Airway Manual Flight Procedures Noise Abatement Procedures.

> **FOM 5.4.7 Noise Abatement Departure Procedure (NADP)**
>
> *"Normal takeoff procedures satisfy either close-in or distant NADP profile requirements."*
>
> That quote is (AS) bannered. This matters for the JCAB ramp-check question in section 1.2.3 about whether HA may perform NADP 1: the OpSpec is C068. confirm the HA-specific position and the OpSpec paperwork the JCAB inspector wants to see.
>
> *Workbook wording: What is NADP1? Describe associated procedures...*

4.1.1 PERFORM DEPARTURE AND CLIMB

What is NADP2?

The distant noise abatement profile

NADP2 is the distant profile. Per FOM 5.4.7, normal takeoff procedures satisfy either close-in or distant NADP profile requirements, so no separate profile is flown unless a specific published procedure requires it. Gradients and configuration-change altitudes come from the aircraft-specific manuals, the 10-7, or the Jeppesen Airway Manual.

> **FOM 5.4.7 Noise Abatement Departure Procedure (NADP)**
>
> *"Flight Crews must comply with published noise abatement procedures when operating from noise-sensitive airports"*
>
> The distinction between the two is where thrust reduction and acceleration occur relative to the airport. the 787 profile detail is FCTM/FCOM, not FOM.
>
> *Workbook wording: What is NADP2? Describe associated procedures...*

Climb Airspeed / Mach Control

4.1.1 PERFORM DEPARTURE AND CLIMB

How do you find max rate of climb speed?

Approximately VREF 30 plus 140 knots until 0.84 Mach

The FMC does not provide maximum rate climb speeds, so this is a Boeing technique approximation for the 787-9. Maximum rate climb gives both high climb rates and minimum time to cruise altitude.

> **FCTM 4.4 Maximum Rate Climb**
>
> *"VREF 30 + 140 knots until intercepting 0.84M"*
>
> Do not confuse it with maximum angle climb speed, which the FMC does provide and is used for obstacles or minimum distance to altitude.
>
> *Workbook wording: How do you determine max rate of climb speed?*

4.1.1 PERFORM DEPARTURE AND CLIMB

Where is max angle climb speed?

Control display unit climb page, MAX ANGLE prompt

Maximum angle of climb speed sits on the climb page of the control display unit, right side, labeled MAX ANGLE. Line select it and the speed drops into the scratchpad, then line select it up to the speed line to fly it.

> **FCOM 11.41.6 FMC Takeoff and Climb - Climb Page**
>
> *"Displays maximum angle of climb speed. Select – displays speed in the scratchpad for pilot selection to the speed line."*
>
> Use it when you need altitude in the shortest distance, such as a steep obstacle departure or a terrain problem. It costs you time and fuel compared with economy climb.
>
> *Workbook wording: Where can you find max angle climb speed?*

SID Departure

4.1.1 PERFORM DEPARTURE AND CLIMB

What else gets briefed on a SID beyond legs and restrictions?

Taxi and runway, takeoff performance, route, return plan

Before pushback the pilot flying briefs the plan: taxi and departure runway, takeoff performance and configuration, route with the ATC clearance compared against the programmed route, and the return plan for an emergency or takeoff alternate. Pilot monitoring calls the threats first, and considerations close it out.

FOM 5.2.7 Threat-Based Departure Briefing

"Taxi, departure runway – Takeoff performance/configuration – Route (ATC clearance vs. programmed route)"

Keep it interactive and big picture. Threats come from the pilot monitoring using the Crew Briefing Reference Card and any airport specific items on the 10-7 page.

Workbook wording: Though the SID legs/restrictions get briefed from the plan view vs the CDU/FMS, what other items should...

4.1.1 PERFORM DEPARTURE AND CLIMB

Is a SID runway specific?

Not always. Runways served are listed in the heading

Jeppesen chart guidance, not company manual text. A SID may serve one runway, several, or all of them. The chart heading lists every runway the procedure serves, for example SID OVERVIEW RWYS 15, 18L/R, and shows ALL RWYS when it applies to every runway. Confirm your departure runway is listed before you brief the SID.

Jeppesen Airline Chart Series

"All runways that have SID/STAR routes are shown in the heading"

Quote describes the SID/STAR overview chart heading. Read the specific SID plate heading and text for the runways it serves.

Workbook wording: Is the SID for a specific runway(s) only?

4.1.1 PERFORM DEPARTURE AND CLIMB

Obstruction climb gradient restrictions?

From runway analysis data, checked before every takeoff

Climb gradient and obstacle numbers come off the runway analysis, which has to be available and checked for every takeoff. It rides in the Flight Information Packet or on the Dispatch Release. If the data is invalid, get new takeoff data by datalink or from Dispatch.

FOM 5.4.5 Runway Analysis

"Runway analysis data must be available and checked for each takeoff."

If an engine fails on takeoff from a runway that has a published engine failure procedure, fly that procedure. That is what guarantees the obstacle clearance, not raw climb performance.

Workbook wording: Obstruction climb gradient restrictions.

4.1.1 PERFORM DEPARTURE AND CLIMB

What are the red boxed notes on a SID?

Not defined in our manuals. Ask the check airman.

The FOM, FCOM and QRH never define what a red boxed chart note is. It comes off the chart, not out of a company manual. The FOM does send you to the chart text block for departure information, so read the box and brief whatever it says.

FOM 5.5.2 Terminal Departures (IFR/VFR)

"(ODP) described in the text block of the Jeppesen chart for the airport."

Boxed notes are there because they change how you fly the procedure, for example a speed hold until advised. Check the Jeppesen chart legend for the convention.

Workbook wording: Red Boxed notes e.g., speed restriction until further advised, etc.

4.1.1 PERFORM DEPARTURE AND CLIMB

Cleared CLIMB VIA. What is the SID top altitude?

The charted maintain altitude in the procedure description

FAA AIM guidance, not company manual text. Every SID has a top altitude, the charted maintain altitude in the procedure description, or an altitude ATC assigns. Cleared climb via, climb to that top altitude and meet all published altitude and speed restrictions along the way.

FAA AIM 5-2-8

> *"the 'top altitude' is the charted 'maintain' altitude contained in the procedure description or assigned by ATC"*

AIM also defines climb via as an abbreviated clearance requiring compliance with the lateral path and all charted speed and altitude restrictions.

Workbook wording: SID top altitude (if given "CLIMB VIA" from ATC)?

4.1.1 PERFORM DEPARTURE AND CLIMB

ATC says climb and maintain. Do SID restrictions still apply?

Yes, until ATC explicitly cancels them

Published restrictions stay in force. A controller cancels them with explicit words like cancel level restriction, climb unrestricted, or disregard SID altitude restrictions. Without that phrase, keep flying the published vertical profile up to the newly assigned altitude. If the clearance is ambiguous, ask.

FOM 23.16.2 SID and STAR Altitude Restrictions

> *"European controllers explicitly cancel intermediate altitude restrictions using standardized phraseology such as "Cancel level restriction," "Climb unrestricted to [level]," or "Climb to [level], disregard SID/STAR altitude restrictions" when these restrictions on a SID or STAR are no longer necessary."*

The FOM says the same for Central America, where an altitude clearance still requires adherence to charted altitude restrictions unless you are specifically relieved.

Workbook wording: While on the SID, if ATC issues a "Climb and Maintain XXX" clearance, does the flight crew still need to...

4.1.1 PERFORM DEPARTURE AND CLIMB

Transition altitude and transition level?

18,000 ft MSL in the US, charted elsewhere

Below the transition altitude you set QNH and state altitudes in thousands and hundreds. At and above it you set 29.92 and say flight level. Outside the US the transition altitude is charted and the transition level comes from the ATIS or the approach controller.

FOM 7.1.2.3 Flight Altitudes

> *"Up to, but not including the transition altitude (18,000 ft MSL in the US and some other areas), state the separate digits of the thousands plus the hundreds, (e."*

Central America runs transition level FL200 and transition altitude 19,000 ft MSL unless charted otherwise. Europe is generally lower than the US, so read the chart every time.

Workbook wording: Transition Altitude/LVL?

Manual Aircraft Control

4.1.1 PERFORM DEPARTURE AND CLIMB

Is the autopilot required on an RNAV SID?

No

FOM 5.1.4.1 is the only autopilot requirement tied to RNAV/RNP procedures. It applies to RNP AR with RNP less than 0.3 or Radius to Fix legs, where autopilot or flight director driven by the RNAV system shall be used. RNP AR departure capability must be authorized by the fleet-specific manuals (FOM 5.1.4.3).

FOM 5.1.4.1 RNP Authorization Required (RNP AR) Procedures

> *"RNP AR procedures with RNP values less than 0.3 or with Radius to Fix (RF) legs shall use autopilot or Flight Director driven by the RNAV system."*

Only RNP AR procedures with RNP below 0.3 or RF legs require autopilot or flight director driven by the RNAV system. Nothing in the FOM, FCOM, or QRH requires the autopilot on a standard RNAV SID.

Workbook wording: Is the autopilot required for an RNAV SID?

Cruise Altitude Considerations

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

OPT vs MAX vs RECOMMENDED altitude?

Optimum ignores winds; maximum is the ceiling; recommended uses forecast winds

Optimum comes from gross weight, cost index, speed schedule and center of gravity, with no wind or temperature input. Maximum is the highest level you can hold at current weight, temperature and number of engines running. Recommended is the most economical altitude for the next 250 to 500 nm using entered forecast winds.

FCOM 11.42.24 FMC Cruise

> *"OPT – calculation of OPT altitude is based on gross weight, cost index, cruise speed schedule, and cruise cg. Forecast winds and temperature deviations from standard day are not used in the calculation."*

Recommended reverts to your cruise altitude within 200 nm of top of descent or 500 nm of destination, so late in the flight it stops telling you anything useful.

Workbook wording: What is the difference between OPT/Max/Recommended altitudes?

In Trail Procedures (OpSpec A354)

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How do you request an ITP climb or descent?

Select the level on the ITP display, CREATE REQUEST, then SEND

Pick the destination altitude on the In Trail Procedure display. CREATE REQUEST auto builds the ITP LEVEL REQUEST page with the reference airplane flight IDs, ITP distances, and maneuver type filled in. Review it, add free text if needed, then SEND transmits it to ATC by datalink.

FCOM 11.35 ITP Page and 5.40 ITP Level Request

> *"When the flight crew selects a destination altitude and selects the CREATE REQUEST key, a ITP level request (COMM) page is automatically created"*

ATC approval or denial comes back by datalink. Crew and ATC remain responsible for the final decision to change altitude.

Workbook wording: How do you (specifically) request an ITP CLB/DES clearance?

Cruise and Diversion (11)

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Review PA usage

No PA in critical phases except emergency announcements

FOM 7.1.18 keeps the PA out of critical air and ground phases except for emergencies, because scanning and flight deck duties come first; delay routine announcements until established in cruise. FOM 7.1.19 lists the announcements the crew is required to make, and crews may vary the wording except where the manual says verbatim.

FOM 7.1.18 Public Address System

> *"Flight Crewmembers should not use the Public Address System during any critical air or ground phase of flight for other than emergency announcements."*

Required PAs are listed in FOM 7.1.19 (seat belt sign on/off, anticipated turbulence, turbulence, stopping short of the jetway). Customer service PA guidance and sample scripts are in FOM 7.1.21.

Workbook wording: Review PA usage (see 12.1.3.16 Know FOM PA policies and Perform Effective Communication from the...

FOM Electronic Plotting Chart Procedure

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

When is an electronic plot required?

On all ORCN segments

An electronic plotting chart should be used on all Oceanic and Remote Continental Navigation segments. The Flight Crew should load the entire cleared route into the approved charting services app, such as FD Pro X, on the EFB.

> **FOM 5.7.3.15 Electronic Plotting Chart**
>
> *“An electronic plotting chart should be used on all ORCN segments.”*
>
> Regulatory basis AC 91-70 and Ops Spec B036. Note the word is should, and the requirement is the ENTIRE cleared route, not just the oceanic portion.
>
> *Workbook wording: When and under what conditions are we required to do an electronic plot?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

How do you plot on the EFB?

Load the entire cleared route into the charting app

The Flight Crew should load the entire cleared route into the approved charting services app on the EFB. FD Pro X is the app named in the FOM.

> **FOM 5.7.3.15 Electronic Plotting Chart**
>
> *“load the entire cleared route into the approved charting services app (such as FD Pro X) on the EFB.”*
>
> The FD Pro X screen-by-screen plotting workflow is vendor documentation, not FOM content. Confirm with the FD Pro Pubs Quick Reference guide in Comply365.
>
> *Workbook wording: How do you plot using the EFB?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Must you save or submit plots after every flight?

No published requirement to save or submit

FOM 5.7.3.15, written against AC 91-70 and Ops Spec B036, requires an electronic plotting chart on all ORCN segments and requires the full cleared route be loaded into the approved charting services app on the EFB. It stops there. Nothing in the section, or in the surrounding oceanic material, requires the crew to save, export or submit the plot afterward.

> **FOM 5.7.3.15 Electronic Plotting Chart**
>
> *“An electronic plotting chart should be used on all ORCN segments.”*
>
> 5.7.3.15 requires the plot be used and the entire cleared route loaded into the approved charting app such as FD Pro X, but it sets no retention, save or submission requirement. Screenshotting a plot is personal practice, not policy.
>
> *Workbook wording: Are you required to save/submit your plots and route after every flight?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

When is it prudent to screenshot your plotted route?

When aware of a possible lateral or vertical deviation report

ICAO North Atlantic guidance says keep records that let the flight be reconstructed weeks later, and specifically retain flight data whenever the crew suspects a lateral or vertical deviation report. A saved image of the plotted route is exactly that kind of record.

> **NAT007 13.2.2 Monitoring by the Operators**
>
> *"retention of aircraft flight data records whenever a flight crew or operator are aware of a possible report of a vertical or lateral deviation"*
>
> The FOM requires the electronic plot in flight but does not direct saving it. This retention guidance is ICAO NAT material, not a company requirement.
>
> *Workbook wording: When would it be prudent to save your plotted route (take screenshot postflight)?*

SATCOM / HF Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

How do you make a SATCOM call?

Select SAT, choose the number from the directory, push MAKE CALL

Push SAT on the tuning and control panel to bring up the satellite phone pages. Line select a number from the directory or type one into the scratchpad. Set the priority, then push MAKE CALL. Talk with the SAT transmitter switch selected on your audio panel. MAKE CALL then reads END CALL.

> **FCOM 5.20.5 Satellite Communication (SATCOM) System**
>
> *"The SATCOM control pages are displayed by selecting SAT on the TCP. Directories of airline-defined numbers are line-selectable or numbers may be manually entered if function is enabled by the operator."*
>
> Priorities are emergency, high, low and public. Emergency sets off an alarm at the ground station. High is for regulatory and flight safety calls, public is non safety correspondence.
>
> *Workbook wording: How do you make a SATCOM (specifically which buttons, and in what order, must be pressed in/on which...*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Where are the preprogrammed phone numbers on the 787?

DIRECTORY key on the satellite phone page, listed by region

On the satellite phone main menu, the DIRECTORY key opens the airline loaded numbers sorted by geographic region. Page 2 of 2 has a directory details key listing the phone number identifiers. Line select what you want, then push MAKE CALL.

> **FCOM 5.10.10 SATCOM Operation**
>
> *"SATCOM DIRECTORY Key Push – Displays directory of available SATCOM phone numbers by geographic region."*
>
> You are not stuck with the list. Type a number into the scratchpad and the call key changes to MANUAL ENTRY, which dials what you entered.
>
> *Workbook wording: In the ACARS/TCP (787) where can you find a list of telephone numbers that are preprogrammed to call?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Total VHF and HF failure. Any other way to reach ATC?

Yes. SATCOM voice, and CPDLC datalink

Satellite voice is the fallback: it is not authorized for routine ATC use, but the FOM specifically permits it when HF or CPDLC are not available (FOM 7.1.7), and all areas of the route structure have satellite voice coverage. CPDLC over SATCOM datalink is the other path, and it is the primary means in oceanic airspace when paired with ADS-C.

> **FOM 7.1.7 SATCOM (if installed)**
>
> *"Routine satellite voice communication with ATC is not authorized; however, it may be used when HF or CPDLC are not available."*
>
> If it is a datalink failure rather than voice, FOM 5.7.4.9 says revert to HF position reporting and tell ATC. For contingencies, FOM 5.7.4.2 also has you alert nearby aircraft on 121.5, backup 123.45, for relay.
>
> *Workbook wording: If you have a total VHF/HF radio failure, is there another way to contact ATC?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Where are the current Pacific HF frequencies?

JetPack HF Frequencies tab, or radio.arinc.net/pacific

The primary source is the HF Frequencies tab in JetPack. ARINC publishes the same list at radio.arinc.net/pacific, and Jeppesen FD Pro carries it under Pubs, North America, General Airway Manual, Enroute Data General, ARINC Services. Frequencies also print in the communications boxes on the enroute chart.

FOM 6.2.7 HF Frequency Assignments

> *"(AS) The primary reference for HF frequencies is from the "HF Frequencies" tab in JetPack."*

ARINC gives you the primary and secondary predicted for that day at check in. If neither works, try any listed frequency, and try several times on one before changing.

Workbook wording: Where can you find the most current (pacific) HF FREQUENCIES in use for your flight?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Rule of thumb for HF frequencies and the sun?

Lower frequencies at night, higher frequencies during the day

Ionization of the refractive layers varies with the sun, so HF propagation shifts through the day. In the North Atlantic, frequencies below 6 MHz work at night and those above 5 MHz work in daylight. Expect the same behavior on Pacific HF families.

NAT007 5.1.7 HF Frequencies

> *"frequencies from the lower HF bands tend to be used for communications during night-time and those from the higher bands during day-time"*

Workbook wording: What is the rule of thumb with HF frequencies vs the Sun?

16.1.3.2 ANALYZE AND ASSESS TERRAIN CRITICAL ROUTES

Western US depressurization strategy?

Fly the charted decompression polygon escape procedure

FOM 11.2.9 covers the western high-terrain depressurization strategy: polygons show where a direct descent to 10,000 ft is not possible before passenger oxygen depletes.

FOM 11.2.9 Decompression Polygon Procedures

> *"Decompression polygon procedures provide a calculated safe method for descending to 10,000 ft before passenger oxygen supplies are depleted following a rapid decompression over high terrain."*

Polygons live on the Jeppesen FD Pro IFR High/Low charts; click the polygon label to open the detail drawer. The three escape types are Exit Any Dir, Exit Twd <fix>, and To Fix then Route.

Workbook wording: Western U.S. Depressurization Strategy...

ETOPS (30)

Dispatch Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

ETOPS alternate weather before departure?

C055 minima, earliest to latest landing time

The Dispatcher will not list an airport as an ETOPS alternate unless weather is forecast at or above the ETOPS alternate minima in Ops Spec C055 (see 8.2.2). That must hold for the whole window it might be used, earliest to latest possible landing time. Field condition reports must indicate a safe landing can be made.

> **FOM 6.2.2 ETOPS Alternate Airport**
>
> *"Field condition reports indicate that a safe landing can be made."*
>
> Each designated ETOPS alternate must have RFFS equivalent to ICAO Category 4 or higher, with a 30-minute response time, which means equipment available on arrival of the diverting airplane, not equipment parked there in advance.
>
> *Workbook wording: Weather required for ETOPS alternate airport(s) prior to departure?*

Enroute Weather Minimums

1.1.2.9 KNOW WEATHER MINIMA & LIMITATIONS FOR DISPATCH, TAKEOFF, APPROACH, AND LANDING

ETOPS alternate weather in flight?

Crew operating minima

From after takeoff until approaching the EEP, alternate weather must be at or above the crew's operating minima. That means what the operating crew needs to fly the approach, given their minima and any applicable MEL. The alternate must remain suitable to conduct an instrument approach.

> **FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements**
>
> *"Note that these weather requirements are less restrictive than the criteria used by Dispatch in planning."*
>
> This is a favourite oral question because the answer changes at the EEP. If a METAR for an ETOPS alternate cannot be obtained after the EEP, a new ETOPS alternate must be designated.
>
> *Workbook wording: Weather required for ETOPS alternate airport(s) in flight?*

ETOPS Verification Flight

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Where are the ETOPS verification flight procedures?

Flight Operations Manual, ETOPS chapter, Verification Flight section

The Hawaiian procedure sits in the FOM ETOPS chapter under ETOPS Verification Flight, with the Alaska system verification flight section just ahead of it. Maintenance notifies you and makes a logbook entry. You report the result to Dispatch between 30 and 60 minutes after takeoff.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"An ETOPS Verification Flight (EVF) is a revenue or non-revenue flight during which the Flight Crew is asked to assist Maintenance to confirm that an ETOPS Significant System is operating normally or that an ETOPS significant problem has been resolved."*
>
> That report and the Dispatch acknowledgment must be complete before the ETOPS entry point. If the item is not operating normally you cannot go past it, and the Captain writes the result in the logbook.
>
> *Workbook wording: Where can the ETOPS verification flight procedures be found?*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Any Dispatch call required after takeoff, before the EEP?

Yes. 30 to 60 minutes after takeoff

Between 30 and 60 minutes after takeoff and prior to reaching the EEP, the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"This report and the return acknowledgment from Dispatch must be complete before reaching the EEP."*
>
> Two-way requirement. Your report alone is not enough; Dispatch's acknowledgment must also be received before the EEP.
>
> *Workbook wording: Is there a communication requirement to Dispatch between takeoff and prior to the EEP?*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

How do you send the pre-EEP component status report?

Phone call or ACARS

The Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system, between 30 and 60 minutes after takeoff and prior to the EEP.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system."*
>
> *Workbook wording: How can/should you make that communication of system performance?*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Component fails before the EEP. Can you continue?

No

If the item being verified is not operating normally, the aircraft cannot proceed beyond the EEP and must land for repairs.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"the aircraft cannot proceed beyond the EEP and must land for repairs."*
>
> Unambiguous. The verification failing before the EEP removes ETOPS entry as an option.
>
> *Workbook wording: If the component being verified fails prior to EEP, can you continue?*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Component fails after the EEP. Can you continue?

Yes. The restriction is on proceeding beyond the EEP

FOM 6.5.10 states the aircraft cannot proceed BEYOND the EEP if the item is not operating normally. Once past the EEP the verification-flight restriction has already been satisfied or failed at that gate. A subsequent failure is handled as a normal in-flight system failure, using ETOPS contingency guidance at 6.4.4 and the PIC/Dispatcher judgment framework.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"the aircraft cannot proceed beyond the EEP and must land for repairs."*
>
> INFERRED from the wording of the EEP gate, not an explicit post-EEP statement. Confirm with a check airman. Treat a post-EEP failure on its own merits under 6.4.4 In-Flight System Failures and the diversion logic in 6.4.1.
>
> *Workbook wording: If the component being verified fails after EEP entry, can you continue?*

ETOPS APU Verification Flight

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

What is the APU verification procedure?

Cold soak 2 hours, start within 1 hour of top of descent

Either stay clear of ETOPS airspace, or complete the verification before the ETOPS entry point if you can hold cruise altitude two hours or more beforehand. Cold soak the APU at least two hours shut down at cruise, then start it within one hour of top of descent.

> **FOM 6.5.10 ETOPS APU Verification Flight**
>
> *"Initiate the APU start within 1 hour of top of descent."*
>
> The cold soak requires the APU shut down at least 2 hours at a normal ETOPS cruising altitude, ideally between FL270 and FL350. That altitude band is the point of the test: it proves a cold start at altitude.
>
> *Workbook wording: What are the APU verification flight procedures?*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

How does APU verification differ from a single component EVF?

A scheduled cold soak and start, not a status report

A single-component EVF is a monitor-and-report task: confirm the system works, report to Dispatch between 30 and 60 minutes after takeoff and before the EEP, and land if it is not normal.

FOM 6.5.10 ETOPS Verification Flight (HA) and ETOPS APU Verification Flight

> *"Allow the APU to cold soak for at least two hours."*

If the APU verification is unsuccessful, the Dispatcher and PIC must confer to determine the best course of action, as ETOPS entry may not be possible. That is a joint decision, unlike the single-component case where the EEP rule is automatic.

Workbook wording: How do those procedures differ from an ETOPS Verification (single component) flight?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

How many APU start attempts?

Three attempts maximum

FOM 6.5.6 sets up the APU In-Flight Start Program. After engine start or takeoff the APU is shut down and cold soaked at least two hours at ETOPS cruise altitude. The crew then attempts a start during the 1-hour window before top of descent.

FOM 6.5.6 APU In-Flight Start Program (HA)

> *"• If the APU starts within the maximum allowed three attempts, the start is deemed to be successful."*

Subsequent attempts after a failed first attempt may be made at lower altitudes, minimum FL270. Do not confuse this with the ground start rules.

Workbook wording: How many start attempts?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Must the APU be shut down before descent?

No. It may be shut down, not required

Under FOM 6.5.6 the test ends once the APU finishes its start sequence or after three failed attempts, and the APU may then be shut down; it is permissive, not required.

FOM 6.5.6 APU In-Flight Start Program (HA)

> *"Once the APU finishes its start sequence or after three failed start attempts, the test is complete, and the APU may be shut down."*

FOM 6.5.10 ETOPS APU Verification Flight sets the same start window (within 1 hour of top of descent) and likewise imposes no shutdown requirement before descent.

Workbook wording: After APU start at altitude, is it required to shut the APU down before descent?

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

APU shut down early, then fails to start after landing. What then?

Make a separate Maintenance Logbook entry

A post-landing APU start attempt should be made as workload permits to avoid delaying the next flight (FOM 6.5.5). If it will not start on the ground, that ground failure gets its own Maintenance Logbook entry, separate from the in-flight start or verification entry the Captain already owes per FOM 6.5.6 / 6.5.10.

FOM 6.5.5 APU Cold Soaked Start (CSS) (AS)

> *"an APU start attempt should be made after landing as workload permits. If the APU fails to start on the ground, a separate Maintenance Logbook entry must be made."*

FOM 5.1.8.1 defines a failed start as three consecutive failed attempts, and 12.4.6 lists APU failure to start as a required logbook entry. The (HA) 787 in-flight start program is 6.5.6; the ETOPS APU verification flight is 6.5.10.

Workbook wording: If APU is shutdown prior to top of descent, then subsequently the APU fails to start after landing,...

Assess Diversion and Alternate Airport Weather Conditions and Minima

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Airborne, what weather do you need at destination and alternates?

Crew operating minima

In flight, alternates must be at or above the crew's operating minima and remain suitable to conduct an IAP. To START the approach, do not begin the final approach segment unless the latest reported RVR, visibility, or ceiling if required, is at or above the authorized minimums.

FOM 6.3.5 / FOM 5.6.11 Visibility Requirements Approach

> *"Do not begin the final approach segment of an instrument approach procedure unless the latest reported RVR, visibility, or ceiling (if required) is at or above the authorized minimums."*

Three different standards get confused here: dispatch planning minima (8.2.2.1), in-flight suitability (crew operating minima, 6.3.5), and the approach-start gate (5.6.11). Keep them separate in an oral.

Workbook wording: Once airborne, what does the weather have to be at your destination, destination alternate, ETOPS...

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

How do you get current METAR and TAF oceanic?

Ask Dispatch by datalink, satellite phone or high frequency radio

Dispatch is your weather shop once you are off very high frequency radio range. They send updates by datalink, satellite phone or high frequency radio, and they must relay anything that would keep you out of an ETOPS alternate. Ask for the airport and the time window you care about.

FOM 6.3.1 Dispatcher Duties

> *"Relay via ACARS, SATCOM, or HF any significant changes in airport status, forecasted weather conditions, or any other conditions that would preclude a safe approach and landing."*

The alternates page in the flight computer also has a weather request prompt that sends a datalink request for alternate airport weather.

Workbook wording: How can you get the most current METAR/TAF when in oceanic airspace?

CRITICAL OBSERVABLE ITEM

5.1.2.6 KNOW CONTINGENCY PLAN (I.E., ALTERNATES, WEATHER MONITORING, FUEL, ETC.)

Who monitors destination and alternate weather?

You and Dispatch, in parallel

The FOM frames this as joint PIC and Dispatcher responsibility rather than a fixed interval.

FOM 6.3.5 In-Flight ETOPS Alternate Airport Weather Requirements

> *"Dispatch will update ETOPS alternates if any conditions preclude safe approach and landing at existing alternates."*

No fixed interval is published. The practical standard is that you should never be surprised by destination or alternate weather at top of descent. Tie to the descent gouge: ATIS, Build, Bug, Brakes, Brief, RNP/Recall.

Workbook wording: How often should you be checking the weather at your destination and alternates?

Fuel Management

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is the ETOPS critical fuel scenario?

Worst-case diversion fuel required from the critical point

The critical fuel scenario is the most fuel-critical of three diversion cases from the critical point to the ETOPS alternate. The cases: immediate descent to 10,000 ft with both engines (depressurization), descent to best single-engine altitude with one engine out, and immediate descent to 10,000 ft with one engine out (FOM 6.2.5).

FOM 6.2.5 Critical Fuel Planning

> *"ETOPS (AS) FUEL REQD OVER CP (HA) CRITF is the most critical fuel scenario from the critical point of the flight (CP/ETP to the diversion airport) chosen from the fuel scenarios outlined above."*

It is a dispatch requirement, not an inflight limit (FOM 6.2.6 area, around FOM 6.4). If before the EEP it looks like actual FOB at the CP/ETP will be below the critical fuel requirement, coordinate with Dispatch.

Workbook wording: What is the ETOPS Critical Fuel Scenario?

APU In-Flight Start Program (APU component currency)

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

EVF vs the routine APU in-flight start program?

EVF confirms a specific repair; the start program is routine currency

FOM 6.5.10 defines an EVF as a flight where the crew assists Maintenance in confirming an ETOPS Significant System works normally, or that an ETOPS significant problem has been fixed. Maintenance makes an AML release entry which the crew must verbally acknowledge with Dispatch.

> **FOM 6.5.10 ETOPS Verification Flight (HA) vs FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *"An ETOPS Verification Flight (EVF) is a revenue or non-revenue flight during which the Flight Crew is asked to assist Maintenance to confirm that an ETOPS Significant System is operating normally or that an ETOPS significant problem has been resolved."*
>
> EVF adds a required report to Dispatch between 30 and 60 minutes after takeoff and before the EEP; if the item is not operating normally the flight may not proceed past the EEP. The APU In-Flight Start Program has no such Dispatch report requirement.
>
> *Workbook wording: How does this differ from the APU In-Flight Start Program?*

Verification flight vs APU in-flight start program

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

What do you log during the APU start?

Flight level, outside air temperature, attempts, result, seconds to available.

The Captain writes the result in the logbook in a set format: flight level and outside air temperature at the attempt, the number of attempts, the word successful or unsuccessful, and the seconds from start sequence to APU available. Successful means it started within three attempts. List any discrepancies if it failed.

> **FOM 6.5.6 APU In-Flight Start Program**
>
> *"APU IN-FLIGHT START FL ___ (flight level during start attempt) OAT ___°C (Outside Air Temperature)"*
>
> Shut the APU down after start or takeoff, let it cold soak at least two hours at cruise, then attempt the start in the hour before top of descent.
>
> *Workbook wording: What items must be logged/noted during the APU start?*

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

How many start attempts are allowed?

Three

FOM 6.5.6 allows a maximum of three start attempts; a start within three attempts is successful. The first attempt is made during the 1-hour period prior to top of descent after at least a two-hour cold soak at ETOPS cruise altitude, and follow-on attempts may be flown at lower altitude but no lower than FL270.

> **FOM 6.5.6 APU In-Flight Start Program (HA)**
>
> *"• If the APU starts within the maximum allowed three attempts, the start is deemed to be successful."*
>
> If the first attempt fails, subsequent attempts may be made at lower altitudes, minimum FL270. FOM 5.1.8.1 uses the same three-attempt rule to define an APU start failure.
>
> *Workbook wording: How many start attempts can be made?*

Flight crew actions prior to the EEP

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

Crossing the EEP. What must be complete?

ETOPS alternate forecast at crew minima; complete ETOPS verification checks

FOM 6.3.2 lists what is required before crossing the EEP. For the 787 the two that apply are forecast weather at the ETOPS alternate at or above the operating crew's minima (accounting for crew minima and any MEL), and completion of any ETOPS verification flight checks.

> **FOM 6.3.2 Flight Crew Duties, Prior to Crossing ETOPS Entry Point (EEP)**
>
> *"• Forecast weather at the ETOPS alternate must be at or above crew's operating minima (this is the weather required for the operating crew to conduct the approach, given the crew's minima and any applicable MEL)."*
>
> Applicable items are CAT II and CAT III, RNAV, GPS, RNP and RNP AR currency, plus autoland and APU in flight start when Maintenance asks. Check the release for the request before you plan the leg.
>
> *Workbook wording: What steps are required?*

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

How is the EEP determined and shown to the crew?

60 minutes from an adequate airport, 410 nm; listed on the release

Per FOM 6.3.3 the EEP is the point 60 minutes from an adequate airport at the approved one-engine-inoperative cruise speed in still air under standard conditions; for the 787 that is 410 nm (FOM 6.2.3 ETOPS Area of Operations table). The EXP is the same fixed distance on the way out.

> **FOM 6.3.3 ETOPS Entry/Exit Point/Equal Time Point**
>
> *"When approaching the ETOPS area of operation, the aircraft reaches the ETOPS Entry Point (EEP) when it is 60 minutes from the adequate airport."*
>
> 787 60-minute distance is 410 nm at.84M/290 KIAS per the FOM 6.2.3 table. Adequate airports defining EEP/EXP must be listed on the Dispatch Release, and crew and Dispatch must use the same ones.
>
> *Workbook wording: How is the EEP determined and depicted to the flight crew?*

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

Max diversion distance for ETOPS 120 and 180?

806 nm at 120 minutes, 1207 nm at 180 minutes

Table 6.2.3(1) in FOM 6.2.3 lists the 787 at.84M/290 KIAS single engine speed, giving 410 nm for 60 minutes, 806 nm for 120 minutes, and 1207 nm for 180 minutes.

> **FOM 6.2.3 ETOPS Area of Operation, Table 6.2.3(1)**
>
> *"787 .84M/290 KIAS 410 nm 806 nm 1207 nm"*
>
> Same table gives the 60-minute EEP/EXP distance of 410 nm. These are Maximum Diversion Distances derived from the approved one-engine-inoperative cruise speed (FOM 6.2.9).
>
> *Workbook wording: What is the maximum diversion distance, ETOPS 120? ETOPS 180?*

CRITICAL OBSERVABLE ITEM
16.1.1.7 PERFORM ETOPS PROCEDURES

ETOPS diversion speed?

.84M/290 KIAS, the one-engine-inoperative cruise speed

FOM 6.4.5 ties the ETOPS diversion speed to the approved One-Engine-Inoperative Cruise Speed in FOM 6.2.3, which for the 787 is.84M/290 KIAS. That speed is what the critical fuel scenario, the EEP/EXP distance, and the Maximum Diversion Distance are all built on, so flying materially off it invalidates the fuel plan.

> **FOM 6.4.5 Diversion Speed Strategy / FOM 6.2.3 ETOPS Area of Operation**
>
> *"Engine inoperative ETOPS fuel requirements are based upon the approved One-Engine-Inoperative Cruise Speed (see 6.2.3 – ETOPS Area of Operation), which should be followed in case of a diversion following an actual engine failure."*
>
> The Captain may deviate from the planned speed profile after assessing the emergency and fuel remaining. For a two-engine decompression diversion the flight plan one-engine diversion speed or LRC is used when level (FOM 6.4.3).
>
> *Workbook wording: ETOPS Diversion speed?*

ETOPS, ORCN and FIR cards

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Terrain polygons with no own-ship position on the EFB?

Show boundary fix coordinates on the ND with FIX INFO

The FOM publishes this method for the magnetic unreliability area border when ownship is not available: pull latitude and longitude from a waypoint on the charted boundary, then display them on the ND through the FIX INFO page. The same method works for polygon boundaries and escape fixes.

> **FOM 22.2.3 Northern Area of Magnetic Unreliability (AMU)**
>
> *"Flight Crew may employ a waypoint placed on the AMU border (as indicated on the Jeppesen FD Pro IFR chart) to derive the latitude and longitude coordinates"*
>
> Polygon escape procedures live in the chart detail drawer. FOM 11.2.9 also lists weather, visibility, ATC, and grid MORAs as aids.
>
> *Workbook wording: Consider techniques for terrain polygons without own-ship position on the EFB...*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Terrain polygon vs ETP considerations?

ETPs are calculated at 10,000 ft, the polygon descent target

The polygon answers how to get down safely over high terrain. The equal time point answers which airport is shortest in time. They meet at 10,000 ft, since that is the altitude the equal time points are built on. Terrain drives the descent path first, then you settle the airport with Dispatch.

> **FOM 6.3.7 Equal-Time Point (ETP)**
>
> *"ETPs are calculated using flight performance and winds at 10,000 ft."*
>
> Equal time points come from planning-time winds, so treat them as a start point. The FOM expects you to weigh weather, visibility, ATC and grid minimum off-route altitudes, and it lets you pick a closer airport that is not a designated ETOPS alternate.
>
> *Workbook wording: Terrain polygon procedure vs ETP considerations (example: ETP is usually west of Greenland and BIKF is...*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

When do you use the HDG REF switch?

When assigned a heading where true north is used

FCOM SP.10.1 is blunt: use TRUE when assigned a heading in a region where true north is used, and NORM in all other regions.

> **FCOM SP.10.1 Supplementary Procedures - Flight Instruments, Displays (Heading Reference Switch Operation)**
>
> *"Use TRUE when assigned a heading in a region where true north is used. Use NORM in all other regions."*
>
> The 787 switches to true automatically in the polar regions with HEADING REF in NORM (FCOM 11.31, FMC Polar Operations), so TRUE is a manual selection you make for an ATC-assigned true heading, e.g. inside the AMU (FOM 22.2.3).
>
> *Workbook wording: AMU, when do you use the HDG REF switch?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

How do you identify the OEP?

Named waypoint on or near the FIR boundary entering the OCA

FOM 22.2.2 defines the OEP as the waypoint from which RCL time estimates are calculated, generally a named waypoint on or near the FIR boundary where the aircraft enters an OCA.

> **FOM 22.2.2 Oceanic Entry Point (OEP)**
>
> *"The OEP is the waypoint from which Request for Clearance (RCL) time estimates are typically calculated (e.g., "X" minutes prior to OEP via data link). It is generally a named waypoint located on or near the FIR boundary where an aircraft enters an OCA."*
>
> If the route has no waypoint sitting on the OCA boundary, the OEP backs up to the last named waypoint before it; on random routing it is the last named waypoint before the first LAT/LONG inside the OCA.

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Entering the GOTA eastbound. Is an RCL required?

Yes, to Gander Oceanic

FOM 22.2.1 states that transition areas (GOTA, NOTA, SOTA, BOTA) are buffers between domestic and oceanic airspace and are not part of the OCAs. Aircraft in them remain under domestic ATC control: GOTA under Gander or Montreal, NOTA and SOTA under Shannon Domestic, BOTA under Brest Domestic.

> **FOM 22.2.1 Transition Areas**
>
> *"transition area. Also, eastbound flights through the GOTA are to send an RCL message to Gander Oceanic. For specific transition area rules and exceptions, see the FIR card and Operational Notes in Jeppesen FD Pro by selecting the airspace."*
>
> This is a named exception. The GOTA is not part of the Gander OCA and stays under Gander or Montreal Center control, yet you still send the RCL to Gander Oceanic.
>
> *Workbook wording: When entering the GOTA eastbound, are you required to send an RCL?*

Airspace ownership vs control

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Is your ETOPS distance 120 or 180?

180 minutes MDT; as dispatched on the Release

FOM 6.2.3 sets the Alaska Airlines authorized Maximum Diversion Time at 180 minutes, computed at one-engine-inoperative cruise speed under standard conditions in still air, and requires the MDT the flight was planned on to be listed in the Dispatch Release.

> **FOM 6.2.3 ETOPS Area of Operation**
>
> *"The Alaska Airlines' authorized MDT is 180 minutes."*
>
> MEL-deferred items can knock Hawaiian down to 120 minutes (FOM 6.2.3), so read the Release rather than assuming 180. Table 6.2.3(1) gives the 787 flying distances at.84M/290 KIAS: 410 nm at 60 min, 806 nm at 120 min, 1207 nm at 180 min.
>
> *Workbook wording: What is your ETOPS distance set to? (120 or 180)*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What are your common ETOPS airports?

Those listed on your Flight Plan/Dispatch Release

FOM 6.2.2 defines an ETOPS alternate as an adequate airport listed in FOM 8.1.3 and designated on the Flight Plan/Dispatch Release for use in an ETOPS diversion. It explicitly says this is a flight-planning definition that does not limit PIC authority in flight.

> **FOM 6.2.2 ETOPS Alternate Airport**
>
> *"An ETOPS alternate airport is an adequate airport that is listed in 8.1.3 – Authorized Airports and is designated on the Flight Plan/Dispatch Release for use in the event of a diversion during ETOPS."*
>
> The master list is FOM 8.1.3 - Authorized Airports, where an E in the fleet column means the field may be listed as an ETOPS alternate. Anything coded A, F, R, P or E may also be planned as an adequate airport to define EEP/EXP.

Oceanic and NAT (35)

Dispatch Release: Review all sections

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Named vs unnamed waypoint?

Named is in the database. Unnamed is lat/long or hand entered

Non-defined or user-named waypoints are waypoints that are either not in the aircraft database, unnamed such as a latitude/longitude like 26N170W, or manually entered by the crew. These may require specific verification procedures, and the FOM sends you to fleet-specific manuals for the route verification procedure.

> **FOM 5.2.11.5 Non-Defined or Unnamed Waypoints**
>
> *“waypoints that are either not in the aircraft database, unnamed (a latitude/longitude, such as “26N170W” or “26N70”), or a waypoint that the Fight Crew manually enters.”*
>
> Oceanic tie-in: FOM 22.3.4 Unnamed Waypoint Verification carries the NAT-specific check.
>
> **NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS**
>
> NAT Doc 007 6.2.7: "independently verify the full latitude and longitude coordinates of “unnamed” (Lat/Long) waypoints" ICAO draws the same line, calling lat/long waypoints unnamed and requiring both pilots to verify their full coordinates.
>
> *Workbook wording: What is the difference between a named vs non-defined/unnamed waypoint?*

Company Reporting

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Are company position reports required?

Yes, and holding always requires one

FOM 7.1.12 Reports to Dispatch carries the requirement set. One that is explicit: through ACARS or radio the Captain will ensure Dispatch is contacted with the holding fix, Expected Further Clearance time, and fuel on board, and diversion plans should be discussed if applicable. Abnormalities require notification under 11.1.2.

> **FOM 7.1.12 Reports to Dispatch / 5.5.15.1 Dispatch Notification of Holding**
>
> *“the Captain will ensure that Dispatch is contacted with the holding fix, Expected Further Clearance (EFC) time, and fuel on board.”*
>
> Three items for the holding report: fix, EFC, fuel. read 7.1.12 in full for the complete routine position-reporting requirement.
>
> *Workbook wording: Are we required to make company position reports?*

RVSM Requirements

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What altitudes are RVSM airspace?

FL290 through FL410

RVSM airspace is any airspace or route where aircraft are separated by 1000 ft vertically, between FL290 and FL410 inclusive.

FOM 5.5.6.5 Enroute RVSM Altitude Monitoring Requirements

“between FL290 and FL410, inclusive), pilots shall verify that all altimeters are set to 29.”

At initial level off, pilots shall verify all altimeters are set to 29.92 and crosscheck them. Use of the standby altimeter as a primary is PROHIBITED in RVSM airspace.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 Chapter 11: "This is defined as any airspace between FL 290 - FL 410 inclusive" ICAO defines NAT RVSM airspace the same way as the company, FL290 through FL410 inclusive.

Workbook wording: What altitudes define RVSM airspace?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

RVSM equipment requirements?

Both primary altimeters, one autopilot, altitude alerting

The following shall be operating normally prior to entry into RVSM airspace: both primary altimeters, one autopilot altitude-control system, and the Altitude Alerting System. The autopilot should be engaged during altitude capture and level cruise, except where circumstances such as the need to retrim or turbulence require disengagement. Report reaching assigned altitudes if not in radar contact.

FOM 5.5.6.3 Entry Into RVSM Airspace

“Both primary altimeters • One autopilot altitude-control system 295 Chapter: 5 Flight Operations Manual Rev: 125.”

Three items only. A common wrong answer adds transponder or TCAS to the list.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.12: "Required equipment includes two primary independent altimetry sources, one altitude alert" The sentence continues with one automatic altitude control system, so ICAO matches the company list of altimeters, autopilot and altitude alerting.

Workbook wording: Are there aircraft equipment requirements to operate in RVSM airspace?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

RVSM equipment fails in flight. What do you do?

Notify ATC and Dispatch

In the event of required RVSM equipment failure prior to or within RVSM airspace, notify ATC and Dispatch to coordinate a prudent course of action.

FOM 5.5.6.4 System Failure Enroute

“ATC may not require any action; however, a reroute or diversion may be required.”

Both ATC and Dispatch, not one or the other. The FOM deliberately leaves the outcome open rather than mandating an immediate descent out of RVSM.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 7.2.1: "The following equipment failures must be reported to ATC as soon as practicable" ICAO matches the notify ATC step; adding Dispatch is company procedure.

Workbook wording: What are you expected to do if you have a failure of required equipment while in RVSM airspace?

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Diverting without a clearance. What altitude do you plan?

Below FL290

Under the oceanic in-flight contingency procedures (FOM 5.7.4.2), if a revised clearance cannot be obtained, turn at least 30 degrees left or right. Establish a same-direction parallel track offset 5 nm. Be below FL290 before diverting from the assigned clearance.

FOM 5.7.4.2 General Procedures

"The aircraft should be below FL290 prior to diverting from the assigned clearance unless a revised ATC clearance has been obtained."

Same guidance is repeated in FOM 5.7.4.3. First turn at least 30 degrees off the cleared track to establish a parallel same-direction offset of 5 nm, then take a 500 ft vertical offset once established.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 10.3.2: "descend below FL 290, and establish a 150 m (500 ft) vertical offset from those flight levels" ICAO matches planning below FL 290 and adds a 500 ft vertical offset from levels normally used.

Workbook wording: When diverting to an alternate without a clearance, what altitude should you plan to be below when...

CPDLC Logon Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

How early do you log on to CPDLC leaving the contiguous 48?

About 30 minutes prior to departure

CPDLC DCL becomes available roughly 30 minutes before departure at DCL airports (FOM 5.2.14), so that is the window to log on and pull the clearance. Where DCL is available and the aircraft is equipped, CPDLC DCL is expected to be used for the preflight clearance; if the logon or the DCL fails, revert to Clearance Delivery on VHF.

FOM 5.2.14 CPDLC Clearances - ATC Data Link Communications

"Controller Pilot Data Link Communications Departure Clearance (CPDLC DCL) is available approximately 30 minutes prior to departure at airports with DCL operational."

FOM 5.2.15 gives the logon table: CONUS (Lower 48) is 'During preflight' with logon address KUSA. Contrast Hawaii, which is above 10k and 15-45 min prior to entry, logon KZAK.

Workbook wording: When departing the 48 Contiguous states, to assist ATC, how many minutes prior to departure is an...

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Airborne CPDLC logon window before the FIR?

A time window before entry, not an altitude

If equipped with CPDLC, log on either on the ground during preflight or after takeoff above 10,000 ft MSL, or as required by the specific FIR/OCA Pilot Reference Card.

FOM 5.2.15 CPDLC Logon Policy

"Oceanic Control Areas typically have a time window prior to entry that defines the logon window, not a specific altitude."

The direct answer to the second half is no, it is not the same for all FIRs. Use the FIR/OCA Pilot Reference Card. For KZAK Oakland the portal phase_flows page carries logon 45 to 15 minutes prior to entry and not below 10,000 ft.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.41: "the pilot should log on to the appropriate FIR" ICAO sets the window at 10 to 25 minutes before the boundary, matching the company time based rather than altitude based trigger.

Workbook wording: Once airborne, what is the CPDLC logon time window prior to the FIR ("Flight Information Region")? Is it...

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

How do you log on for a departure clearance? Enroute?

Flight Information menu for departure clearance; ATC Logon/Status page enroute

Departure clearance comes off the multifunction display: FLIGHT INFORMATION, then DEPARTURE CLEARANCE REQUEST. Enter flight number, the four letter facility, departure, destination, gate and the automatic terminal information service letter, then send. Enroute, use ATC LOGON/STATUS: enter the center identifier, flight number, origin and destination, then push SEND.

> **FCOM 5.40.42 ATC Logon/Status**
>
> *"The ATC LOGON/STATUS page allows logon to the desired ATC facility for establishment of a datalink connection and displays ATC datalink connection status information."*
>
> Entering a new flight number on the clearance request page ends the current ATC connection. Status reads SENDING, then CONNECTING, and the page closes itself five seconds later.
>
> *Workbook wording: How, specifically in your aircraft, do you logon for a departure clearance? Enroute?*

FOM Waypoint Transition Procedure

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Two minutes before a waypoint, what do you do?

Confirm the next waypoint and the one after it

Approximately 2 minutes prior to reaching a waypoint, confirm that the upcoming waypoint and the subsequent waypoint (next plus one) agree with the Flight Plan and ATC clearance.

> **FOM 5.7.3.16 Waypoint Transition**
>
> *"confirm that the upcoming waypoint and the subsequent waypoint (next + 1) agrees with the Flight Plan/ATC clearance."*
>
> The next-plus-one check is the whole point. Verifying only the active waypoint will not catch a gross navigation error that begins one leg downstream.
>
> **NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS**
>
> NAT Doc 007 6.3.51: "Approaching an oceanic waypoint, pilots should crosscheck the coordinates of the next" ICAO ties the check to approaching the waypoint and includes the next plus one, matching the company two waypoint confirmation.
>
> *Workbook wording: Approximately 2 minutes prior to reaching a waypoint, what crew actions are required?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

What gets checked after waypoint transition?

Time and fuel recorded, difference computed, navigation mode verified, waypoint slashed

After crossing, record time and fuel from the flight management computer on the flight plan or release and compare with the estimate. Note the difference: plus is underburn, minus is overburn. Confirm the navigation mode on the flight mode annunciator, slash the waypoint, and send the position report if required.

> **FOM 5.7.3.16 Waypoint Transition**
>
> *"Record the time and fuel remaining from the FMC/FMS on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review and compare them with the estimates. Record the difference (+ is ahead/underburn and - is behind/overburn)."*
>
> Two minutes before the fix you confirm the next waypoint and the one after it against the clearance. Ten minutes after passage you run the navigation crosscheck.
>
> **NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS**
>
> NAT Doc 007 6.3.54: "Pilots should also note and record their fuel status at each oceanic waypoint." ICAO matches the fuel recording at each waypoint; 6.3.55 adds the LNAV tracking check 10 minutes after passage.
>
> *Workbook wording: What items are required to be checked on the flight plan after waypoint transition?*

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Where is fuel remaining recorded at waypoint transition?

The flight plan, the release, or the ACARS HOWGOZIT

Time and fuel remaining from the FMC are recorded on the Flight Plan/Dispatch Release, or alternatively on the ACARS HOWGOZIT/Flight Plan Review, and compared against the estimates.

FOM 5.7.3.16 Waypoint Transition

> *"Record the time and fuel remaining from the FMC/FMS on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review"*

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.54: "Pilots should also note and record their fuel status at each oceanic waypoint." ICAO matches the company practice; where to write it, release or HOWGOZIT, is company guidance.

Workbook wording: On which document is the fuel remaining (at waypoint transition) recorded?

CRITICAL OBSERVABLE ITEM
5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Ten minutes after waypoint transition, what do you do?

Crosscheck navigation performance and course

Approximately 10 minutes after waypoint passage, crosscheck navigational performance and course compliance.

FOM 5.7.3.17 10 Minutes After Waypoint Passage

> *"Take immediate corrective action for any anomalies or deviations."*

Cross-reference 5.7.4.11 Position Doubtful if the crosscheck does not resolve. If you are on a weather deviation, the FOM requires this process to be completed immediately rather than waiting for the 10-minute point.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.55: "pilots should confirm that proper lateral (LNAV/NAV) mode is engaged and the aircraft is tracking to the proper waypoint" ICAO matches the company crosscheck and names exactly what to verify at the 10 minute point.

Workbook wording: Approximately 10 minutes after waypoint transition, what crew actions are required?

CRITICAL OBSERVABLE ITEM
5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Why track fuel score?

It reveals fuel leaks and burn trends early

Scoring fuel at every waypoint is the primary in-flight leak detector: a growing negative trend that fuel flows do not explain is the signature of a leak or fuel system malfunction (FOM 5.5.3).

FOM 5.5.3 Flight Plan Time and Fuel Monitoring - Enroute

> *"The difference between the actual and the estimated fuel remaining is an important factor in identifying and troubleshooting fuel leaks or other fuel system malfunctions."*

FOM 7.1.12 requires a Dispatch call when actual fuel differs from planned by 3000 lbs. Call early rather than at the gate, so Dispatch can rerun the numbers while you still have options.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.54: "Pilots should also note and record their fuel status at each oceanic waypoint. This is especially important if the cleared route and flight level differ significantly from the filed flight plan." ICAO requires the record; the leak detection rationale is the company addition.

Workbook wording: Why is keeping track of the actual fuel remaining vs planned fuel remaining ("FUEL SCORE") important?

CRITICAL OBSERVABLE ITEM
5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Which fuel figure goes on the release, tank total or PROG?

Progress page fuel from the flight management computer, not tank total

The FOM says record fuel remaining from the flight management computer, so the progress page figure goes on the release, not the tank total off the fuel synoptic. Keep the same source every waypoint or the trend means nothing. Progress page 2/4 shows totalizer and calculated side by side.

FOM 5.7.3.16 Waypoint Transition

> *"Record the time and fuel remaining from the FMC/FMS on the Flight Plan/Dispatch Release or the ACARS HOWGOZIT/Flight Plan Review and compare them with the estimates."*

Totalizer is sensed tank fuel. Calculated is what the computer has burned down from the starting quantity. Totalizer lower than calculated is one of the fuel leak clues in the QRH.

Workbook wording: (787) Which fuel remaining is noted on the dispatch release, the EICAS fuel panel (tank total) or the PROG...

SATCOM / HF Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Where is the HF position report template?

Jeppesen FD Pro X, Pubs, Pacific, Oceanic Position Reporting Procedures

For the Oakland FIR the full format sits in Jeppesen FD Pro X under Pubs, Pacific, Oceanic Position Reporting Procedures. The FOM carries the same skeleton: present position, estimated next position, ensuing fix, then preplanning flight levels. Oakland also wants a meteorological report with every position report.

FOM 5.7.3.19 Position Reporting, Oceanic (non-CPDLC)

> *"For operations in the Oakland FIR, oceanic position reporting procedures are clearly delineated in the Jeppesen FD Pro X app > Pubs > Pacific > Oceanic Position Reporting Procedures Oakland Oceanic FIR."*

Present position is callsign, the word position, the fix, time in four digits and altitude. Ensuing fix is name only. The meteorological part is temperature, wind and any significant weather.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 5.3.6: "position should be expressed in terms of latitude and longitude except when flying over named reporting points" For the NAT the report format sits in this chapter of Doc 007; the company Pacific reference covers the Oakland FIR.

Workbook wording: Where can you find a 'template' to make an HF Radio Position report?

FOM and FCTM guidance on Weather avoidance

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

Where are the ICAO oceanic weather deviation procedures?

Our Flight Operations Manual oceanic chapter, reprinted from ICAO Doc 4444

The oceanic weather deviation procedure sits in the oceanic chapter of our Flight Operations Manual, taken from ICAO Doc 4444. It covers the request, what to do when a clearance cannot be obtained, and the offset table. Under 5 nm hold your level. At 5 nm or more, change level 300 ft.

FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace

> *"When weather deviation is required, the pilot should initiate communications with ATC via voice or CPDLC. A rapid response may be obtained by either:"*

Eastbound, deviating left descend 300 ft, right climb 300 ft. Westbound is the opposite. Start the offset about 5 nm off track and be back at level within 5 nm of centerline.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 10.4: "The following procedures are intended for deviations around adverse meteorological conditions." Doc 007 Chapter 10.4 carries the same ICAO weather deviation procedures for the NAT that the FOM reprints.

Workbook wording: Where can you find the ICAO Weather Deviation procedures for Oceanic Controlled Airspace?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

How do you request a weather deviation?

State WEATHER DEVIATION REQUIRED by voice, or send a datalink request

Two ways to get a quick answer. Say WEATHER DEVIATION REQUIRED on the frequency, which tells the controller you want priority, or send a lateral deviation request over datalink. Tell ATC how far you need. If no clearance is coming and the weather is dangerous, use PAN PAN or the urgency downlink.

FOM 5.7.4.4 Weather Deviation Procedures for Oceanic Airspace

> *"Stating "WEATHER DEVIATION REQUIRED" to indicate that priority is desired on the frequency and for ATC response; or • Requesting a weather deviation using a CPDLC lateral downlink message."*

Tell ATC when the deviation is over and you are back on the cleared route. On the 787 the datalink option is the controller pilot datalink lateral request.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 10.4.1: "stating "WEATHER DEVIATION REQUIRED" to indicate that priority is desired on the frequency and for ATC response" ICAO matches the company procedure and adds the urgency call PAN PAN when needed.

Workbook wording: How, specifically to your aircraft, do can you make a weather deviation request?

15.1.2.3.4 ANALYZE WEATHER CONDITIONS USING WEATHER RADAR SYSTEM

How do you insert the offset in the CDU?

LATERAL OFFSET page, enter L or R and distance, execute

Open the LATERAL OFFSET page with the OFST function key, the OFFSET prompt on INIT REF INDEX, the OFFSET prompt on RTE page 1, or LOAD SLOP from the SLOP TRANSITION help window (FCOM 11.42.18).

FCOM 11.42.18 Route Offset and Strategic Lateral Offset Procedure (SLOP)

> *"The LATERAL OFFSET page can be accessed by: selecting the OFST function key, selecting the OFFSET prompt on the INIT REF INDEX page or on the RTE page 1, or by selecting the LOAD SLOP command button on the SLOP TRANSITION Help Window."*

Valid DIST entries are L or R with XX.X, or XX.X then L or R, from 0.1 to 99.0 nm in 0.1 nm steps. SLOP itself is right of course only, 0.0 to 2.0 nm (FOM 5.7.4.7).

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.49: "Pilots should make sure the "TO" waypoint is correct after entering SLOP." Doc 007 leaves the CDU steps to the aircraft manuals but adds this check because some FMS sequence past a waypoint.

Workbook wording: Once received, how (aircraft CDU/FMS specific) can you insert the offset distance?

Oceanic Contingency Procedures

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Engine failure in oceanic airspace. What do you do?

Nearest suitable airport, after getting a revised oceanic clearance

Land at the nearest suitable airport where a safe landing can be made. Get a revised clearance before leaving the track if you can. If you cannot, turn at least 30 degrees left or right, establish a parallel track offset 5 nm, and get below FL290. Tell air traffic control what you have and ask for expeditious handling.

FOM 6.4.2 Engine Failure

> *"14 CFR 121.565 requires the Captain of a two-engine aircraft with one engine inoperative to land at the nearest suitable airport where, in the Captain's judgment after considering all relevant factors, a safe landing can be made."*

Squawk 7700, exterior lights on, and broadcast identification, condition, position and intentions on 121.5. Fuel to go farther, passenger convenience and maintenance are never reasons to pass the nearest suitable airport.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 10.2.1: "a revised clearance shall be obtained, whenever possible, prior to initiating any action" ICAO matches the company procedure of getting a revised clearance before leaving the track.

Workbook wording: What are the engine failure and diversion procedures while in oceanic airspace?

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Immediate action items for engine failure in cruise?

None; ENG FAIL L, R has no memory items

The QRH Quick Action Index marks memory-item checklists with an asterisk; ENG FAIL L, R (QRH 7.14) is not listed, and the checklist has no dashed memory-item line.

QRH Quick Action Index / QRH 7.14 ENG FAIL L, R

*"*indicates a checklist which includes memory items."*

ENG FAIL L, R does not appear on the QRH Quick Action Index at all, so there are no memory items to recite. Reference actions start with A/T ARM switch (affected side) confirm OFF and thrust lever (affected side) confirm idle.

Workbook wording: Review the immediate action items for engine failure in cruise (787, back of the QRC).

CRITICAL OBSERVABLE ITEM

13.1.1.1.1 KNOW ENGINE FAILURE IN CRUISE PROCEDURE

Driftdown off the route. Which way do you turn?

At least 30 degrees, left or right per listed factors

FOM 5.7.4.2 has you leave the cleared track by initially turning at least 30 degrees left or right to establish a parallel, same-direction track offset 5 nm. Direction is a judgment call driven by where you sit in the track system, what is allocated on adjacent tracks, which way the alternate lies, the SLOP you are already flying, and terrain.

FOM 5.7.4.2 General Procedures (Oceanic Emergency/Abnormal)

"The direction of the turn should be based on one or more of the following factors:"

The five factors are position relative to any organized track or ATS route system, direction of flights and levels on adjacent tracks, direction to an alternate airport, any strategic lateral offset being flown, and terrain clearance.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 10.2.1: "leave the cleared route or track by initially turning at least 30 degrees to the right or to the left" Same turn as the FOM, and Doc 007 lists the same direction factors, adjacent tracks, alternate, terrain.

Workbook wording: When turning off the route during a drift-down, which direction will you turn? Why?

SLOP

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is SLOP for?

Collision risk and wake turbulence

The Flight Crew should use SLOP as standard operating practice in normal oceanic operations to mitigate collision risk and wake turbulence. It addresses both wake vortex encounters and the heightened collision risk when non-normal events such as operational altitude deviation errors and turbulence-induced altitude deviations occur.

FOM 5.7.4.7 Strategic Lateral Offsets in Oceanic Airspace

"to mitigate collision risk and wake turbulence."

SLOP is generally accepted but disallowed in specific regions; check the AIP. Any aircraft overtaking another should offset within the procedure to create the least wake turbulence for the aircraft being overtaken.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.48: "This procedure distributes traffic between the centreline and 2 NM right of centreline and greatly reduces collision risk in the airspace by virtue of randomness" ICAO gives the same purposes and also directs SLOP when avoiding wake turbulence.

Workbook wording: What is the purpose of SLOP?

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How far can you offset with SLOP?

Up to 2 nm right of centerline. Never left

SLOP is applied in relation to a route or track. The aircraft may fly zero, or up to a maximum of 2 nm right of course only, from 0.0 to 2.0 nm right of centerline in 0.1 nm increments.

FOM 5.7.4.7 Strategic Lateral Offsets in Oceanic Airspace

> *"SLOP left of course is not authorized."*

Three legal positions in practice: centerline, 1 nm right, 2 nm right, though the FOM permits any 0.1 increment. No ATC clearance is needed. Must be back on centerline by the oceanic exit point.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.49: "Operators that have an automatic offset capability should fly up to 2 NM right of the centreline." ICAO matches the company limit, prohibits left offsets, and with 0.1 nm capability the 787 may fly any tenth up to 2 nm right.

Workbook wording: Up to what distance/direction from the route is SLOP used/acceptable?

16.1.2 PERFORM LONG RANGE COMMUNICATION PROCEDURES

When is an HF position report required?

Only when you are not on CPDLC

When using CPDLC you must contact ARINC after a successful CPDLC logon but prior to the initial Gateway Fix to obtain a SELCAL check on HF, establishing a backup if CPDLC fails. No other HF reports need be made while using CPDLC. Non-CPDLC position reporting varies by area and is specified on the enroute chart.

FOM 5.7.3.18 Position Reporting Oceanic (CPDLC)

> *"Flight Crews should not make HF position reports when using CPDLC procedures."*

REV 125.1 ADDED: if SELCAL is not functioning, one member of the crew must listen to HF continuously. That is the new sentence in this revision and a likely oral question.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 5.1.17: "flight crews of flights using ADS-C should not additionally submit position reports via voice unless requested by aeronautical radio operator" ICAO matches the company rule and separately requires the SELCAL check before oceanic entry.

16.1.2 PERFORM LONG RANGE COMMUNICATION PROCEDURES

Can you enter oceanic airspace with both HF radios out?

No. At least one HF radio is required

FOM 20.4.1 sets dispatch minimums of two HF radios, or a single HF plus SATVOICE. After pushback and prior to entering oceanic airspace the requirement is dual LRNS plus two HF radios, or a single HF with SATCOM data link, or a single HF with SATVOICE.

FOM 20.4.1 Required Equipment for OAK, ANC, and FUK FIRs

> *"After pushback, prior to entering oceanic airspace: • Dual Long Range Navigation Systems (to include 2 MCDUs) • Two HF radios, or – A single HF radio and SATCOM data link, or – A single HF radio and SATVOICE"*

Every authorized combination for OAK/ANC/FUK includes at least one working HF. Same story in the NAT: FOM 22.4.3 requires two independent LRCS, one of which must be HF. FOM 20.4.4.1 covers what to do if both HF fail after you are already established oceanic.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 5.1.1: "Operations in the NAT outside VHF coverage require the carriage of two long range communication systems, one of which must be HF." ICAO matches the company requirement and allows SATVOICE or CPDLC over SATCOM as the second system.

Workbook wording: Can you enter Oceanic airspace with both HF radios not working?

Procedure selection, waypoint coding, navigation performance

8.3.5.23 [PM] DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

Diversion. How do you select a new destination?

Select the airport on the FMS alternates page, then DIVERT NOW

Go to the alternates page on the control display unit, put the airport in the selected slot, then push DIVERT NOW. That builds a route from present position to that airport. Look it over and change what you need, then execute. Executing changes the destination and drops the rest of the old route.

FCOM 11.43.27 FMC Descent and Approach, Alternates Page

> *"Selecting DIVERT NOW displays the route from the present position to the selected alternate using the route displayed on the XXXX ALTN page for the diversion airport."*

The divert also deletes any existing descent constraints. Read the modified route before you execute, because everything from the old route that is not part of the diversion goes away.

Workbook wording: Diversion Procedures/Select New Destination...

8.3.5.23 [PM] DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is the alternate navigation system?

TCP backup navigation after loss of all displays or all FMCs

FCOM 11.50 describes the alternate navigation system as a backup capability hosted in the Tuning and Control Panels, reached by pushing the NAV key on any TCP; on standby power only the left (Captain's) TCP works. It runs two pages, ALTN NAV and ALTN NAV RADIO.

FCOM 11.50.1 Flight Management, Navigation - Alternate Navigation System Description

> *"In the unlikely event that an unrecoverable loss of all displays or loss of all three FMCs occurs, a backup navigation capability is available through the Tuning and Control Panel (TCP)."*

Push NAV on any of the three TCPs under normal power; on standby power only the Captain's left TCP is available. LNAV and VNAV are lost, but autopilot and autothrottles may still be available.

Workbook wording: Alternate Navigation System...

Airspace ownership vs control

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

NUUK FIR. Who controls what, and what do you do?

Nuuk owns it; fly Reykjavik OCA north, Gander OCA south

FOM 22.4 makes the ownership-versus-control distinction operational. In North Atlantic operations CPDLC and ADS coverage is based on Oceanic Control Areas, not FIRs. The logon must be directed to the controlling authority for the relevant OCA regardless of which FIRs you transit.

FOM 22.4 Communication and Surveillance

> *"For example, the north section of the Nuuk FIR is in the Reykjavik OCA, but the south section of the Nuuk FIR is in the Gander OCA. CPDLC should sequence and a new SELCAL check is required."*

The practical action is the SELCAL check: CPDLC coverage in the NAT follows OCAs, not FIRs, so crossing inside one FIR you still change controlling OCA, CPDLC re-sequences, and a new SELCAL check is required on the new HF frequencies.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.41: "the pilot should log on to the appropriate FIR 10 to 25 minutes prior to the boundary" ICAO gives the generic logon timing; the company chapter names the controlling OCA for the Nuuk FIR.

Workbook wording: NUUK FIR: Owned by Nuuk but controlled by Gander South of N6330 and Reykjavik to the North. What are...

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Can you cross the NAT-HLA without HF?

No. At least one HF radio is required

FOM 22.4.3 requires two independent LRCS outside VHF coverage in the NAT, one of which must be HF. FOM 22.1.10 lists the NAT HLA equipment as dual IRS, dual long range nav with two MCDUs, dual NDs, one GPS receiver, ADS-C, CPDLC, SATCOM data link, and two HF radios or a single HF radio and SATVOICE.

FOM 22.4.3 HF Radio Requirements

> *"Operations in the NAT outside of VHF coverage require aircraft to be equipped with two independent Long Range Communications Systems (LRCS), one of which must be HF."*

Cross-check FOM 7.1.8, which lists the only two NAT HLA ETOPS dispatch options: 2 HF radios plus SATDATA/CPDLC, or 1 HF plus SATDATA/CPDLC plus SATVOICE. Both contain HF. FOM 22.1.10's note softens it only after departure: a single HF radio and SATCOM data link.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 4.2.12: "When operating outside of VHF coverage the carriage of fully functioning HF is mandatory" ICAO matches the company rule that HF is required to cross the NAT outside VHF coverage.

Workbook wording: Can you Fly across the NAT-HLA without HF radios? If so, how?

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What is your oceanic exit point?

Flight specific; the point re-entering domestic airspace

FOM 22.1.6 identifies the Oceanic Exit Point (OXP) as the point of re-entry into domestic airspace, and directs crews to request Normal Speed via CPDLC or voice after it when ATC has assigned a fixed Mach.

FOM 22.1.6 Mach Number Technique and Cost Index (ECON)

> *"When ATC assigns a fixed Mach number in Oceanic Airspace, crews should request Normal Speed (via CPDLC or voice) after the Oceanic Exit Point (OXP) upon re-entry into domestic airspace."*

The OXP is taken off your flight plan, not defined in the FOM.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.61: "If ATC assigns a fixed Mach number in oceanic airspace, request NORMAL SPEED (via CPDLC or voice) after the OXP in Domestic airspace." ICAO matches the company procedure for resuming ECON after the OXP.

Workbook wording: What is your oceanic Exit point?

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Initial call to Gander or Shanwick. Exact words?

Callsign, SELCAL check, and the next OCA

FOM 22.4 publishes the model call for an eastbound flight entering the Gander OCA: Gander Radio, Alaska 123, SELCAL Check, Shanwick Next. The response format shown is Alaska 123, Gander Radio, HF Primary and Secondary, at 30W contact Shanwick Radio HF Primary and Secondary.

FOM 22.4 Communication and Surveillance

> *"Example (Initial contact from an eastbound flight entering GANDER OCA): "Gander Radio, Alaska 123, SELCAL Check, Shanwick Next."*

Expect the reply to hand you HF primary and secondary frequencies plus the next transfer point, e.g. at 30W contact Shanwick Radio. A new SELCAL check is required before entering each individual OCA, not each FIR (FOM 22.4).

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 5.1.13: "after the radio operator responds, request a SELCAL check and state the next OCA" ICAO matches the company model call word for word, including the Gander Radio example.

Workbook wording: What does your initial call to Gander or Shanwick on VHF or HF require? (Exact verbiage (not RCL call))

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

When do you request a SELCAL check?

After CPDLC logon, before the initial gateway fix

When using CPDLC, contact ARINC after a successful CPDLC logon but prior to the initial Gateway Fix in order to obtain a SELCAL check on HF. This establishes a backup means of communication if CPDLC fails.

FOM 5.7.3.18 Position Reporting Oceanic (CPDLC)

> *"Flight Crews must contact ARINC after a successful CPDLC logon, but prior to the initial Gateway Fix in order to obtain a SELCAL check on HF."*

REV 125.1: if SELCAL is not functioning, one crew member must listen to HF continuously, which is a real workload consideration on a long Pacific crossing.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 5.1.22: "check the operation of the SELCAL equipment, at or prior to entry into oceanic airspace, with the appropriate radio station." The company timing, after CPDLC logon and before the gateway fix, satisfies this ICAO deadline.

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Review the airspace around eastbound oceanic entry

GOTA transition area, then the Gander OCA

FOM 22.2.1 describes the transition areas adjacent to the major OCAs, GOTA, NOTA, SOTA and BOTA, as buffers that are not part of the OCAs. Aircraft in them remain under domestic ATC: GOTA under Gander or Montreal Centers, NOTA and SOTA under Shannon Domestic, BOTA under Brest Domestic.

FOM 22.2.1 Transition Areas

> *"Transition areas such as the Gander Oceanic Transition Area (GOTA), Northern Oceanic Transition Area (NOTA), Shannon Oceanic Transition Area (SOTA), and Brest Oceanic Transition Area (BOTA)"*

Two GOTA traps for eastbound entry: OEPs sit in the middle of the airspace so SLOP starts halfway through, and you send the RCL to Gander Oceanic while still talking to a domestic center (FOM 22.2.1).

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 3.3.12: "Air Traffic service is provided by the Gander ACC, call sign GANDER CENTRE." ICAO confirms GOTA traffic stays with domestic Gander Centre, matching the company description.

Workbook wording: Review airspace surrounding Eastbound oceanic entry...

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Westbound into BIRD at RATSU. Oceanic clearance needed?

No. Only Shanwick still issues oceanic clearances

FOM 22.6.1 states that except for the Shanwick OCA, oceanic clearances are no longer issued in North Atlantic OCAs. For all other OCAs the pre-departure clearance serves as the oceanic clearance. The crew still transmits an RCL message as a notification.

FOM 22.6.1 Request for Clearance (RCL) and Oceanic Clearances

> *"Except for the Shanwick OCA, oceanic clearances are no longer issued in North Atlantic OCAs."*

You still send an RCL message as a notification into Reykjavik; the pre-departure clearance serves as the oceanic clearance.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 6.3.35: "Enroute aircraft shall enter the oceanic airspace in accordance with their existing ATC clearance. No oceanic clearance is required." ICAO matches the company answer; a separate note keeps Shanwick issuing clearances until further notice.

Workbook wording: Do westbound routes that enter BIRD from Scottish at RATSU need an oceanic clearance?

International Theaters (6)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Where is the aircraft Sanitation Certificate?

Not named in our manuals; check Comply365 international documents

The FOM names only the Airworthiness Certificate, Aircraft Registration and radio license as always required on board. Country-specific operating documents are added for international flights. Those are electronic, on the electronic flight bag in Comply365. No sanitation certificate appears anywhere in the FOM, FCOM or QRH.

FOM 5.2.1.2 International Documents

> *"These documents are electronic and located on the EFB in Comply365."*

The closest named item is the Certificate of Residual Disinsection, filed in Comply365 under Flying, Regulatory and Operating Docs, International Documents. Confirm the Japan ramp check list with the check airman.

Workbook wording: Where is the aircraft Sanitation Certificate located? (inspected during Japanese JCAB Ramp Checks)

1.2.3 PERFORM INTERIOR/EXTERIOR INSPECTION

Is HA approved for the NADP 1 departure?

Yes. Normal takeoff procedures satisfy it

FOM 5.4.7 carries the noise abatement departure procedure under Ops Spec C068. Compliance with published noise abatement procedures is required at noise-sensitive airports. Because normal takeoff procedures already satisfy both the close-in (NADP 1) and distant (NADP 2) profiles, No separate NADP profile to fly.

FOM 5.4.7 Noise Abatement Departure Procedure (NADP), Ops Spec C068

> *"(AS) Normal takeoff procedures satisfy either close-in or distant NADP profile requirements."*

The FOM tags this (AS), and post-merger it is the governing procedure. Specific gradients and configuration-change altitudes come from the fleet manuals, the 10-7, or the Jeppesen Airway Manual noise abatement pages.

Workbook wording: Is HA allowed to perform the NADP 1 Departure Procedure? (Japanese JCAB Ramp Check question; they...

Dispatch Release: Review all sections

CRITICAL OBSERVABLE ITEM

1.1.1 PERFORM FLIGHT PLANNING

Where is country-specific guidance?

FOM theater chapters plus Jeppesen State Rules and Procedures

FOM chapters 19 through 24 are organized by theater and carry country-specific airspace, navigation, communication and procedural rules.

FOM chapters 19-24 Theater chapters; example FOM 19.3.1.2 Holding Airspeeds

> *"For holding procedures and airspeeds, see Jeppesen FD Pro app at the following location: Pubs > North America > North America Airway Manual > State Rules and Procedures – Canada > Instrument Flight Rules – Holding Procedures."*

The six theater chapters are 19 North America, 20 Pacific, 21 Caribbean, 22 Atlantic, 23 Europe, 24 Asia. They repeatedly hand off to Jeppesen FD Pro State Rules and Procedures, and airport-specific items live in the Jeppesen 10-7 pages.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 Foreword: "This Document is for guidance only. Regulatory material relating to North Atlantic aircraft operations" For the NAT theater this manual sits behind the FOM chapter; the regulatory material lives in ICAO Annexes, State AIPs and NOTAMs.

Workbook wording: International Planning: Where can you find country-specific guidance?

Holding (Procedures, speeds)

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are international holding speeds?

ICAO Doc 8168 and the theater chapters

FOM 5.5.15 cites AIM 5-3-8 and ICAO Doc 8168. Country-specific holding speeds and procedures appear in the theater chapters 19 through 24 and in the Jeppesen country pages, for example the Heathrow holding restriction in the Europe chapter.

FOM 5.5.15 Holding, ICAO DOC 8168

> *"ICAO DOC 8168 Flights anticipating holding at an enroute fix or flights that have not received clearance beyond a fix should begin a speed reduction no earlier than three minutes prior to reaching such fix."*

Rev 125.1 renumbering: the United Kingdom chapter is 23.17 with LHR subsections under 23.17.1, and Italy is 23.18.

Workbook wording: Where can you find international country-specific holding speeds?

Airspace ownership vs control

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

GPS jamming reporting requirements?

Report to ATC; file a safety report

FOM 5.1.21.3 says GPS/GNSS disruptions cluster around conflict zones, military operations areas and counter-UAS protection areas, and directs the crew to report any GPS/GNSS malfunction to ATC and submit a safety report for the listed discrepancies.

FOM 5.1.21.3 GPS/GNSS Disruptions

> *"Report any GPS/GNSS malfunction to ATC. Submit a safety report for any of the following discrepancies or anomalies:"*

The safety report is required for any of five listed anomalies: inability to use GPS for navigation, loss of RNAV/RNP capability, unreliable EGPWS triggering, inaccurate position on the ND, and loss of or erroneous ADS-B outputs (FOM 5.1.21.3). Background on the threat is in 5.1.21.2.

NAT DOC 007, REFERENCE ONLY. COMPANY PROCEDURE GOVERNS

NAT Doc 007 9.1.10: "Flight crews should inform ATC of any GNSS malfunction. ATC aircraft separation minimums may be affected by the GNSS malfunction." ICAO matches the report to ATC; the safety report is a company addition.

Workbook wording: GPS Jamming reporting requirements.

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Arrival and departure considerations on the Jepps

Transition altitude charted; transition level from ATIS or ATC

FOM 23.4 lists the ICAO chart and weather differences that bite on arrival and departure. Visibility comes in meters or kilometers and QNH in hectopascals, with some fields defaulting to QFE. Transition altitude and level are typically low and country-dependent, and STAR charts may just say TRANS LEVEL: BY ATC.

FOM 23.4 Weather (Europe Theater)

> *"• Transition Altitude/Level: Typically, low in Europe and can vary by country. Vigilance is required for correct altimeter settings on arrival and departure. STAR charts in Europe may indicate "TRANS LEVEL: BY ATC."*

European transition altitudes run lower than US values and vary by country (FOM 22.1.7, 23.10). Altimeter settings come in hectopascals, visibility in meters or kilometers, and wind speed can be in meters per second. Jeppesen FD Pro search term for this is TRANSITION LEVEL.

Workbook wording: Arrival/Departure considerations – Jepps...

TCAS and Traffic (6)

TCAS

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are TCAS procedures in the FOM?

TCAS Policy, in the Operating - General chapter

FOM 2.3.17 TCAS Policy is the FOM home for TCAS. It requires compliance with an RA unless the pilot considers it unsafe, prohibits maneuvering opposite an RA unless visually determined to be the only means of safe separation, and reminds crews the aircraft in sight may not be the intruder.

> **FOM 2.3.17 TCAS Policy**
>
> *"Compliance with a TCAS Resolution Advisory (RA) is required, unless the pilot considers it unsafe to do so."*
>
> The FOM carries policy only. The actual maneuver is the QRH recall item Traffic Avoidance (QRH MAN.1.5), and the system description is FCOM 15.20.17. Oceanic TCAS guidance is at FOM 5.7 (TA/RA mode whenever possible).
>
> *Workbook wording: Where can you find TCAS procedures in the FOM?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What is a TA?

Traffic advisory: traffic predicted to conflict in 20-48 seconds

Per FCOM 15.20.18, a Traffic Advisory predicts another airplane entering the conflict airspace in 20 to 48 seconds, and exists to help the crew get visual contact. Indications are the amber ND message TRAFFIC plus a single 'TRAFFIC, TRAFFIC' voice annunciation; No PFD/HUD vertical guidance for a TA.

> **FCOM 15.20.18 Traffic Advisories (TA) and Display**
>
> *"A TA is a prediction that another airplane will enter the conflict airspace in 20 to 48 seconds."*
>
> A TA gets an amber TRAFFIC message on the ND and one 'TRAFFIC, TRAFFIC' aural, no vertical guidance. TA symbol is a filled amber circle (amber circle with black chevron for ADS-B traffic).
>
> *Workbook wording: What is an "TA"?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What is an RA?

Resolution advisory: conflict predicted in about 15-35 seconds

FCOM 15.20.18 defines an RA as a prediction that another airplane will enter the TCAS conflict airspace within approximately 15 to 35 seconds; without altitude data from the intruder, no RA can be generated.

> **FCOM 15.20.18 Resolution Advisories (RA) and Display**
>
> *"An RA is a prediction that another airplane will enter the TCAS conflict airspace within approximately 15 to 35 seconds."*
>
> No RA is possible if the intruder is not reporting altitude. RA gives a red ND TRAFFIC message, a voice annunciation, and PFD/HUD vertical guidance; the symbol is a filled red square.
>
> *Workbook wording: What is an "RA"?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What is a phantom TA?

Not published in our manuals. Ask the check airman.

The term phantom traffic advisory appears nowhere in the FOM, FCOM or QRH, so No company definition to recite. What the FOM does warn about is the related trap: the airplane you can see out the window may not be the one the system is alerting on. Do not brief a definition you cannot source.

FOM 2.3.17 TCAS Policy

> *"Flight Crews are cautioned that the aircraft in visual contact may not necessarily be the intruder aircraft causing the TCAS RA."*

The traffic surveillance description in the FCOM covers how targets are built from transponder and ADS-B data, but it never uses that phrase either.

Workbook wording: What is Phantom TA?

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Where are the TCAS symbology definitions?

FCOM Warning Systems chapter, TCAS traffic display pages

Symbol definitions are in FCOM Chapter 15 Warning Systems: the annotated display legend at 15.10.16-15.10.17 and the same set described in prose at 15.20.17-15.20.19. TCAS-only traffic uses a filled red square for an RA, a filled amber circle for a TA, a filled white diamond for proximate traffic, and an unfilled white diamond for other traffic.

FCOM 15.10.16 TCAS Traffic and Alert Message TRAFFIC Display

> *"filled red square indicates a resolution advisory (RA)"*

Controls and Indicators at 15.10.13-15.10.20 has the annotated ND picture and symbol legend; the narrative repeat is in System Description at 15.20.17-15.20.19. ADS-B (In) symbology is separate, at FCOM 10.40.

Workbook wording: Where can you find TCAS symbology definitions?

Airspace ownership vs control

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What transponder setting monitors traffic?

XPDR or above; that enables ADS-B (In)

FCOM 11.10 lists the Transponder/TCAS Mode Selector positions. STBY leaves the transponder inactive. ALT RPTG OFF enables modes A and S, disables altitude reporting, enables ADS-B out but disables ADS-B (In).

FCOM 11.10 Flight Management, Navigation - Controls and Indicators (Transponder/TCAS Mode Selector)

> *"• altitude reporting disabled • ADS-B out enabled • ADS-B (In) disabled XPDR (transponder) – • transponder enabled in modes A, C, and S"*

ADS-B (In) is disabled in STBY and in ALT RPTG OFF, so anything below XPDR blinds you to traffic.

Workbook wording: What should you set your transponder to in order to monitor traffic?

Medical (7)

MedLink

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

What is MedLink?

Contract ER physicians giving 24-hour inflight medical advice

FOM 11.1.9.1 describes MedLink as emergency room physicians available 24 hours a day with aviation-specific experience, providing recommendations on onboard medical issues, and notes contact with MedLink removes Company liability for passenger health.

FOM 11.1.9.1 MedLink Overview

> *"MedLink utilizes emergency room physicians 24 hours a day to provide crewmembers with informed recommendations regarding on-board medical issues."*

Company policy is to follow MedLink's recommendation, but the Captain keeps final authority on operational requests. The one-way exception: if MedLink says a passenger is not fit to fly, the Captain will not override MedLink.

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

When do you contact MedLink?

Any medical support need, aircraft oxygen use, fitness-to-fly doubt

FOM 11.1.9.2 makes MedLink contact mandatory, except during critical phases of flight, for any medical situation where the crew requires support or advice. It is also mandatory for any passenger or crew use of aircraft oxygen, and any uncertainty about whether a passenger is contagious or fit to fly (11.1.8.2).

> **FOM 11.1.9.2 When to Contact MedLink**
>
> *"MedLink shall be contacted for the following, except during critical phases of flight:"*
>
> Note the carve-out: not during critical phases of flight (sterile flight deck per 2.3.12). Passenger or crew use of aircraft oxygen is a mandatory trigger even if the person seems fine.
>
> *Workbook wording: When should you contact MedLink?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Who contacts MedLink on the ground?

CSA at the gate; after door close, F/As or crew

FOM 11.1.9.3 On Ground has the Flight Crew, F/As, station personnel, and MedLink work together to decide fitness to fly, and requires MedLink contact to clear a passenger even if paramedics already responded. At the gate or prior to door closure the CSA should initiate contact, and may deplane the passenger for privacy while assessing fitness to fly.

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - On Ground**
>
> *"The CSA should initiate contact with MedLink."*
>
> After door closure and after landing the split is by equipment: with operational broadband or a SATCOM cabin device the F/As initiate; without it the Flight Crew initiates. Advise Operations if medical assistance is expected at the gate.
>
> *Workbook wording: Who is responsible for contacting MedLink while on the ground?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Who contacts MedLink in flight?

F/As if cabin SATCOM works; otherwise the Flight Crew

Per FOM 11.1.9.3 In Flight, on HA aircraft with available and operational cabin SATCOM handsets the F/As advise the Flight Crew and then call MedLink directly; if the cabin device is unavailable, the Flight Crew initiates.

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight**
>
> *"On aircraft with operational SATCOM handsets in the cabin, the F/As call MedLink directly."*
>
> MedLink is reached directly, with Dispatch conferenced in when a diversion looks possible. Get the flight attendants gathering vitals while you set up the call, so the doctor has data on first contact.
>
> *Workbook wording: Who is responsible for contacting MedLink while in flight?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Do you call MedLink direct or Dispatch first?

Directly. Dispatch is conferenced in if diversion possible

FOM 11.1.9.3 has you go to MedLink first, not Dispatch. With operational cabin SATCOM the F/As call MedLink directly; with SATCOM but no cabin handset the 787 Flight Crew calls MedLink directly at (602) 282-4910 and conferences the F/As on the cabin headset.

> **FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight**
>
> *"The Flight Crew will call MedLink. MedLink Inflight Medical Assistance may be reached at (602) 282-4910."*
>
> Only with no operational SATCOM does the call route through a relay: ARINC patch, or Dispatch if ARINC fails. Dispatch always gets notified for a potential diversion, either by MedLink conferencing them in or by you calling after.
>
> *Workbook wording: Do we as flight crew contact MedLink directly, or do we call dispatch first?*

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

How do you make a MedLink call from the aircraft?

SATCOM to (602) 282-4910, or ARINC phone patch

Per FOM 11.1.9.3, the 787 primary method is a SATCOM call to MedLink Inflight Medical Assistance at (602) 282-4910, with the Flight Crew conferencing in the F/As using the cabin headset.

FOM 11.1.9.3 Responsibility for Contacting MedLink - In Flight

> *"The Flight Crew will contact ARINC for a patch to MedLink Inflight Medical Assistance at (602)"*

On the 787 conference the F/As in on the cabin headset so the physician talks to the person with eyes on the patient. Ground Medical Assistance is (602) 282-6613 or (800) 961-4847; the (AS) number listed elsewhere in Ch 11 is (602) 282-4967.

11.1.10.4 KNOW AIRPLANE SHUTDOWN AND SECURING PROCEDURES

Medical event on an international flight. Any special notifications?

Yes. General Declaration entry; Australia notified 30 minutes before landing

FOM 18.1.15 uses the US Public Health Service ill person definition. Temperature of 100F (38C) or greater persisting more than 48 hours counts. So does fever with rash, swollen glands, jaundice, persistent cough, vomiting, difficulty breathing, muscle pain, headache with stiff neck, decreased consciousness, or unexplained bleeding, and persistent diarrhea.

FOM 18.1.15 Death or Illness Reporting - International

> *"For Flights to Australia: The Captain must report to the Australian authorities the presence on board the aircraft of any apparent death or any ill person among passengers or crew at least 30 minutes before landing."*

The General Declaration requirement applies to countries other than the US and Canada. For a suspected communicable disease inbound to the US, 11.1.8.2 adds the CDC/public health authority path, usually conferenced in by MedLink or set up through Dispatch.

Workbook wording: If on an international flight, are there special procedures/notifications that must be made by the flight...

HAZMAT (6)

Load Closeout/TPR Update and review procedure

1.2.2 PERFORM ACARS AND FMS INITIALIZATION AND VERIFICATION

Dangerous goods on board. What governs?

Review and acknowledge the eNOTOC, then retain it

Dangerous goods on board means the PIC must receive a NOTOC (FOM 14.1.1).

FOM 14.1.6 Captain Responsibilities (HA)

> *"• Acknowledging the eNOTOC by selecting either "ACCEPT*" on the MCDU or "ACKNOWLEDGE NOTOC" on the Comm page (depending on fleet type)."*

HA rules differ from AS: hand edits to the eNOTOC are not permissible (AS allows Ramp Lead edits to tail number and DG location on the paper NOTOC, FOM 14.1.5). eNOTOC arrives at departure minus 30 min international, minus 15 min interisland (FOM 14.1.4).

Workbook wording: Describe the Dangerous goods procedures if you have any on your flight...

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

Where is the dangerous goods policy in the FOM?

The HAZMAT chapter, under Dangerous Goods

Dangerous goods policy lives in FOM Chapter 14, HAZMAT, section 14.1 Dangerous Goods, starting with 14.1.1 Regulatory Compliance (Ops Spec A055, 49 CFR Parts 171-175, IATA DGR). Key subsections for a 787 Captain: 14.1.4 eNOTOC (HA), 14.1.6 Captain Responsibilities (HA), 14.1.8 NOTOC acceptable versions (HA), 14.1.17 rejected shipments (HA), 14.1.20 ICAO ERG drill codes, and 14.1.26 damaged or leaking DG.

FOM 14.1 Dangerous Goods (Chapter 14 HAZMAT)

> *"• The Company accepts Dangerous Goods (DG) in accordance with International Air Transport Association (IATA) Dangerous Goods Regulations and the domestic Code of Federal Regulations (CFR) Title 49."*

FOM Chapter 14 is titled HAZMAT and section 14.1 is Dangerous Goods, running 14.1.1 through 14.1.28. eNOTOC and HA Captain responsibilities are 14.1.4 and 14.1.6.

Workbook wording: Where can the dangerous goods policy be found in the FOM?

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

What is an eNOTOC?

The electronic Notice to Captain of dangerous goods, sent by ACARS

Per FOM 14.1.4 the eNOTOC is the Hawaiian electronic Notice to Captain for dangerous goods, uplinked at departure minus 30 minutes for trans-Pacific and international flights and departure minus 15 minutes for inter-island, and it prints automatically.

FOM 14.1.4 eNOTOC (HA)

> *"The eNOTOC will be sent to the Flight Crew at departure time minus 30 minutes for trans-Pacific and international flights and at departure time minus 15 minutes for inter-island flights."*

It prints automatically on all aircraft. Versions per FOM 14.1.8: eNOTOC (primary), hard-copy eNOTOC from the Load Master if ACARS/printer is inop but FM Amadeus works, and HAL Form 902 (freighter DG COMAT only).

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

How do you acknowledge an eNOTOC?

Select ACKNOWLEDGE NOTOC on the Comm page

FOM 14.1.6 requires the HA Captain to acknowledge the eNOTOC by selecting ACCEPT* on the MCDU or ACKNOWLEDGE NOTOC on the Comm page, depending on fleet type, which on the 787 means the MFD COMPANY/communications page.

FOM 14.1.6 Captain Responsibilities (HA)

> *"• Acknowledging the eNOTOC by selecting either "ACCEPT*" on the MCDU or "ACKNOWLEDGE NOTOC" on the Comm page (depending on fleet type)."*

Acknowledge the NOTOC on the Comm page. If digital acknowledgment is impossible, acknowledge by voice to Dispatch and note it, because the acknowledgment is the record that you saw the load.

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

How do you fight a lithium battery fire?

HALON, then water. Never ice

FIGHT THE FIRE with a HALON extinguisher. Water may be used to fight a lithium ION battery fire but should NOT be used on a lithium METAL battery fire. It is very important to prevent the fire spreading to adjacent flammable materials.

FOM 11.1.10.2 PED Fire Containment Bag (Yellow Bag)

> *"AVOID USING ICE to cool the device. Ice will act as an insulator."*

It may take 15 or more minutes to completely cool the device. REV 125.1: this is the YELLOW bag section, renumbered from 11.1.10.

15.2.1 DEMONSTRATE DANGEROUS GOODS POLICIES AND PROCEDURES

Where are the cabin lithium battery limits?

The Alaska Airlines onboard PED and prohibited-items web pages

FOM 13.4 sends you to the alaskaair.com onboard-policies portable electronic devices page and the prohibited-items page for the list of permitted and restricted devices and battery types/sizes.

FOM 13.4 Portable Electronic Devices / FOM 13.4.1 Lithium Batteries in PEDs

> *"See the list of portable electronic devices that are permitted and restricted on board at: https://www.alaskaair.com/content/travel-info/policies/onboard-policies-portable-electronic-devices and https://www."*

The FOM itself does not publish watt-hour numbers; it points to the alaskaair.com PED policy and prohibited-items pages. FOM 13.4.1 gives the carry-on vs checked rules, and 11.1.10 covers the containment bags.

Workbook wording: Where can you find lithium battery type/size limitations to be carried in the cabin?

Descent and Approach (37)

In Trail Procedures (OpSpec A354)

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is ITP?

In Trail Procedure, an ADS-B based flight level change request

ITP uses ADS-B In data to list nearby traffic and designate up to two reference airplanes ahead or behind on a similar track. The request goes to ATC by datalink, and ATC approves or denies the altitude change. Available from FL240 to the service ceiling.

FCOM 11.35 In Trail Procedure (ITP)

> *"the ITP application lets the flight crew select a desired altitude and create an ATC request to climb or descend to that altitude"*

Reference airplanes need at least 15 NM with 20 knots or less closure, or 20 NM with 30 knots or less.

Workbook wording: What is "ITP"?

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

When would you use ITP?

When traffic between levels blocks a normal climb or descent

Use it in oceanic airspace when you want a different level, say for turbulence, winds, or fuel, but airplanes at intermediate levels would normally block the clearance. ITP lets ATC approve the climb or descent past up to two properly spaced reference airplanes.

FCOM 11.35 In Trail Procedure (ITP)

> *"If the desired flight level is available, but airplanes are between the current and desired flight levels, the application determines if these airplanes can be used as ITP reference airplanes"*

Workbook wording: When might you consider using the ITP?

Diversion Procedures/Select New Destination

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How do you select a new destination?

Pick the airport on the alternates page, then DIVERT NOW

Line select the airport on the alternates page so it shows selected, then push DIVERT NOW. Review the route on the modification page and execute. Executing changes the destination, folds the diversion into the active flight plan, drops the rest of the old route, and deletes any descent constraints.

FCOM 11.43.27 Alternate Airport Diversions

> *"Selecting DIVERT NOW displays the route from the present position to the selected alternate using the route displayed on the XXXX ALTN page for the diversion airport."*

Typing a new airport into the list does not select it. Watch for the DESCENT PATH DELETED message and rebuild the arrival for the new field.

Workbook wording: How (aircraft specific) do you "Change" or "Select" a new destination?

Alternate Navigation System

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

What is alternate navigation?

TCP backup navigation after losing all displays or all FMCs

Alternate navigation is a backup capability in the Tuning and Control Panel used after unrecoverable loss of all displays or all three FMCs (FCOM 11.50). Under normal power the NAV key on any of the three TCPs brings it up; on standby power only the Captain's TCP works.

> **FCOM 11.50 Alternate Navigation System Description**
>
> *“In the unlikely event that an unrecoverable loss of all displays or loss of all three FMCs occurs, a backup navigation capability is available through the Tuning and Control Panel (TCP).”*
>
> LNAV and VNAV are not available in alternate navigation, but autopilot and autothrottle may be. On standby power only the left (Captain's) TCP is available.
>
> *Workbook wording: What is Alternate Navigation?*

8.3.4.26 DEMONSTRATE ABILITY TO CORRECTLY PROGRAM AIRCRAFT GUIDANCE SYSTEMS INCLUDING APPROACH

How do you fly alternate navigation to a point?

Push NAV, enter waypoint latitude and longitude, fly the displayed track

This is the backup for losing all displays or all three flight computers. Push NAV on a tuning and control panel and enter the target latitude and longitude. The panel then shows true track to that point, ground track, groundspeed, distance and time to go. Fly the track manually or with the autopilot.

> **FCOM 11.50.2 Alternate Navigation Page**
>
> *“The ALTN NAV page provides a stand-alone single waypoint navigation capability. Once the waypoint’s latitude and longitude have been entered at LSK 2 L/R, flight progress information to the waypoint is then displayed.”*
>
> One waypoint at a time, and you update it yourself. On standby power only the Captain panel works. LNAV and VNAV are gone, but autopilot and autothrottle may still be available.
>
> *Workbook wording: How do you perform Alternate Navigation to a point or destination?*

Descent Preparation

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What goes into preparing for descent?

Review messages, check landing performance, set minimums, brief the approach

Start it before leaving cruise altitude and finish it by 10,000 feet above field elevation. Review alerts, memos and operational notes, check landing performance and set VREF, set minimums, set the navigation radios and the autobrake. Crosscheck the flight computer against the chart, brief, then call for the descent checklist.

> **FCOM NP.21.38 Descent Procedure (Amplified Procedures)**
>
> *“Start the Descent Procedure before the airplane descends below the cruise altitude for arrival at destination. Complete the Descent Procedure by 10,000 feet AFE.”*
>
> Minimums depend on approach type: minimum descent altitude plus 50 for an MDA approach, the published decision altitude for a DA approach, radio for CAT II, blank for CAT III.
>
> *Workbook wording: What items are included in preparing for descent?*

Descent Plan Execution

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

How do you start the descent?

Set a lower altitude, then push the altitude selector or DES NOW

Set the lower altitude in the mode control panel window. At top of descent the airplane starts down on its own in VNAV. To go early, push the altitude selector when inside 50 nm of top of descent, or page to the descent page, line select DES NOW, and execute.

FCOM 11.42.36 FMC Cruise, Early Descent

> *"During cruise, setting an altitude below the current altitude in the MCP altitude window and pushing the altitude selector activates the DES NOW function"*

Outside 50 nm from top of descent, pushing the selector gives a cruise descent to that altitude instead of a path descent. DES NOW gives about 1250 feet per minute at economy speed.

Workbook wording: How do you get the aircraft to begin the descent?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Cleared the arrival vs cleared to descend via?

Not spelled out in our manuals. Ask the check airman.

The only place our manuals use the phrase is the Mexico section, where a descend via clearance means fly the published lateral routing and the published altitudes. Neither the FOM nor the FCOM defines it against a plain arrival clearance. That distinction lives in the Aeronautical Information Manual, not in our books.

FOM 21.3.3.5 Mexico Phraseology for Climbs and Descents

> *"ATC expects the pilot to navigate on the lateral routing and altitudes as published on a procedure. Typically, the controller will issue a "Descend Via" or "Climb Via" clearance to keep the radio transmission short and concise."*

Central America does not use climb via or descend via at all. Charted altitude restrictions still apply there unless the controller specifically relieves you of them.

Workbook wording: What is the difference between "Cleared XX arrival" vs "Cleared to descend via XX arrival"?

Analysis of Airport Conditions

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Does the FOM table CAT II and III limitations?

No. It points to the fleet manuals, PRC and QRH

FOM 5.6.17 directs Flight Crews to use the approach procedures in the fleet-specific manuals for all CAT II/III approaches. Missed approach criteria for CAT II/III are also in the fleet-specific manuals.

FOM 5.6.17 Category II/III Operational Requirements

> *"The approach briefing references and required equipment tables in the PRC/QRH contain complete listings of the Flight Crew procedures and required equipment."*

Ops Spec C060. Also: Alaska may be restricted by variant from CAT II/III at certain runway/airport combinations, see the tailored Jeppesen chart notes (5.6.17.1).

Workbook wording: Is there a table in the FOM categorizing CAT II/III Approach and Landing Limitations?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

When can you autoland?

Category II or III runways only, with an autoland qualified Captain

The runway has to be a Category II or III runway, and on the 787 the Captain must be autoland qualified. At Hawaiian that means one automatic landing completed in the aircraft. Advise the tower so the ILS critical area is protected, since it is only protected automatically below 800 feet or 2 miles.

FOM 5.6.18 Autoland Operations

> *"Autoland approaches and landings are permitted on CAT II/III runways only. Pilots should advise the tower anytime Autoland operations are conducted to ensure the ILS Critical Zone is protected."*

Before committing, check tower wind against the autoland wind limits and check the autoland maximum altitude for the field.

Workbook wording: Under what conditions can you perform an autoland?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What qualifies a 787 Captain for autoland?

One automatic landing in the aircraft

For CAT II and CAT III, autoland is the approved means. The Captain must be autoland qualified and must have completed one automatic landing in the aircraft. Record every autoland attempt on the ACARS Autoland Report page.

> **FOM 5.6.17 / 5.6.18**
>
> *"(HA) The Captain must complete one automatic landing in the aircraft to be qualified for CAT II/III (Autoland) operations."*
>
> This is a currency item that bites new Captains: without that one automatic landing logged in the aircraft, you are not CAT II/III qualified regardless of sim performance. Autoland reporting requirements are at FOM 7.1.14.
>
> *Workbook wording: Under what conditions must you perform an autoland?*

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

How do you log an autoland?

Fill in every field on the datalink Autoland Report page

Every autoland attempt gets reported on the ACARS Autoland Report page, logbook page information included. If the datalink is inoperative, write a logbook entry giving airport, landing runway, category, and whether it was successful. An unsuccessful autoland also needs an Info to Maintenance entry explaining the failure.

> **FOM 5.6.21.2 Autoland Reporting, ACARS**
>
> *"For all Autoland attempts, the Captain is responsible for ensuring that all applicable fields on the ACARS Autoland Report page are filled in, including the logbook page information."*
>
> Autoland landings are also recorded on the flight summary page. A successful autoland that maintenance did not request needs no logbook entry at all.
>
> *Workbook wording: What are the requirements to log/report an autoland?*

Diversion Considerations

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

When do you start the divert?

Early enough to reach the alternate with 30 minutes fuel

Decide with Dispatch, and decide before you are into minimum fuel. Ask air traffic control for delay information early so you do not hold, then divert, and land at the alternate under 30 minutes. Minimum fuel is only declared once you are committed to a field, so it is not a decision tool.

> **FOM 11.2.5 Minimum Fuel**
>
> *"The decision whether or not to divert to an alternate or other diversion airfield should be made in coordination with Dispatch prior to the declaration of minimum fuel."*
>
> If it becomes clear you will land with less than 30 minutes of fuel, declare MAYDAY MAYDAY MAYDAY FUEL. Thirty minutes is a floor, not a target.
>
> *Workbook wording: What factors should you consider when deciding when to start diverting to your alternate?*

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

Not using your listed alternate. What do you weigh?

Aircraft state and fuel, enroute weather, runways and approaches, rescue and firefighting

Weigh configuration, weight, systems status and fuel remaining, then wind and weather at the diversion altitude, minimum enroute altitudes, and fuel burn to get there. At the field, weigh terrain, weather, wind, runways and surface condition, approach aids and lighting, and rescue and firefighting. Your familiarity with the airport counts too.

> **FOM 11.2.7.1 Nearest Suitable Airport**
>
> *"Aircraft configuration, weight, systems status, and fuel remaining • Wind and weather conditions enroute at the diversion altitude • Minimum altitudes enroute to the diversion airport • Fuel burn to the diversion airport"*
>
> With one engine out, having the fuel to go farther, passenger accommodation, and where maintenance sits are never justification for flying past the nearest suitable airport.
>
> *Workbook wording: If you are not diverting to your listed alternate, what factors should you consider before selecting the...*

LAHSO Policy

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is LAHSO?

Land and Hold Short Operations

LAHSO involves landing and holding short of an intersecting runway, an intersecting taxiway, or some other designated point on a runway. The Available Landing Distance (ALD) is measured from the landing threshold to the hold short point, and the hold short point is marked by a painted holding position marking and red and white hold short position signs.

> **FOM 5.6.32.1 Definitions LAHSO**
>
> *"A point on the runway beyond which a landing aircraft with a LAHSO clearance is not authorized to proceed."*
>
> Revisions to this section require FAA approval, which signals it is regulatory text, not company preference.

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What must you check before accepting LAHSO?

Stopping within the ALD, using AFM distance plus 1000 ft

The Captain may accept a LAHSO clearance only if every condition in 5.6.32.2 is satisfied. The HA determination is that existing conditions allow the aircraft to safely land and stop within the Available Landing Distance, where calculated landing distance equals the FAA-approved AFM distance plus 1000 ft for the actual configuration, environment and weight.

> **FOM 5.6.32.2 Operational Information LAHSO**
>
> *"The calculated landing distance will be the FAA-approved Aircraft Flight Manual (AFM) distance plus 1000 ft"*
>
> RECONCILE: Rev 125 revised LAHSO guidance to prohibit LAHSO on wet runways and require tailwind to be calm, less than 3 kts. Confirm the current four-factor list against the Rev 125.1 PDF before quoting it on the line.
>
> *Workbook wording: What are the 4 different factors that must be considered prior to accepting a LAHSO clearance?*

Complete Landing Performance Assessment

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What is a TLR?

Takeoff and Landing Report

FOM 5.6.22.1 sets the surrounding policy. Captains shall ensure landing is allowed and executed within the appropriate performance data. The landing data used for in-flight analysis is based on input conditions, which is what makes an accurate runway assessment critical.

> **FOM 5.6.22.1 Landing Policy**
>
> *"The landing data used for in-flight analysis is based on input conditions, this highlights the importance of an accurate runway assessment."*
>
> Acceptance is not weight-only: environmental conditions matter equally. Do not interchange the takeoff and landing crosswind tables, they serve different functions and the takeoff table is CG and weight sensitive while the TALPA landing table is not.

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

How do you request landing performance data?

Fill in Landing Conditions, press SEND, read Landing Data

Go to the Performance menu, open Landing Conditions, and enter airport, runway, wind, temperature, altimeter, runway condition, speed additive and reverse credit. Press SEND. When the data arrives you get a chime and a LANDING DATA READY message, and the LANDING DATA button appears.

FCOM PI.19.33 ACARS Landing Performance Request

> *"Once all required entries have been made, the SEND prompt is available for the flight crew to select on the LANDING CONDITIONS page."*

If the datalink request is unavailable, use the onboard performance tool on the electronic flight bag, or call Dispatch and ask for an analysis of the intended runway.

Workbook wording: How can you request landing data performance?

6.1.1.4 ASSESS OPERATIONAL AND ENVIRONMENTAL FACTORS FOR DESCENT AND ARRIVAL

What do HXX/TXX and LXX/RXX mean on the printout?

Headwind or tailwind knots, then crosswind from left or right

That field is the wind broken into components for the landing runway. H means headwind and T means tailwind, in knots. L means the crosswind is from the left and R means it is from the right, also in knots. So H12/R08 is 12 knots down the runway and 8 knots from the right.

FCOM PI.19.36 Landing Data Report (LDR)

> *"15. Headwind and crosswind (R = right, L = left)"*

The report legend spells out only R for right and L for left. Check these numbers against the reported surface wind before you accept the landing distances.

Workbook wording: On the landing data analysis printout, what do the designator/s "HXX/TXX" and "LXX/RXX" represent?

Altimeter procedures

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

When do you set QNH in the descent?

Descending below the transition level

Set the local altimeter as you descend below the transition level, as part of the Approach Procedure. Above the transition level you fly on standard, 29.92 in Hg or 1013.25 hectopascals. Outside North America the transition level is usually much lower than 18,000 ft, so it comes up early.

FCOM NP.21.40 Approach Procedure

> *"When descending below the transition level, set the altimeters."*

Transition altitudes are charted. Transition levels normally come from the arrival information broadcast or the approach controller, so listen for it and brief it.

Workbook wording: When, in the descent, do you change your altimeter setting to QNH (local Altimeter)?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

Transition level vs transition altitude?

Altitude is charted, flown on QNH; level is assigned, flown on standard

Transition altitude is the climb boundary. At and below it you use the local altimeter setting and call the number an altitude. Transition level is the descent boundary. At and above it you use standard and call the number a flight level. The gap between them is the transition layer.

FOM 18.1.6 International Altimeter Settings

> *"The aircraft shall be operated with the appropriate QNH value set below the Transition Altitude and STD (1013.25 hPa/29.92 in Hg) above the Transition Level."*

Outside North America both are usually lower than the 18,000 ft you are used to, and they vary by country. Central America uses FL200 and 19,000 ft unless charted otherwise.

Workbook wording: What is the difference between a transition level vs transition altitude?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

What altimeter setting defines a level?

Standard, 29.92 in Hg or 1013.25 hectopascals

A flight level is a height flown on the standard setting. Set 29.92 in Hg or 1013.25 hectopascals and the number you read is a flight level, spoken as flight level three five zero. It is a pressure surface, not a true height above the ground.

FOM 18.1.6 International Altimeter Settings

> *"The aircraft shall be operated with the appropriate QNH value set below the Transition Altitude and STD (1013.25 hPa/29.92 in Hg) above the Transition Level."*

Two airplanes on standard stay separated from each other even when the actual pressure is far from standard, which is the whole point of using it in the upper airspace.

Workbook wording: What (altimeter setting) defines a "level"?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

What altimeter setting defines an altitude?

QNH, the local altimeter setting

An altitude is a height above mean sea level flown on the local setting. Set the reported QNH, in inches of mercury or hectopascals, and the number you read is an altitude. You use it at and below the transition altitude, and it is what puts your indicator near field elevation on the ground.

FOM 18.1.6 International Altimeter Settings

> *"The aircraft shall be operated with the appropriate QNH value set below the Transition Altitude and STD (1013.25 hPa/29.92 in Hg) above the Transition Level."*

Outside North America most settings come in hectopascals. Confirm the unit before you set it, because 1002 hectopascals and 30.02 inches are very different worlds.

Workbook wording: What (altimeter setting) defines an "altitude"?

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

What is QFE?

Setting referenced to field elevation; altimeter reads height above the field

QFE is the pressure at the airport itself. Set it and the altimeter reads zero on the runway and height above the field in the air, not height above sea level. We do not fly it. If a controller gives you QFE, ask for QNH and set that instead.

FOM 18.1.6 International Altimeter Settings

> *"Where QFE is given by ATC, the Flight Crews shall request QNH and set in the normal manner."*

Integrated approach navigation and the automatic landing procedures are not authorized using QFE. Where charted altitudes are built on QFE, use the correction tables published on the arrival or approach chart.

Manually Perform Descent and Arrival (optional)

6.1.1 PERFORM DESCENT AND TERMINAL ARRIVAL

Is the autopilot required on an RNAV1 STAR?

No, but recommended in high workload environments

FOM 2.3.9.2 makes autopilot use a recommendation, not a requirement, and FOM 2.3.9.3 encourages hand flying in lower workload terminal environments to keep proficiency. The one place the FOM mandates automation for PBN is 5.1.4.1: RNP AR procedures with RNP values less than 0.3 or with RF legs shall use autopilot or flight director driven by the RNAV system.

FOM 2.3.9.2 Autopilot Basic Operating Procedures

> *"In high workload, high traffic environments, autopilot use is strongly recommended."*

INFERRED from absence: no autopilot requirement for an RNAV 1 STAR exists anywhere in FOM, FCOM or QRH. The only PBN autopilot mandate is FOM 5.1.4.1, and that applies to RNP AR only.

Workbook wording: Is the autopilot required to be engaged on an RNAV1 STAR?

8.2.1.5 CONDUCT APPROACH BRIEFING

What is in the arrival and approach TPC briefing?

Threats, Plan, Considerations. The PF starts it before descent

Prior to beginning the descent the PF initiates the approach briefing, which should be interactive, concise and focused on the big picture. T, Threats, shared: the PM identifies relevant threats using the threat list on the Crew Briefing Reference Card and any airport-specific threats in the 10-7, and the PF briefs additional threats as needed.

> **FOM 5.5.16 Threat-Based Approach Briefing/TPC**
>
> *"Pilots jointly identify any relevant threats, followed by the PF presenting the arrival plan"*
>
> Threats is the only shared element; Plan and Considerations are PF. Management strategies are discussed as threats are identified rather than saved to the end. The departure equivalent is briefed the same way, see FOM 1.3.1 in the workbook.
>
> *Workbook wording: What items compose the Arrival and Approach TPC Briefing?*

Flap and Gear Management

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

Normal flap and gear schedule on approach?

Flaps 1, 5, 20 at each speed bug; gear at glideslope alive

Work the flap schedule off the speed tape: at the UP bug call flaps 1, at the 1 bug call flaps 5, at the 5 bug call flaps 20. Gear down and flaps 20 go together at glideslope alive. At glideslope capture call landing flaps, 25 or 30.

> **FCOM NP.21.42 Landing Procedure ILS or GLS**
>
> *"At glideslope alive, call: • "GEAR DOWN" • "FLAPS 20" Set the landing gear lever to DN."*
>
> Flying an approach without a glideslope, gear and flaps 20 come about 2 nm before the final approach fix, once altitude hold or the vertical path mode is showing.
>
> *Workbook wording: What is the normal flap/gear approach configuration schedule?*

Stabilized Approach Criteria

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

Where are the stabilized approach criteria?

Fleet-specific manuals; FOM only points you there

FOM 5.6.5 gives the philosophy, speed, descent rate, lateral and vertical guidance, and configuration, plus the warning not to land from an unstable approach. The gate numbers themselves live in the flight crew training manual.

> **FOM 5.6.5 Stabilized Approach**
>
> *"Comply with the stabilized approach criteria and procedures contained in the fleet-specific manuals. An approach is stabilized only if all criteria are met."*
>
> FOM 5.6.5 carries the policy and the warning but explicitly defers the numeric criteria to the fleet manual (787 FCTM/AOM). The one hard number the FOM does give is runway alignment by 500 ft AFE, with 300 ft AFE exceptions annotated on the airport 10-7.
>
> *Workbook wording: Where is the Stabilized Approach Criteria parameters located?*

CRITICAL OBSERVABLE ITEM

8.1.1.5 CONFIGURE AIRPLANE FOR APPROACH/LANDING

Why go around if not stabilized?

Unstable approaches must not be landed from

FOM 5.6.5 pairs the WARNING with the no-fault go-around policy: either crewmember may call for a go-around and the professional judgment of the flight crew is used to assess the probability of meeting the criteria.

> **FOM 5.6.5 Stabilized Approach**
>
> *"Do not land from an unstable approach."*
>
> This is a WARNING in FOM 5.6.5, not a recommendation. Either crewmember may call the go-around under the Alaska no-fault policy, so an FO go-around call is not a challenge to your authority as PIC.
>
> *Workbook wording: Emphasize the importance of going around if the stabilized approach criteria are not met and the no fault...*

8.2.1 PERFORM CAT 1 ILS APPROACH WALKTHROUGH ITEM, PERFORM RATHER THAN RECALL

Demonstrate the callouts and configuration steps

Flaps 1, 5, 20, then landing flaps at displayed maneuver speeds

Boeing technique: when the speedtape displays the maneuver speed for your current flaps, select the next position, so UP to 1, 1 to 5, 5 to 20, then flaps 25 or 30 with VREF plus wind additives. Callouts and trigger points come from the FCOM normal procedures.

> **FCTM 5.5 Flap Extension Schedule**
>
> *"Flight profiles should be flown at, or slightly above, the recommended maneuver speed for the existing flap configuration"*
>
> Flying at or above the current flap maneuver speed keeps full 25 degree bank margin with a 15 degree overshoot.
>
> *Workbook wording: Discuss/demonstrate the callouts and configuration steps...*

8.2.1 PERFORM CAT 1 ILS APPROACH

What shows on the FMA with the approach armed?

Localizer and glideslope in small white letters below the active green modes

Push the approach switch and the armed roll and pitch modes appear at the bottom of the roll and pitch boxes, small and white on the primary flight display, green on the head up display. On an instrument landing system approach that is LOC and G/S. Flying integrated approach navigation you get FAC and G/P instead.

> **FCOM 4.20.7 AFDS Flight Mode Annunciations**
>
> *"Armed modes (except for TO/GA in the air) display in smaller letters (white on the PFD) at the bottom of the flight mode annunciation boxes."*
>
> Armed modes move up into large green letters as each one captures. Glideslope or glidepath capture is blocked until the localizer or course captures first.
>
> *Workbook wording: Discuss what is shown on FMA when the approach is armed...*

8.3.1 PERFORM MANAGED LATERAL/MANAGED VERTICAL APPROACH

Which modes are eligible for a non-precision approach?

IAN or VNAV

IAN drives the APP mode with an ILS-look-alike FAC and G/P presentation, so it is armed with the APP switch; VNAV flies the FMC path with LNAV or LOC as the roll mode. FCOM NP.21.45 adds that IAN is not recommended when the approach has a visual maneuver segment not in the FMC database.

> **FCOM SP.4 Instrument Approach - RNAV (RNP) AR (note); NP.21.45 and NP.21.48**
>
> *"Note: For RNAV (GPS) and RNAV (GNSS) procedures use the Landing Procedure - Instrument Approach using IAN or Landing Procedure - Instrument Approach using VNAV in Normal Procedures."*
>
> FCOM NP.21.45 (IAN) and NP.21.48 (VNAV) are the two non-precision landing procedures. Both require a published GP angle on the LEGS page, an RWxx waypoint at the approach end, or a missed approach waypoint before the runway end. Neither is authorized using QFE.
>
> *Workbook wording: What different approach modes(s) are eligible to perform a non-precision approach?*

8.3.2 PERFORM MANAGED LATERAL/SELECTED VERTICAL APPROACH (OPTIONAL)

Which approaches are flown managed or selected?

Vertical is always the FMS glide path; lateral is FMS or localizer

Integrated approach navigation flies every non precision approach like an instrument landing system approach, and vertical guidance is always the computed glide path. Lateral guidance comes from the final approach course mode on area navigation, satellite, omnidirectional range and beacon approaches. It comes from the localizer mode on localizer, directional aid and simplified directional facility approaches.

> **FCOM 4.20.19 Integrated Approach Navigation**
>
> *"Approaches flown with IAN procedures always use FMC computed glidepath (G/P) for vertical path guidance. Depending on the type of approach flown, lateral guidance can come from the FMC or localizer."*
>
> Arm both modes with the approach switch. Integrated approach navigation does not support an automatic landing, so disconnect the autopilot and land manually or you get a NO AUTOLAND alert at 100 ft.
>
> *Workbook wording: What approaches would be flown in Managed/Selected?*

8.1.1 PERFORM VISUAL APPROACH

Can you fly a visual approach at night?

Yes, unless that airport restricts night visuals

FOM 5.6.6.1 permits visual approaches with 1000 ft ceiling and 3 sm visibility or better, VFR cloud clearance, within 35 nm of the destination, under ATC control, with the airport or preceding aircraft in sight.

> **FOM 5.6.6.1 Visual Approach or RNAV Visual Flight Procedure (without CVFP)**
>
> *"Airports that restrict night visual approaches require that an instrument approach procedure be flown to guarantee adequate terrain clearance."*
>
> Night restrictions are published in the airport 10-7 (see FOM 5.6.6.3 note). FOM 5.6.6.1 also tells you to consider terrain a threat any time a night visual is planned or even possible.
>
> *Workbook wording: Can we fly a visual approach at night?*

Airspeed Control

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Is the autothrottle required on approach and landing?

No, recommended for all phases, not required

This is Boeing technique, not a limitation. Keep the autothrottle engaged even when hand flying, though manual thrust is fine for proficiency. On approach it protects speed in gusts by adjusting thrust quickly, with command speed at VREF plus 5 knots.

> **FCTM 1.40 Autothrottle Use**
>
> *"Autothrottle use is recommended during all phases of flight. When in manual flight, autothrottle use is also recommended, however manual thrust control may be used to maintain pilot proficiency"*
>
> If you plan to disconnect it before landing, apply the gust based wind additive to VREF instead.
>
> *Workbook wording: Is the auto-thrust/auto-throttle system required during approach and/or landing?*

Bounced Landing Recovery

8.5.1 PERFORM GO-AROUND/MISSED APPROACH

Key to bounced landing recovery without a tail strike?

Hold a normal landing attitude, do not increase pitch

Boeing technique: after a bounce, hold the normal landing attitude and add thrust only to control descent rate. Speedbrakes may still be extended after a shallow bounce, and raising the nose further in that state can strike the tail on the second touchdown. High hard bounce means go around.

> **FCTM 6.13 Bounced Landing Recovery**
>
> *"hold or re-establish a normal landing attitude and add thrust as necessary to control the rate of descent"*
>
> On a go around from a bounce, push the thrust levers up manually since TO/GA is inhibited near the ground.
>
> *Workbook wording: What is a key component during a bounced landing recovery to avoid a tail strike?*

10.1.2 PERFORM POSTFLIGHT

ACARS post-flight report pages?

All Autoland attempts and all RNP AR procedures

FOM 5.6.21.2 makes the Captain responsible for completing the ACARS Autoland Report page for every Autoland attempt, including the logbook page information. If ACARS/ATSU is inoperative, the Captain makes a logbook entry stating Autoland successful or unsuccessful with airport, runway and category, and the reason for any failure.

> **FOM 5.6.21.2 Autoland Reporting - ACARS; FOM 7.1.13.1 RNP AR Required Reporting**
>
> *"For all Autoland attempts, the Captain is responsible for ensuring that all applicable fields on the ACARS Autoland Report page are filled in, including the logbook page information."*
>
> Autoland reporting is every attempt, satisfactory or not. RNP AR reporting covers results of all RNP AR procedures plus any RNP degradation, and excludes visual approaches even if an RNP AR approach was displayed as reference (FOM 7.1.13.1 note).
>
> *Workbook wording: ACARS Post-Flight Report (Autoland and RNP-AR Report pages)*

Landing and Rollout (6)

Airspeed Control

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

What would make you turn the autothrottle off?

Approach category, wind and gust protection, plus the speed additive you lose

Think about three things. Approach category: the autothrottle is mandatory for CAT III and recommended for CAT II. Wind and gusts: with it active, VREF plus 5 already gives you gust protection, and you give that up when you turn it off. Windshear risk: if you plan to fly manual thrust, add a speed correction up to 15 knots.

> **FCOM 4.20.19 Automatic Flight, Approach and Landing**
>
> *"With a command speed of VREF+5 knots and landing flaps, there is sufficient wind and gust protection available with the autothrottle active."*
>
> Manual thrust also raises workload right when you need capacity for the runway and the go-around decision. Brief it and tell the First Officer what you want called.
>
> *Workbook wording: What factors should be considered if deciding to turn off the auto-thrust/auto-throttle system for the...*

Thrust Reverse Usage

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Is max reverse required on landing?

No, use reverse as required, up to maximum

Boeing technique, not a requirement. Raise the reverse levers to interlock right after touchdown and use the amount needed for the runway. Getting the desired reverse level in early matters most, since minimum reverse can double brake energy. Start reducing to idle by 60 knots.

> **FCTM 6.25 Reverse Thrust Operation**
>
> *"Maintain reverse thrust as required, up to maximum, until stopping on the remaining runway is assured"*
>
> With autobrakes, prompt full reverse lets the system cut brake pressure, saving brake temperature and wear.
>
> *Workbook wording: Is max thrust reverser required upon landing?*

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

When do you stow the thrust reversers?

Start at 60 knots, reverse idle before taxi speed

By 60 knots indicated, start moving the reverse thrust levers so they reach the reverse idle detent before taxi speed. Once the engines are at reverse idle, move the levers full down to stow. If you used maximum reverse and performance allows, begin reducing at 80 knots.

FCOM NP.21.52 Landing Roll Procedure

> *"reverse thrust levers to reach the reverse idle detent before taxi speed."*

Reducing early cuts engine wear from debris ingestion. Once the reverse levers are raised you are committed to a full stop landing.

Workbook wording: When should the thrust reversers be stowed?

Brakes Usage

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Braking technique with carbon brakes?

Fewer, steady applications with the pedals fully released between them

Carbon brake wear tracks the number of applications, not how hard you press. So brake in fewer, firmer, steady applications and let the airplane roll between them rather than riding the pedals. Below 30 knots the airplane applies half the brakes on each main gear and alternates wheel pairs, but only if you fully release the pedal.

FCOM 14.20.5 Taxi Brake Release

> *"The system sequences through alternating wheel pairs at each brake application, reducing the number of brake applications by each brake. The brake pedal must be fully released to transition to the other two brakes on a main landing gear."*

Dragging the brakes defeats the alternating logic and costs brake life. The full braking technique writeup is in the 787 flight crew training manual, not the operating manual.

Workbook wording: What braking technique is used with carbon brakes?

CRITICAL OBSERVABLE ITEM

9.1.1 PERFORM MANUAL LANDING

Factored vs unfactored landing data?

Factored adds margin. The 787 adds 15 percent

Unfactored landing distance is the raw computed distance. Factored distance applies a safety margin on top. Rev 125 moved the 787 to a TALPA landing model using a 7-second air/flare distance and a 15 percent margin on TOTAL landing distance including air distance, calculated to a computed rather than fixed touchdown point.

FOM 5.6.22.2 / 5.6.22.3 Landing Performance

> *"this distance includes air-run to touchdown plus ground roll and a 15% safety factor for degraded braking action requests"*

Read 5.6.22.3 in full for the landing distance wording. The model adds a 7 second air and flare distance and a 15 percent margin on the total landing distance, calculated rather than assuming a fixed touchdown point.

Workbook wording: What is the difference between factored vs unfactored landing data?

9.1.1.14 PERFORM FCOM PROCEDURE(S) FOLLOWING LANDING

What are the FCOM procedures after landing?

After Landing, Shutdown, then Secure Procedure

After Landing (NP.21.53): speedbrake down verified by the Captain, APU as needed, engine anti-ice as needed, exterior lights, weather radar off, autobrake OFF, flaps UP, transponder to XPDR, and the COMPANY ARRIVAL REPORT page ON/IN times and landing pilot.

FCOM NP.21.53 After Landing Procedure

> *"Start the After Landing Procedure when clear of the active runway."*

After Landing starts when clear of the runway (FCOM NP.21.53), Shutdown starts after taxi is complete (NP.21.54), Secure is run any time the flight deck will be left unattended (NP.21.56).

MEL and Maintenance (17)

ETOPS Verification Flight

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Must the verification request be in the logbook before departure?

Yes

The Flight Crew is notified of an open item on the MIC sheet. Maintenance places this entry in the Aircraft Maintenance Logbook: AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF ______.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF"*
>
> *Workbook wording: Does the verification flight item request have to be in the maintenance logbook prior to departure?*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

Must Maintenance talk to the crew before an EVF?

Yes, verbally

Maintenance will verbally notify the Flight Crew and place the release entry in the Aircraft Maintenance Logbook. The notification is therefore threefold: the MIC sheet open item, the verbal notification, and the logbook entry.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"Maintenance will also verbally notify the Flight Crew"*
>
> *Workbook wording: Is Maintenance required to communicate with the flight crew prior to departing on an ETOPS verification...*

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

After talking to Maintenance, must the PIC call Dispatch?

Yes, a verbal acknowledgement

That closes the loop between Maintenance, the crew and Dispatch before departure.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"The Flight Crew must verbally acknowledge the EVF requirement with Dispatch."*
>
> *Workbook wording: Does the PIC have to communicate with dispatch after speaking with Maintenance?*

ETOPS APU Verification Flight

16.1.1.7.6 ETOPS VERIFICATION FLIGHT PROCEDURE

What logbook entry is required after the verification flight?

Verification flight completed satisfactorily or unsatisfactorily

The Captain must make an entry in the Aircraft Maintenance Logbook detailing the results: VERIFICATION FLIGHT FOR ____ COMPLETED SATISFACTORILY or UNSATISFACTORILY. If the test was unsuccessful the Captain must list any discrepancies. For the APU specifically, a successful entry must state: VERIFICATION FLIGHT FOR APU COMPLETED, APU OPERATES NORMALLY.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"VERIFICATION FLIGHT FOR ____ COMPLETED "SATISFACTORILY" or "UNSATISFACTORILY"*
>
> Captain writes this, not the FO and not Maintenance. The APU wording is distinct from the generic component wording, so know both.
>
> *Workbook wording: What maintenance logbook entry (specific verbiage) must be entered after the verification flight?*

Company Reporting

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

How do you report a discrepancy before landing?

ACARS or radio to Dispatch, who relays to Maintenance

Report through Dispatch using ACARS or radio. FOM 11.1.2 requires contacting Dispatch for abnormalities as safety of flight permits, and if time does not permit in flight, after gate arrival. Reporting early lets Maintenance position parts and personnel at the gate rather than starting the troubleshooting clock on block-in.

> **FOM 11.1.2 Abnormal Operations Dispatch Notification**
>
> *"If time does not permit contact with Dispatch while in flight, contact shall be made after gate arrival."*
>
> That is also the answer to the advantages question in the maintenance logbook section: enroute notification converts ground time from diagnosis into repair.
>
> *Workbook wording: How can you report a discrepancy to Maintenance in advance, prior to landing?*

SATCOM / HF Procedures

CRITICAL OBSERVABLE ITEM

5.1.1.5 PERFORM COMMUNICATIONS/ADMINISTRATIVE FUNCTIONS DURING CRUISE

Who do you call on SATCOM for maintenance or a guest issue?

Dispatch

Inflight, the Dispatcher is the primary contact and shares operational control with the Captain (FOM 7.1.1.2). ACARS messages sent to Maintenance Control go automatically to the Dispatcher, and Maintenance does not normally communicate directly with an airborne aircraft because their diagnostic guidance differs from operational guidance for crews.

> **FOM 7.1.1.2 In-Flight Operations**
>
> *"The Dispatcher is the primary contact for requests and notifications, both managing information flow and sharing operational control with the Captain."*
>
> Maintenance will not normally talk directly to an airborne aircraft; the FODO provides the link to Maintenance and Engineering. For a medical event, MedLink is the SATCOM call (FOM 7.1.7).
>
> *Workbook wording: When we make a SATCOM call for Maintenance or guest issue, who do we call?*

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

Where are the logbook review procedures?

Maintenance Logbook Preflight Review, in the FOM maintenance chapter.

Hawaiian procedures sit under Maintenance Logbook Procedures, Preflight Review. Every pilot reviews at least the last 10 pilot entered discrepancies, not counting No Items or Info to Maintenance entries. Alaska has its own preflight review paragraph a few pages earlier in the same chapter.

> **FOM 12.4.12.1 Maintenance Logbook Preflight Review**
>
> *"Prior to the first flight on a particular aircraft, all Flight Crew shall review the Aircraft Maintenance Logbook."*
>
> Deferred items get a second look. Check the MEL, CDL or SDL for restrictions, operational procedures and performance penalties, and confirm every one is on the release.
>
> *Workbook wording: Where in the FOM can you find the maintenance logbook review/checking procedures?*

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

How long is the PDSC good for?

Four hours from the Airworthiness Release signature.

The clock starts when the Airworthiness Release block is signed in the M258 logbook, not at scheduled departure. If four hours pass and boarding has not started, the check is invalid and Maintenance must redo it. If boarding started in time you may continue, but tell Maintenance Control.

FOM 6.5.4 ETOPS Pre-Departure Service Check, PDSC Policy

"An ETOPS PDSC Airworthiness Release will be valid for four (4) hours from the time the Airworthiness Release block was signed in the M258 Aircraft Maintenance Logbook."

An aircraft swap always needs a new check. The check belongs to that tail and that flight number and cannot follow the airplane to another flight.

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

White vs yellow MEL placard?

White: standard deferral placard. Yellow: deferral needing repetitive maintenance action.

Both placards go in the plastic placard page of the logbook, with an INOP sticker on or near the inoperative component. White HAL 46-1333 covers any deferred item. Yellow HAL 46-1333b is used when the deferral carries a repetitive maintenance action in the REMARKS or EXCEPTIONS column.

MEL 2.9.4 and 2.9.5 Aircraft Placards

"When an item is deferred and requires a repetitive maintenance action in the REMARKS or EXCEPTIONS column"

The FOM separately uses the number HAL 46-1333 for the Project Operation and Maintenance Information Form. For deferral paperwork, the MEL placard descriptions govern.

Workbook wording: What is the difference between a white (HAL 46-1333) vs yellow (HAL 46-1333b) MEL placard?

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

What are the four repair intervals?

Categories A, B, C, and D.

Repair categories run A through D. Category A carries its own limit written into the MEL item itself, while B, C and D are fixed day limits published in the MEL general pages. All four are dispatch limits, not limits on the flight once it is under way.

FOM 8.5.4.1 Dispatch Release Sample and Description

"The repair categories of A, B, C, or D in an MEL are dispatch time limits not flight operational time limits."

Dispatch happens at pushback or engine start, whichever comes first. Start inside the limit and you may finish that flight, but you may not continue past the next termination point.

Workbook wording: What are the 4 different repair intervals...

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

What is an (O) procedure in the MEL?

A crew operational procedure required with that item inoperative.

An (O) means the flight crew has something to do to operate with that item deferred, such as a limit, a workaround or an extra check. Read it during the logbook review, before you accept the airplane. An (M) is the maintenance side, done by mechanics.

FOM 12.4.12.1 Maintenance Logbook Preflight Review

"all Flight Crew shall review the aircraft MEL/CDL/SDL for applicability, restrictions, MEL Operational Procedures (O), and performance decrements."

The full wording lives in the MEL introduction. The FOM requires you to review deferred items for applicability, restrictions, operational procedures and performance decrements.

Workbook wording: What is an "(O)" procedure listed in an MEL?

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

Where is the crew placarding procedure?

Maintenance chapter of the FOM, and Crew Placarding in the MEL.

The FOM paragraph sets the conditions: a discrepancy after pushback but before takeoff, with Dispatch and Maintenance Control both concurring, and no return to the gate needed. The step by step placarding text is in the MEL under Crew

Placarding. Remote placarding follows right after it.

> **FOM 12.4.12.9 Crew Placarding Procedure**
>
> *"If the item is Crew Placardable in the MEL/CDL/SDL, follow the appropriate procedures as described in the MEL/CDL/SDL under Crew Placarding and a return to gate (BTB) is not required."*
>
> Maintenance Control normally sends the whole item by ACARS. Print it, check the blocks, do the placarding and any procedures, and note the item on the release.
>
> *Workbook wording: Where can you find the Crew Placarding Procedure?*

Maintenance Communication Procedure

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

EVF communication requirements with Maintenance?

Call or ACARS Dispatch 30-60 minutes after takeoff, before EEP

FOM 6.5.10 (HA): the crew is notified of an open item on the MIC sheet and Maintenance also verbally notifies the crew. The logbook entry reads AIRCRAFT RELEASED TO ETOPS SERVICE IAW WITH HA ETOPS MAINTENANCE MANUAL FOR VERIFICATION FLIGHT DUE TO REPAIR OF ___. The crew verbally acknowledges the EVF requirement with Dispatch.

> **FOM 6.5.10 ETOPS Verification Flight (HA)**
>
> *"Between 30 and 60 minutes after takeoff and prior to reaching the EEP, the Flight Crew will call or send an ACARS message to Dispatch to report the status of the affected component or system."*
>
> Gotcha: the in-flight report goes to Dispatch, not to Maintenance. Maintenance's role is the up-front verbal notification plus the logbook entry, and the crew verbally acknowledges the EVF requirement with Dispatch. The report AND Dispatch's acknowledgment must both be complete before the EEP.
>
> *Workbook wording: What are the ETOPS verification flight communication requirements regarding contacting Maintenance...*

CRITICAL OBSERVABLE ITEM

1.2.1.2.5.3 KNOW MAINTENANCE LOGBOOK PROCEDURES

Why tell Maintenance enroute rather than at the gate?

The arrival station can have Maintenance ready before you block in.

Airborne, put the discrepancy in the logbook and pass it to Maintenance and Dispatch by the ACARS maintenance report page or company radio. Telling them early is what lets the inbound station plan for the airplane. Waiting until block in gives them no lead time.

> **FOM 12.4.5.3 After Takeoff**
>
> *"Communicate defects via ACARS MAINT REPORT page (primary) or Company radio. Ensure the inbound station is aware of the defect."*
>
> MEL and crew placarding procedures are not used in flight. Enroute you report and coordinate. The deferral or repair happens on the ground.
>
> *Workbook wording: What are the advantages of notifying MX of a discrepancy enroute (time permitting) vs after blocking in?*

APU In-Flight Start Program (APU component currency)

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What release items matter for the APU in-flight start program?

The APU Administrative Control Item and any conflicting MEL items

Per FOM 12.4.12.5, a Maintenance request for an APU in-flight start is issued as an Administrative Control Item (ACI) found in Section 05 of the fleet MEL, and that ACI is reflected on the Dispatch Release.

FOM 12.4.12.5 Maintenance Requests / FOM 6.5.6 APU In-Flight Start Program (HA)

> *"These requests will be in the form of an Administrative Control Item (ACI) as found in Section 05 of the fleet MEL. The associated ACI will be applied to the aircraft and will be reflected on the Dispatch Release."*

The ACI comes from Section 05 of the fleet MEL. Crew is also notified by an open item on the MIC sheet and a placard card (form HAL-46-1333) in the AML per FOM 6.5.6.

Workbook wording: What items on the dispatch release must you take note of when planning an APU In-Flight Start Program...

Verification flight vs APU in-flight start program

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What goes in the logbook after the flight?

No Items, Info to Maintenance, or a discrepancy, plus signature.

If you are done with the airplane, the Captain closes out the logbook. Record No Items, Info to Maintenance, or the mechanical discrepancy, then sign and add your employee number on the proper lines. Info to Maintenance is information only, so never use it to preface a real discrepancy.

FOM 12.4.12.4 Post-Flight Logbook Closeout

> *"This can be accomplished by recording "No Items," "Info to Maintenance," or a mechanical discrepancy as appropriate and entering their signature and employee number on the appropriate lines."*

Date entries by current GMT wherever you are, one discrepancy per block, and leave the Action Taken column alone.

Workbook wording: What items must be entered into the maintenance logbook after flight completion?

CRITICAL OBSERVABLE ITEM

16.1.1.7 PERFORM ETOPS PROCEDURES

What aircraft currency items are the ACIs?

Max thrust takeoff, autoland, and APU in-flight start.

Maintenance asks for these through an Administrative Control Item in Section 05 of the MEL, and it shows up on the release. Typical requests are a max thrust takeoff, an autoland, or an APU in flight start.

FOM 12.4.12.5 Maintenance Requests

A completed item can still appear on a later release that was built before you finished it. That is normal, not a second request.

Workbook wording: What A/C Currency Items are listed in the "Administrative Control Items" (ACI's)?

General Operations (1)

12.1.3.16 KNOW FOM PA POLICIES AND PERFORM EFFECTIVE COMMUNICATION FROM THE FLIGHT DECK

PA PRIO vs PA ALL?

PRIO is priority. ALL includes crew rest

The 787 audio control panel provides selectable PA transmit options. PA ALL keys the passenger address announcement from that station to all PA areas, which includes the crew rest facilities. The priority selection is used when the announcement must override lower-priority audio.

FCOM 5.30 Communications, Passenger Address System

> *"keys passenger address announcement from this station to all PA areas 7 Receiver Lights Illuminated (green) – indicates the respective receiver volume control is manually selected on."*

Confirm the exact PRIO versus ALL behaviour and the crew-rest exclusion selection against FCOM 5.30.2 in the PDF.

Workbook wording: What is the difference between PA PRIO and PA ALL selections?

North Atlantic: NAT Doc 007 Notes (30)

Every place NAT Doc 007, the ICAO North Atlantic Operations and Airspace Manual V.2026-1, speaks to a question in this workbook. The company manuals govern in every case; these are the ICAO references behind them, collected for review before a North Atlantic crossing. Each note also appears on its card.

Named vs unnamed waypoint?

Named is in the database. Unnamed is lat/long or hand entered

NAT Doc 007 6.2.7: "independently verify the full latitude and longitude coordinates of “unnamed” (Lat/Long) waypoints" ICAO draws the same line, calling lat/long waypoints unnamed and requiring both pilots to verify their full coordinates.

Where is country-specific guidance?

FOM theater chapters plus Jeppesen State Rules and Procedures

NAT Doc 007 Foreword: "This Document is for guidance only. Regulatory material relating to North Atlantic aircraft operations" For the NAT theater this manual sits behind the FOM chapter; the regulatory material lives in ICAO Annexes, State AIPs and NOTAMs.

What altitudes are RVSM airspace?

FL290 through FL410

NAT Doc 007 Chapter 11: "This is defined as any airspace between FL 290 - FL 410 inclusive" ICAO defines NAT RVSM airspace the same way as the company, FL290 through FL410 inclusive.

RVSM equipment requirements?

Both primary altimeters, one autopilot, altitude alerting

NAT Doc 007 6.3.12: "Required equipment includes two primary independent altimetry sources, one altitude alert" The sentence continues with one automatic altitude control system, so ICAO matches the company list of altimeters, autopilot and altitude alerting.

RVSM equipment fails in flight. What do you do?

Notify ATC and Dispatch

NAT Doc 007 7.2.1: "The following equipment failures must be reported to ATC as soon as practicable" ICAO matches the notify ATC step; adding Dispatch is company procedure.

Diverting without a clearance. What altitude do you plan?

Below FL290

NAT Doc 007 10.3.2: "descend below FL 290, and establish a 150 m (500 ft) vertical offset from those flight levels" ICAO matches planning below FL 290 and adds a 500 ft vertical offset from levels normally used.

Airborne CPDLC logon window before the FIR?

A time window before entry, not an altitude

NAT Doc 007 6.3.41: "the pilot should log on to the appropriate FIR" ICAO sets the window at 10 to 25 minutes before the boundary, matching the company time based rather than altitude based trigger.

Two minutes before a waypoint, what do you do?

Confirm the next waypoint and the one after it

NAT Doc 007 6.3.51: "Approaching an oceanic waypoint, pilots should crosscheck the coordinates of the next" ICAO ties the check to approaching the waypoint and includes the next plus one, matching the company two waypoint confirmation.

What gets checked after waypoint transition?

Time and fuel recorded, difference computed, navigation mode verified, waypoint slashed

NAT Doc 007 6.3.54: "Pilots should also note and record their fuel status at each oceanic waypoint." ICAO matches the fuel recording at each waypoint; 6.3.55 adds the LNAV tracking check 10 minutes after passage.

Where is fuel remaining recorded at waypoint transition?

The flight plan, the release, or the ACARS HOWGOZIT

NAT Doc 007 6.3.54: "Pilots should also note and record their fuel status at each oceanic waypoint." ICAO matches the company practice; where to write it, release or HOWGOZIT, is company guidance.

Ten minutes after waypoint transition, what do you do?

Crosscheck navigation performance and course

NAT Doc 007 6.3.55: "pilots should confirm that proper lateral (LNAV/NAV) mode is engaged and the aircraft is tracking to the proper waypoint" ICAO matches the company crosscheck and names exactly what to verify at the 10 minute point.

Why track fuel score?

It reveals fuel leaks and burn trends early

NAT Doc 007 6.3.54: "Pilots should also note and record their fuel status at each oceanic waypoint. This is especially important if the cleared route and flight level differ significantly from the filed flight plan." ICAO requires the record; the leak detection rationale is the company addition.

Where is the HF position report template?

Jeppesen FD Pro X, Pubs, Pacific, Oceanic Position Reporting Procedures

NAT Doc 007 5.3.6: "position should be expressed in terms of latitude and longitude except when flying over named reporting points" For the NAT the report format sits in this chapter of Doc 007; the company Pacific reference covers the Oakland FIR.

Where are the ICAO oceanic weather deviation procedures?

Our Flight Operations Manual oceanic chapter, reprinted from ICAO Doc 4444

NAT Doc 007 10.4: "The following procedures are intended for deviations around adverse meteorological conditions." Doc 007 Chapter 10.4 carries the same ICAO weather deviation procedures for the NAT that the FOM reprints.

How do you request a weather deviation?

State WEATHER DEVIATION REQUIRED by voice, or send a datalink request

NAT Doc 007 10.4.1: "stating “WEATHER DEVIATION REQUIRED” to indicate that priority is desired on the frequency and for ATC response" ICAO matches the company procedure and adds the urgency call PAN PAN when needed.

How do you insert the offset in the CDU?

LATERAL OFFSET page, enter L or R and distance, execute

NAT Doc 007 6.3.49: "Pilots should make sure the “TO” waypoint is correct after entering SLOP." Doc 007 leaves the CDU steps to the aircraft manuals but adds this check because some FMS sequence past a waypoint.

Engine failure in oceanic airspace. What do you do?

Nearest suitable airport, after getting a revised oceanic clearance

NAT Doc 007 10.2.1: "a revised clearance shall be obtained, whenever possible, prior to initiating any action" ICAO matches the company procedure of getting a revised clearance before leaving the track.

Driftdown off the route. Which way do you turn?

At least 30 degrees, left or right per listed factors

NAT Doc 007 10.2.1: "leave the cleared route or track by initially turning at least 30 degrees to the right or to the left" Same turn as the FOM, and Doc 007 lists the same direction factors, adjacent tracks, alternate, terrain.

What is SLOP for?

Collision risk and wake turbulence

NAT Doc 007 6.3.48: "This procedure distributes traffic between the centreline and 2 NM right of centreline and greatly reduces collision risk in the airspace by virtue of randomness" ICAO gives the same purposes and also directs SLOP when avoiding wake turbulence.

How far can you offset with SLOP?

Up to 2 nm right of centerline. Never left

NAT Doc 007 6.3.49: "Operators that have an automatic offset capability should fly up to 2 NM right of the centreline." ICAO matches the company limit, prohibits left offsets, and with 0.1 nm capability the 787 may fly any tenth up to 2 nm right.

When is an HF position report required?

Only when you are not on CPDLC

NAT Doc 007 5.1.17: "flight crews of flights using ADS-C should not additionally submit position reports via voice unless requested by aeronautical radio operator" ICAO matches the company rule and separately requires the SELCAL check before oceanic entry.

Can you enter oceanic airspace with both HF radios out?

No. At least one HF radio is required

NAT Doc 007 5.1.1: "Operations in the NAT outside VHF coverage require the carriage of two long range communication systems, one of which must be HF." ICAO matches the company requirement and allows SATVOICE or CPDLC over SATCOM as the second system.

NUUK FIR. Who controls what, and what do you do?

Nuuk owns it; fly Reykjavik OCA north, Gander OCA south

NAT Doc 007 6.3.41: "the pilot should log on to the appropriate FIR 10 to 25 minutes prior to the boundary" ICAO gives the generic logon timing; the company chapter names the controlling OCA for the Nuuk FIR.

Can you cross the NAT-HLA without HF?

No. At least one HF radio is required

NAT Doc 007 4.2.12: "When operating outside of VHF coverage the carriage of fully functioning HF is mandatory" ICAO matches the company rule that HF is required to cross the NAT outside VHF coverage.

GPS jamming reporting requirements?

Report to ATC; file a safety report

NAT Doc 007 9.1.10: "Flight crews should inform ATC of any GNSS malfunction. ATC aircraft separation minimums may be affected by the GNSS malfunction." ICAO matches the report to ATC; the safety report is a company addition.

What is your oceanic exit point?

Flight specific; the point re-entering domestic airspace

NAT Doc 007 6.3.61: "If ATC assigns a fixed Mach number in oceanic airspace, request NORMAL SPEED (via CPDLC or voice) after the OXP in Domestic airspace." ICAO matches the company procedure for resuming ECON after the OXP.

Initial call to Gander or Shanwick. Exact words?

Callsign, SELCAL check, and the next OCA

NAT Doc 007 5.1.13: "after the radio operator responds, request a SELCAL check and state the next OCA" ICAO matches the company model call word for word, including the Gander Radio example.

When do you request a SELCAL check?

After CPDLC logon, before the initial gateway fix

NAT Doc 007 5.1.22: "check the operation of the SELCAL equipment, at or prior to entry into oceanic airspace, with the appropriate radio station." The company timing, after CPDLC logon and before the gateway fix, satisfies this ICAO deadline.

Review the airspace around eastbound oceanic entry

GOTA transition area, then the Gander OCA

NAT Doc 007 3.3.12: "Air Traffic service is provided by the Gander ACC, call sign GANDER CENTRE." ICAO confirms GOTA traffic stays with domestic Gander Centre, matching the company description.

Westbound into BIRD at RATSU. Oceanic clearance needed?

No. Only Shanwick still issues oceanic clearances

NAT Doc 007 6.3.35: "Enroute aircraft shall enter the oceanic airspace in accordance with their existing ATC clearance. No oceanic clearance is required." ICAO matches the company answer; a separate note keeps Shanwick issuing clearances until further notice.